SampleSize_n100_10Treatments/table2_nonshrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	96.55	96.55	93.10	96.55	100.00	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	85.71	96.43	92.86	100.00	96.43	85.71	96.43	92.86	92.86	67.86
20%	85.71	96.43	96.43	100.00	85.71	89.29	92.86	92.86	92.86	100.00	64.29
30%	89.66	89.66	96.55	96.55	96.55	96.55	89.66	82.76	96.55	96.55	62.07
40%	93.10	93.10	96.55	100.00	93.10	93.10	93.10	93.10	89.66	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	82.14	92.86	89.29	100.00	96.43	85.71	96.43	96.43	92.86	60.71
20%	82.14	92.86	96.43	100.00	89.29	89.29	89.29	85.71	89.29	96.43	57.14
30%	89.66	86.21	96.55	96.55	96.55	96.55	86.21	86.21	89.66	96.55	58.62
40%	93.10	93.10	93.10	96.55	93.10	93.10	89.66	82.76	89.66	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	89.29	92.86	92.86	100.00	92.86	85.71	96.43	96.43	96.43	64.29
20%	85.71	96.43	96.43	100.00	89.29	92.86	89.29	89.29	92.86	100.00	64.29
30%	89.66	86.21	96.55	96.55	96.55	96.55	89.66	86.21	93.10	96.55	58.62
40%	89.66	93.10	96.55	86.21	89.66	89.66	89.66	86.21	89.66	89.66	51.72
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	93.10	93.10	96.55	96.55	100.00	93.10	96.55	100.00	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_nonshrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	92.86	100.00	96.43	96.43	89.29	92.86	96.43	100.00	92.86	67.86
20%	96.55	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	89.66	96.55	96.55	93.10	96.55	86.21	86.21	89.66	100.00	89.66	68.97
40%	92.86	89.29	96.43	92.86	96.43	96.43	96.43	96.43	92.86	96.43	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	96.43	96.43	92.86	96.43	96.43	100.00	92.86	67.86
20%	93.10	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	82.76	96.55	96.55	89.66	93.10	89.66	89.66	86.21	96.55	86.21	65.52
40%	92.86	85.71	96.43	89.29	96.43	96.43	92.86	96.43	89.29	96.43	57.14
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	96.43	89.29	89.29	96.43	100.00	92.86	64.29
20%	93.10	89.66	100.00	96.55	96.55	89.66	93.10	93.10	89.66	93.10	65.52
30%	86.21	96.55	96.55	93.10	96.55	86.21	86.21	82.76	96.55	86.21	62.07
40%	96.43	85.71	96.43	89.29	96.43	96.43	96.43	96.43	92.86	100.00	57.14

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	96.55	93.10	93.10	96.55	96.55	100	82.76
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	96.43	100.00	100.00	96.43	100	85.71
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	93.10	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_nonshrink_10_treatments_MAR_n100

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	92.86	96.43	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	92.86
20%	100.00	96.55	100.00	100.00	100	100.00	100.00	96.55	93.10	100.00	86.21
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	100.00	96.43	96.43	100	100.00	100.00	100.00	96.43	100.00	89.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	100.00	100.00	100	100.00	96.43	100.00	100.00	100.00	89.29
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100.00	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	96.55	89.66
40%	100.00	96.43	96.43	96.43	100	100.00	96.43	100.00	96.43	100.00	85.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	92.86
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100.00	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	92.86	100.00	100.00	100	100.00	96.43	100.00	96.43	100.00	89.29

SampleSize_n100_10Treatments/table2_shrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	96.55	96.55	93.10	96.55	100.00	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	85.71	96.43	92.86	100.00	96.43	85.71	96.43	92.86	92.86	67.86
20%	85.71	96.43	96.43	100.00	85.71	89.29	92.86	92.86	92.86	100.00	64.29
30%	89.66	89.66	96.55	96.55	96.55	96.55	89.66	82.76	96.55	96.55	62.07
40%	93.10	93.10	96.55	100.00	93.10	93.10	93.10	93.10	89.66	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	82.14	92.86	89.29	100.00	96.43	85.71	96.43	96.43	92.86	60.71
20%	82.14	92.86	96.43	100.00	89.29	89.29	89.29	85.71	89.29	96.43	57.14
30%	89.66	86.21	96.55	96.55	96.55	96.55	86.21	86.21	89.66	96.55	58.62
40%	93.10	93.10	93.10	96.55	93.10	93.10	89.66	82.76	89.66	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	89.29	92.86	92.86	100.00	92.86	85.71	96.43	96.43	96.43	64.29
20%	85.71	96.43	96.43	100.00	89.29	92.86	89.29	89.29	92.86	100.00	64.29
30%	89.66	86.21	96.55	96.55	96.55	96.55	89.66	86.21	93.10	96.55	58.62
40%	89.66	93.10	96.55	86.21	89.66	89.66	89.66	86.21	89.66	89.66	51.72
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	93.10	93.10	96.55	96.55	100.00	93.10	96.55	100.00	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_shrink_10_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	92.86	100.00	96.43	96.43	89.29	92.86	96.43	100.00	92.86	67.86
20%	96.55	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	89.66	96.55	96.55	93.10	96.55	86.21	86.21	89.66	100.00	89.66	68.97
40%	92.86	89.29	96.43	92.86	96.43	96.43	96.43	96.43	92.86	96.43	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	96.43	96.43	92.86	96.43	96.43	100.00	92.86	67.86
20%	93.10	89.66	100.00	96.55	96.55	86.21	93.10	93.10	86.21	93.10	62.07
30%	82.76	96.55	96.55	89.66	93.10	89.66	89.66	86.21	96.55	86.21	65.52
40%	92.86	85.71	96.43	89.29	96.43	96.43	92.86	96.43	89.29	96.43	57.14
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	96.43	89.29	89.29	96.43	100.00	92.86	64.29
20%	93.10	89.66	100.00	96.55	96.55	89.66	93.10	93.10	89.66	93.10	65.52
30%	86.21	96.55	96.55	93.10	96.55	86.21	86.21	82.76	96.55	86.21	62.07
40%	96.43	85.71	96.43	89.29	96.43	96.43	96.43	96.43	92.86	100.00	57.14

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	96.43	100.00	100.00	100.00	100.00	100.00	100	89.29
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	96.55	100.00	100.00	96.55	93.10	93.10	96.55	96.55	100	82.76
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.43	100.00	100.00	96.43	96.43	100.00	96.43	100.00	100	96.43
30%	100.00	100.00	100.00	100.00	96.55	100.00	96.55	96.55	100.00	100	89.66
40%	100.00	93.10	100.00	100.00	100.00	100.00	96.55	96.55	100.00	100	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	100.00	96.55	100.00	100.00	100.00	100.00	96.55	100.00	100	89.66
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_10Treatments/table2_shrink_10_treatments_MAR_n100

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	96.43	96.43	100.00	96.43	100	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	96.55	100.00	100.00	100	100.00	100.00	96.55	93.10	100	86.21
30%	100.00	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100	93.10
40%	100.00	100.00	96.43	96.43	100	100.00	100.00	100.00	96.43	100	89.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	100.00	96.43	100	100.00	96.43	100.00	100.00	100	89.29
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100	93.10
40%	100.00	100.00	96.43	96.43	100	100.00	96.43	100.00	96.43	100	89.29
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	100.00	96.43	100	100.00	100.00	100.00	100.00	100	92.86
20%	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	93.10	100	89.66
30%	96.55	100.00	100.00	100.00	100	96.55	96.55	100.00	100.00	100	93.10
40%	100.00	92.86	100.00	100.00	100	100.00	96.43	100.00	96.43	100	89.29

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.757	1.761	1.770	1.795	1.795	1.796	1.842	1.813	1.839	1.841	2.417
Imputation using linear regression											
10%	1.836 (4.49)	1.863 (5.77)	1.876 (6.00)	1.857 (3.47)	1.853 (3.20)	1.906 (6.11)	1.869 (1.50)	1.905 (5.06)	1.921 (4.46)	1.909 (3.67)	2.533 (4.80)
20%	1.950 (10.97)	1.946 (10.47)	1.958 (10.62)	1.987 (10.70)	1.966 (9.51)	2.024 (12.69)	2.001 (8.63)	2.018 (11.29)	2.012 (9.45)	2.009 (9.12)	2.689 (11.26)
30%	2.104 (19.71)	2.102 (19.32)	2.133 (20.53)	2.142 (19.32)	2.142 (19.31)	2.172 (20.93)	2.152 (16.85)	2.186 (20.56)	2.153 (17.08)	2.204 (19.70)	2.901 (20.03)
40%	2.324 (32.22)	2.293 (30.16)	2.327 (31.47)	2.328 (29.71)	2.338 (30.22)	2.329 (29.64)	2.318 (25.89)	2.384 (31.51)	2.403 (30.70)	2.406 (30.65)	3.201 (32.46)
Imputation using predictive mean matching method											
10%	1.820 (3.54)	1.850 (5.05)	1.854 (4.73)	1.841 (2.57)	1.847 (2.88)	1.883 (4.81)	1.853 (0.63)	1.890 (4.27)	1.905 (3.63)	1.902 (3.30)	2.509 (3.82)
20%	1.912 (8.81)	1.912 (8.57)	1.913 (8.11)	1.930 (7.52)	1.930 (7.53)	1.984 (10.43)	1.945 (5.63)	1.956 (7.87)	1.969 (7.08)	1.976 (7.29)	2.639 (9.19)
30%	1.987 (13.07)	2.013 (14.28)	2.042 (15.39)	2.062 (14.85)	2.029 (13.03)	2.076 (15.58)	2.036 (10.55)	2.081 (14.78)	2.075 (12.86)	2.084 (13.16)	2.740 (13.36)
40%	2.182 (24.15)	2.180 (23.74)	2.178 (23.08)	2.201 (22.61)	2.228 (24.13)	2.239 (24.67)	2.209 (19.94)	2.211 (21.94)	2.258 (22.80)	2.214 (20.24)	3.007 (24.42)
Imputation using propensity−score method											
10%	1.823 (3.71)	1.857 (5.41)	1.854 (4.73)	1.845 (2.78)	1.842 (2.60)	1.895 (5.50)	1.861 (1.04)	1.892 (4.35)	1.922 (4.52)	1.901 (3.25)	2.513 (3.99)
20%	1.915 (8.98)	1.932 (9.68)	1.928 (8.93)	1.951 (8.71)	1.949 (8.60)	2.005 (11.64)	1.963 (6.60)	1.976 (9.01)	1.989 (8.18)	1.997 (8.43)	2.643 (9.36)
30%	2.041 (16.15)	2.063 (17.10)	2.039 (15.21)	2.085 (16.13)	2.079 (15.84)	2.077 (15.65)	2.048 (11.19)	2.105 (16.09)	2.085 (13.39)	2.127 (15.53)	2.818 (16.59)
40%	2.218 (26.22)	2.230 (26.63)	2.228 (25.89)	2.270 (26.46)	2.259 (25.85)	2.293 (27.66)	2.288 (24.23)	2.279 (25.72)	2.318 (26.07)	2.283 (24.00)	3.056 (26.45)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.761	1.765	1.777	1.771	1.808	1.795	1.801	1.821	1.819	1.840	2.418
Imputation using linear regression											

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.825 (3.63)	1.815 (2.82)	1.850 (4.12)	1.879 (6.09)	1.858 (2.82)	1.877 (4.58)	1.859 (3.21)	1.909 (4.84)	1.890 (3.88)	1.892 (2.81)	2.521 (4.26)
20%	1.960 (11.28)	1.950 (10.43)	1.977 (11.28)	1.970 (11.22)	1.981 (9.62)	1.986 (10.64)	2.000 (11.04)	2.005 (10.13)	2.000 (9.95)	2.030 (10.33)	2.704 (11.83)
30%	2.068 (17.39)	2.073 (17.39)	2.046 (15.16)	2.069 (16.81)	2.101 (16.22)	2.103 (17.14)	2.142 (18.94)	2.156 (18.42)	2.193 (20.57)	2.176 (18.25)	2.858 (18.22)
40%	2.250 (27.76)	2.276 (28.89)	2.230 (25.51)	2.276 (28.50)	2.249 (24.41)	2.363 (31.63)	2.316 (28.61)	2.305 (26.61)	2.329 (28.05)	2.389 (29.85)	3.109 (28.59)
Imputation using predictive mean matching method											
10%	1.804 (2.41)	1.800 (1.97)	1.839 (3.53)	1.860 (5.04)	1.839 (1.72)	1.866 (3.96)	1.838 (2.04)	1.888 (3.69)	1.878 (3.26)	1.879 (2.10)	2.490 (2.99)
20%	1.918 (8.92)	1.917 (8.60)	1.932 (8.74)	1.947 (9.95)	1.957 (8.28)	1.952 (8.74)	1.962 (8.94)	1.961 (7.70)	1.961 (7.80)	1.992 (8.27)	2.645 (9.43)
30%	1.949 (10.66)	1.974 (11.78)	1.947 (9.61)	1.966 (11.01)	2.015 (11.47)	1.984 (10.51)	2.048 (13.69)	2.006 (10.16)	2.072 (13.94)	2.036 (10.64)	2.695 (11.47)
40%	2.136 (21.29)	2.159 (22.32)	2.155 (21.29)	2.146 (21.19)	2.162 (19.61)	2.212 (23.20)	2.201 (22.20)	2.209 (21.32)	2.209 (21.44)	2.238 (21.60)	2.945 (21.81)
Imputation using propensity−score method											
10%	1.808 (2.64)	1.799 (1.89)	1.838 (3.48)	1.870 (5.60)	1.848 (2.25)	1.868 (4.03)	1.843 (2.35)	1.896 (4.16)	1.878 (3.27)	1.883 (2.33)	2.496 (3.24)
20%	1.931 (9.64)	1.935 (9.59)	1.950 (9.77)	1.954 (10.35)	1.975 (9.28)	1.976 (10.08)	1.971 (9.44)	1.993 (9.44)	1.978 (8.77)	2.002 (8.81)	2.664 (10.18)
30%	1.999 (13.46)	2.007 (13.69)	2.005 (12.86)	1.984 (12.05)	2.033 (12.48)	2.030 (13.10)	2.080 (15.49)	2.038 (11.95)	2.125 (16.81)	2.068 (12.38)	2.765 (14.37)
40%	2.171 (23.23)	2.208 (25.05)	2.188 (23.18)	2.261 (27.67)	2.236 (23.69)	2.311 (28.74)	2.265 (25.79)	2.295 (26.05)	2.238 (23.05)	2.282 (24.01)	2.990 (23.68)

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	2.422	2.434	2.469	2.469	2.470	2.533	2.493	2.529	2.532
Imputation using linear regression									
10%	2.570 (6.08)	2.587 (6.30)	2.561 (3.76)	2.555 (3.50)	2.629 (6.42)	2.578 (1.78)	2.627 (5.37)	2.649 (4.75)	2.633 (3.97)
20%	2.684 (10.79)	2.699 (10.91)	2.740 (11.00)	2.711 (9.82)	2.791 (13.00)	2.758 (8.91)	2.781 (11.56)	2.775 (9.73)	2.771 (9.42)
30%	2.898 (19.65)	2.940 (20.82)	2.954 (19.67)	2.953 (19.61)	2.995 (21.23)	2.966 (17.11)	3.013 (20.83)	2.969 (17.41)	3.038 (19.98)
40%	3.159 (30.43)	3.206 (31.74)	3.208 (29.97)	3.221 (30.49)	3.209 (29.90)	3.197 (26.21)	3.285 (31.76)	3.311 (30.93)	3.315 (30.90)
Imputation using predictive mean matching method									
10%	2.551 (5.33)	2.556 (5.00)	2.539 (2.84)	2.546 (3.15)	2.596 (5.09)	2.555 (0.89)	2.607 (4.55)	2.627 (3.90)	2.623 (3.57)
20%	2.639 (8.97)	2.640 (8.49)	2.664 (7.90)	2.664 (7.92)	2.738 (10.83)	2.685 (6.01)	2.699 (8.25)	2.717 (7.44)	2.727 (7.68)
30%	2.775 (14.57)	2.814 (15.63)	2.842 (15.14)	2.798 (13.32)	2.861 (15.84)	2.806 (10.80)	2.868 (15.05)	2.861 (13.14)	2.872 (13.42)
40%	3.003 (23.99)	3.001 (23.33)	3.033 (22.85)	3.071 (24.38)	3.085 (24.90)	3.044 (20.19)	3.047 (22.21)	3.110 (22.99)	3.052 (20.53)
Imputation using propensity−score method									
10%	2.560 (5.69)	2.556 (5.01)	2.544 (3.04)	2.540 (2.87)	2.613 (5.78)	2.566 (1.30)	2.608 (4.62)	2.650 (4.79)	2.622 (3.52)
20%	2.666 (10.05)	2.660 (9.30)	2.693 (9.09)	2.690 (8.98)	2.767 (12.01)	2.709 (6.97)	2.727 (9.37)	2.745 (8.54)	2.755 (8.80)
30%	2.848 (17.56)	2.815 (15.68)	2.879 (16.61)	2.870 (16.27)	2.869 (16.14)	2.828 (11.64)	2.906 (16.55)	2.880 (13.88)	2.937 (15.98)
40%	3.072 (26.83)	3.070 (26.14)	3.126 (26.64)	3.112 (26.08)	3.158 (27.83)	3.151 (24.42)	3.140 (25.95)	3.193 (26.26)	3.146 (24.23)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	2.423	2.439	2.431	2.481	2.464	2.472	2.499	2.497	2.526
Imputation using linear regression									

SampleSize_n100_10Treatments/table3_nonshrink_10_treatments_MAR_n100

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.506 (3.44)	2.554 (4.74)	2.594 (6.73)	2.566 (3.44)	2.593 (5.21)	2.567 (3.83)	2.636 (5.47)	2.609 (4.51)	2.612 (3.43)
20%	2.690 (11.00)	2.728 (11.86)	2.717 (11.78)	2.734 (10.18)	2.740 (11.20)	2.758 (11.58)	2.766 (10.68)	2.759 (10.51)	2.801 (10.90)
30%	2.865 (18.22)	2.828 (15.97)	2.859 (17.63)	2.904 (17.04)	2.906 (17.94)	2.961 (19.78)	2.980 (19.22)	3.031 (21.40)	3.006 (19.02)
40%	3.145 (29.77)	3.083 (26.42)	3.145 (29.38)	3.109 (25.32)	3.267 (32.57)	3.201 (29.49)	3.186 (27.48)	3.219 (28.94)	3.302 (30.72)
Imputation using predictive mean matching method									
10%	2.485 (2.54)	2.539 (4.11)	2.568 (5.63)	2.538 (2.30)	2.576 (4.55)	2.537 (2.62)	2.606 (4.28)	2.593 (3.84)	2.593 (2.68)
20%	2.644 (9.12)	2.665 (9.27)	2.685 (10.47)	2.699 (8.78)	2.692 (9.26)	2.706 (9.44)	2.705 (8.23)	2.704 (8.32)	2.748 (8.80)
30%	2.729 (12.61)	2.693 (10.42)	2.718 (11.81)	2.785 (12.25)	2.743 (11.32)	2.831 (14.50)	2.774 (10.98)	2.865 (14.77)	2.815 (11.43)
40%	2.977 (22.86)	2.971 (21.82)	2.960 (21.76)	2.981 (20.16)	3.051 (23.81)	3.035 (22.77)	3.046 (21.87)	3.046 (22.02)	3.086 (22.18)
Imputation using propensity−score method									
10%	2.483 (2.47)	2.538 (4.07)	2.581 (6.20)	2.552 (2.84)	2.578 (4.63)	2.545 (2.94)	2.618 (4.76)	2.593 (3.86)	2.600 (2.92)
20%	2.668 (10.12)	2.690 (10.31)	2.695 (10.88)	2.724 (9.80)	2.726 (10.61)	2.719 (9.97)	2.748 (9.97)	2.729 (9.30)	2.762 (9.35)
30%	2.777 (14.60)	2.774 (13.75)	2.746 (12.96)	2.813 (13.37)	2.809 (14.00)	2.877 (16.39)	2.821 (12.87)	2.939 (17.72)	2.861 (13.28)
40%	3.042 (25.54)	3.016 (23.67)	3.114 (28.10)	3.080 (24.13)	3.184 (29.23)	3.122 (26.29)	3.162 (26.51)	3.085 (23.57)	3.145 (24.52)

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.757	1.761	1.770	1.795	1.795	1.796	1.842	1.813	1.839	1.841	2.443
Imputation using linear regression											
10%	1.836 (4.49)	1.863 (5.77)	1.876 (6.00)	1.857 (3.47)	1.853 (3.20)	1.906 (6.11)	1.869 (1.50)	1.905 (5.06)	1.921 (4.46)	1.909 (3.67)	2.542 (4.06)
20%	1.950 (10.97)	1.946 (10.47)	1.958 (10.62)	1.987 (10.70)	1.966 (9.51)	2.024 (12.69)	2.001 (8.63)	2.018 (11.29)	2.012 (9.45)	2.009 (9.12)	2.695 (10.34)
30%	2.104 (19.71)	2.102 (19.32)	2.133 (20.53)	2.142 (19.32)	2.142 (19.31)	2.172 (20.93)	2.152 (16.85)	2.186 (20.56)	2.153 (17.08)	2.204 (19.70)	2.907 (19.02)
40%	2.324 (32.22)	2.293 (30.16)	2.327 (31.47)	2.328 (29.71)	2.338 (30.22)	2.329 (29.64)	2.318 (25.89)	2.384 (31.51)	2.403 (30.70)	2.406 (30.65)	3.197 (30.90)
Imputation using predictive mean matching method											
10%	1.820 (3.54)	1.850 (5.05)	1.854 (4.73)	1.841 (2.57)	1.847 (2.88)	1.883 (4.81)	1.853 (0.63)	1.890 (4.27)	1.905 (3.63)	1.902 (3.30)	2.521 (3.21)
20%	1.912 (8.81)	1.912 (8.57)	1.913 (8.11)	1.930 (7.52)	1.930 (7.53)	1.984 (10.43)	1.945 (5.63)	1.956 (7.87)	1.969 (7.08)	1.976 (7.29)	2.645 (8.29)
30%	1.987 (13.07)	2.013 (14.28)	2.042 (15.39)	2.062 (14.85)	2.029 (13.03)	2.076 (15.58)	2.036 (10.55)	2.081 (14.78)	2.075 (12.86)	2.084 (13.16)	2.745 (12.37)
40%	2.182 (24.15)	2.180 (23.74)	2.178 (23.08)	2.201 (22.61)	2.228 (24.13)	2.239 (24.67)	2.209 (19.94)	2.211 (21.94)	2.258 (22.80)	2.214 (20.24)	3.001 (22.85)
Imputation using propensity−score method											
10%	1.823 (3.71)	1.857 (5.41)	1.854 (4.73)	1.845 (2.78)	1.842 (2.60)	1.895 (5.50)	1.861 (1.04)	1.892 (4.35)	1.922 (4.52)	1.901 (3.25)	2.524 (3.34)
20%	1.915 (8.98)	1.932 (9.68)	1.928 (8.93)	1.951 (8.71)	1.949 (8.60)	2.005 (11.64)	1.963 (6.60)	1.976 (9.01)	1.989 (8.18)	1.997 (8.43)	2.651 (8.52)
30%	2.041 (16.15)	2.063 (17.10)	2.039 (15.21)	2.085 (16.13)	2.079 (15.84)	2.077 (15.65)	2.048 (11.19)	2.105 (16.09)	2.085 (13.39)	2.127 (15.53)	2.828 (15.77)
40%	2.218 (26.22)	2.230 (26.63)	2.228 (25.89)	2.270 (26.46)	2.259 (25.85)	2.293 (27.66)	2.288 (24.23)	2.279 (25.72)	2.318 (26.07)	2.283 (24.00)	3.054 (25.01)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.761	1.765	1.777	1.771	1.808	1.795	1.801	1.821	1.819	1.840	2.418
Imputation using linear regression											

SampleSize_n100_10Treatments/table3_shrink_10_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.825 (3.63)	1.815 (2.82)	1.850 (4.12)	1.879 (6.09)	1.858 (2.82)	1.877 (4.58)	1.859 (3.21)	1.909 (4.84)	1.890 (3.88)	1.892 (2.81)	2.535 (4.81)
20%	1.960 (11.28)	1.950 (10.43)	1.977 (11.28)	1.970 (11.22)	1.981 (9.62)	1.986 (10.64)	2.000 (11.04)	2.005 (10.13)	2.000 (9.95)	2.030 (10.33)	2.715 (12.25)
30%	2.068 (17.39)	2.073 (17.39)	2.046 (15.16)	2.069 (16.81)	2.101 (16.22)	2.103 (17.14)	2.142 (18.94)	2.156 (18.42)	2.193 (20.57)	2.176 (18.25)	2.867 (18.53)
40%	2.250 (27.76)	2.276 (28.89)	2.230 (25.51)	2.276 (28.50)	2.249 (24.41)	2.363 (31.63)	2.316 (28.61)	2.305 (26.61)	2.329 (28.05)	2.389 (29.85)	3.133 (29.54)
Imputation using predictive mean matching method											
10%	1.804 (2.41)	1.800 (1.97)	1.839 (3.53)	1.860 (5.04)	1.839 (1.72)	1.866 (3.96)	1.838 (2.04)	1.888 (3.69)	1.878 (3.26)	1.879 (2.10)	2.505 (3.59)
20%	1.918 (8.92)	1.917 (8.60)	1.932 (8.74)	1.947 (9.95)	1.957 (8.28)	1.952 (8.74)	1.962 (8.94)	1.961 (7.70)	1.961 (7.80)	1.992 (8.27)	2.657 (9.87)
30%	1.949 (10.66)	1.974 (11.78)	1.947 (9.61)	1.966 (11.01)	2.015 (11.47)	1.984 (10.51)	2.048 (13.69)	2.006 (10.16)	2.072 (13.94)	2.036 (10.64)	2.710 (12.04)
40%	2.136 (21.29)	2.159 (22.32)	2.155 (21.29)	2.146 (21.19)	2.162 (19.61)	2.212 (23.20)	2.201 (22.20)	2.209 (21.32)	2.209 (21.44)	2.238 (21.60)	2.972 (22.87)
Imputation using propensity−score method											
10%	1.808 (2.64)	1.799 (1.89)	1.838 (3.48)	1.870 (5.60)	1.848 (2.25)	1.868 (4.03)	1.843 (2.35)	1.896 (4.16)	1.878 (3.27)	1.883 (2.33)	2.512 (3.86)
20%	1.931 (9.64)	1.935 (9.59)	1.950 (9.77)	1.954 (10.35)	1.975 (9.28)	1.976 (10.08)	1.971 (9.44)	1.993 (9.44)	1.978 (8.77)	2.002 (8.81)	2.676 (10.66)
30%	1.999 (13.46)	2.007 (13.69)	2.005 (12.86)	1.984 (12.05)	2.033 (12.48)	2.030 (13.10)	2.080 (15.49)	2.038 (11.95)	2.125 (16.81)	2.068 (12.38)	2.775 (14.75)
40%	2.171 (23.23)	2.208 (25.05)	2.188 (23.18)	2.261 (27.67)	2.236 (23.69)	2.311 (28.74)	2.265 (25.79)	2.295 (26.05)	2.238 (23.05)	2.282 (24.01)	3.013 (24.60)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	2.445	2.442	2.451	2.454	2.450	2.463	2.460	2.468	2.468
Imputation using linear regression									
10%	2.547 (4.20)	2.568 (5.15)	2.550 (4.02)	2.542 (3.61)	2.581 (5.36)	2.547 (3.45)	2.577 (4.76)	2.582 (4.62)	2.576 (4.40)
20%	2.688 (9.95)	2.705 (10.76)	2.722 (11.04)	2.696 (9.90)	2.747 (12.10)	2.736 (11.10)	2.758 (12.10)	2.743 (11.11)	2.727 (10.51)
30%	2.894 (18.38)	2.934 (20.14)	2.924 (19.32)	2.938 (19.73)	2.969 (21.19)	2.955 (20.00)	2.987 (21.44)	2.935 (18.90)	3.002 (21.65)
40%	3.156 (29.11)	3.198 (30.94)	3.195 (30.36)	3.204 (30.59)	3.197 (30.49)	3.177 (29.01)	3.259 (32.46)	3.284 (33.05)	3.277 (32.80)
Imputation using predictive mean matching method									
10%	2.531 (3.53)	2.540 (3.98)	2.532 (3.30)	2.533 (3.23)	2.547 (3.97)	2.526 (2.57)	2.557 (3.96)	2.558 (3.62)	2.565 (3.92)
20%	2.640 (8.01)	2.650 (8.49)	2.648 (8.05)	2.649 (7.98)	2.695 (10.00)	2.661 (8.06)	2.675 (8.76)	2.689 (8.93)	2.680 (8.59)
30%	2.768 (13.22)	2.812 (15.14)	2.810 (14.64)	2.777 (13.16)	2.838 (15.83)	2.798 (13.63)	2.836 (15.29)	2.822 (14.34)	2.835 (14.86)
40%	2.996 (22.57)	2.995 (22.63)	3.017 (23.09)	3.047 (24.18)	3.063 (25.03)	3.019 (22.61)	3.016 (22.60)	3.083 (24.90)	3.016 (22.20)
Imputation using propensity−score method									
10%	2.538 (3.83)	2.539 (3.97)	2.535 (3.41)	2.527 (3.00)	2.565 (4.71)	2.536 (2.98)	2.560 (4.05)	2.580 (4.52)	2.565 (3.94)
20%	2.669 (9.16)	2.669 (9.29)	2.679 (9.28)	2.675 (9.02)	2.725 (11.22)	2.685 (9.05)	2.705 (9.95)	2.716 (10.03)	2.710 (9.79)
30%	2.845 (16.36)	2.814 (15.21)	2.849 (16.24)	2.855 (16.38)	2.848 (16.25)	2.821 (14.56)	2.878 (17.00)	2.843 (15.18)	2.903 (17.62)
40%	3.068 (25.49)	3.064 (25.46)	3.115 (27.10)	3.096 (26.19)	3.141 (28.19)	3.132 (27.17)	3.114 (26.61)	3.167 (28.29)	3.116 (26.24)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	2.420	2.421	2.424	2.435	2.428	2.436	2.436	2.439	2.445
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.530 (4.55)	2.549 (5.27)	2.559 (5.58)	2.552 (4.82)	2.553 (5.17)	2.547 (4.56)	2.584 (6.05)	2.561 (5.01)	2.568 (5.02)
20%	2.701 (11.61)	2.719 (12.32)	2.714 (11.98)	2.722 (11.77)	2.724 (12.21)	2.736 (12.32)	2.736 (12.31)	2.729 (11.89)	2.752 (12.54)
30%	2.866 (18.44)	2.842 (17.39)	2.863 (18.15)	2.894 (18.85)	2.899 (19.41)	2.934 (20.47)	2.959 (21.46)	2.994 (22.75)	2.980 (21.87)
40%	3.151 (30.22)	3.091 (27.68)	3.146 (29.80)	3.108 (27.63)	3.237 (33.32)	3.190 (30.96)	3.171 (30.14)	3.199 (31.17)	3.274 (33.88)
Imputation using predictive mean matching method									
10%	2.508 (3.65)	2.531 (4.53)	2.532 (4.47)	2.523 (3.63)	2.537 (4.48)	2.519 (3.42)	2.555 (4.88)	2.543 (4.27)	2.546 (4.14)
20%	2.658 (9.82)	2.662 (9.94)	2.684 (10.75)	2.686 (10.32)	2.678 (10.31)	2.682 (10.11)	2.677 (9.89)	2.676 (9.72)	2.701 (10.45)
30%	2.727 (12.69)	2.705 (11.73)	2.726 (12.48)	2.777 (14.06)	2.735 (12.66)	2.808 (15.28)	2.750 (12.88)	2.823 (15.74)	2.786 (13.94)
40%	2.991 (23.58)	2.978 (23.02)	2.962 (22.23)	2.981 (22.41)	3.018 (24.32)	3.021 (24.04)	3.027 (24.25)	3.025 (24.01)	3.056 (24.99)
Imputation using propensity−score method									
10%	2.507 (3.61)	2.531 (4.53)	2.548 (5.14)	2.538 (4.24)	2.538 (4.55)	2.525 (3.68)	2.569 (5.45)	2.546 (4.38)	2.554 (4.46)
20%	2.681 (10.79)	2.686 (10.93)	2.690 (11.01)	2.714 (11.44)	2.710 (11.62)	2.694 (10.59)	2.720 (11.67)	2.699 (10.67)	2.714 (10.99)
30%	2.777 (14.74)	2.787 (15.11)	2.752 (13.55)	2.803 (15.11)	2.799 (15.28)	2.854 (17.16)	2.799 (14.91)	2.899 (18.88)	2.833 (15.84)
40%	3.054 (26.18)	3.022 (24.81)	3.113 (28.45)	3.077 (26.36)	3.153 (29.88)	3.108 (27.62)	3.143 (29.02)	3.065 (25.68)	3.116 (27.44)

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	86.21	89.66	89.66	93.10	93.10	93.10	93.10	96.55	93.10	96.55	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	92.86	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	93.10	93.10	96.55	93.10	68.97
30%	86.21	100.00	79.31	89.66	93.10	89.66	96.55	89.66	86.21	96.55	48.28
40%	89.29	96.43	85.71	100.00	89.29	96.43	89.29	92.86	92.86	92.86	57.14
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	96.43	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	96.55	93.10	96.55	89.66	72.41
30%	82.76	100.00	75.86	89.66	89.66	93.10	100.00	86.21	86.21	86.21	44.83
40%	85.71	100.00	85.71	96.43	85.71	96.43	89.29	92.86	92.86	89.29	53.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	93.10	96.55	100.00	96.55	96.55	93.10	96.55	89.66	96.55	89.66	72.41
30%	79.31	96.55	82.76	86.21	93.10	89.66	100.00	89.66	86.21	93.10	55.17
40%	82.14	96.43	92.86	96.43	85.71	100.00	85.71	92.86	96.43	89.29	46.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	89.29	96.43	89.29	89.29	89.29	92.86	89.29	89.29	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	96.55	96.55	89.66	89.66	96.55	89.66	86.21	93.10	86.21	55.17
30%	100.00	96.55	96.55	100.00	96.55	96.55	100.00	100.00	96.55	96.55	79.31
40%	92.86	92.86	92.86	96.43	89.29	82.14	89.29	92.86	92.86	89.29	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	89.66	93.10	89.66	89.66	89.66	89.66	86.21	93.10	82.76	48.28
30%	100.00	89.66	93.10	96.55	96.55	93.10	100.00	100.00	100.00	96.55	72.41
40%	92.86	92.86	92.86	96.43	89.29	78.57	89.29	89.29	100.00	92.86	50.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	100.00	96.43	92.86	100.00	100.00	100.00	96.43	100.00	100.00	75.00
20%	86.21	93.10	93.10	86.21	89.66	93.10	89.66	89.66	93.10	82.76	51.72
30%	100.00	93.10	100.00	100.00	96.55	100.00	100.00	100.00	100.00	93.10	82.76
40%	96.43	92.86	92.86	96.43	89.29	82.14	92.86	89.29	100.00	92.86	60.71

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	96.55	100.00	96.55	100.00	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	96.43
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	96.55	100.00	100.00	100.00	100.00	93.10	100.00	100.00	100.00	100.00	93.10
40%	96.43	100.00	100.00	100.00	100.00	100.00	96.43	100.00	100.00	100.00	92.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	96.43
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	89.66	100.00	96.55	100.00	100.00	96.55	100.00	100.00	96.55	96.55	89.66
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	96.55	100.00	96.55	100.00	96.55	96.55	100.00	100.00	100.00	96.55	89.66
40%	96.43	100.00	100.00	100.00	96.43	100.00	96.43	100.00	100.00	100.00	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	96.43	100.00	100.00	96.43	100.00	96.43	89.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_nonshrink_10_treatments_MAR_n150

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	96.55	100.00	100	100.00	96.55	96.55	100.00	100.00	89.66
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	100.00	96.43	100	96.43	100.00	100.00	100.00	96.43	92.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	100.00	100.00	100	100.00	96.55	89.66	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	96.43	96.43	100	96.43	92.86	96.43	100.00	100.00	85.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	96.43
20%	96.55	100.00	96.55	100.00	100	100.00	96.55	93.10	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	96.43	100.00	96.43	100	96.43	96.43	96.43	100.00	100.00	89.29

SampleSize_n150_10Treatments/table2_shrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	86.21	89.66	89.66	93.10	93.10	93.10	93.10	96.55	93.10	96.55	65.52
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	92.86	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	93.10	93.10	96.55	93.10	68.97
30%	86.21	100.00	79.31	89.66	93.10	89.66	96.55	89.66	86.21	96.55	48.28
40%	89.29	96.43	85.71	100.00	89.29	96.43	89.29	92.86	92.86	92.86	57.14
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	96.43	92.86	100.00	96.43	96.43	96.43	78.57
20%	96.55	100.00	100.00	96.55	96.55	93.10	96.55	93.10	96.55	89.66	72.41
30%	82.76	100.00	75.86	89.66	89.66	93.10	100.00	86.21	86.21	86.21	44.83
40%	85.71	100.00	85.71	96.43	85.71	96.43	89.29	92.86	92.86	89.29	53.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	89.29	100.00	100.00	92.86	100.00	96.43	96.43	96.43	78.57
20%	93.10	96.55	100.00	96.55	96.55	93.10	96.55	89.66	96.55	89.66	72.41
30%	79.31	96.55	82.76	86.21	93.10	89.66	100.00	89.66	86.21	93.10	55.17
40%	82.14	96.43	92.86	96.43	85.71	100.00	85.71	92.86	96.43	89.29	46.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	89.29	96.43	89.29	89.29	89.29	92.86	89.29	89.29	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_shrink_10_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	96.55	96.55	89.66	89.66	96.55	89.66	86.21	93.10	86.21	55.17
30%	100.00	96.55	96.55	100.00	96.55	96.55	100.00	100.00	96.55	96.55	79.31
40%	92.86	92.86	92.86	96.43	89.29	82.14	89.29	92.86	92.86	89.29	64.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	100.00	100.00	100.00	100.00	100.00	82.14
20%	86.21	89.66	93.10	89.66	89.66	89.66	89.66	86.21	93.10	82.76	48.28
30%	100.00	89.66	93.10	96.55	96.55	93.10	100.00	100.00	100.00	96.55	72.41
40%	92.86	92.86	92.86	96.43	89.29	78.57	89.29	89.29	100.00	92.86	50.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	89.29	100.00	96.43	92.86	100.00	100.00	100.00	96.43	100.00	100.00	75.00
20%	86.21	93.10	93.10	86.21	89.66	93.10	89.66	89.66	93.10	82.76	51.72
30%	100.00	93.10	100.00	100.00	96.55	100.00	100.00	100.00	100.00	93.10	82.76
40%	96.43	92.86	92.86	96.43	89.29	82.14	92.86	89.29	100.00	92.86	60.71

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	96.55	100.00	96.55	100.00	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	96.55	100.00	93.10
30%	96.55	100.00	96.55	100.00	100.00	93.10	100.00	96.55	100.00	100.00	86.21
40%	96.43	100.00	100.00	100.00	100.00	100.00	96.43	100.00	100.00	100.00	92.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	96.55	100.00	93.10
30%	89.66	100.00	96.55	100.00	100.00	96.55	100.00	100.00	96.55	96.55	89.66
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
20%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55	100.00	96.55
30%	96.55	100.00	96.55	100.00	96.55	93.10	100.00	100.00	100.00	96.55	86.21
40%	96.43	100.00	100.00	100.00	96.43	100.00	96.43	100.00	100.00	100.00	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100.00	100.00	96.43	100.00	100.00	96.43	100.00	96.43	89.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_10Treatments/table2_shrink_10_treatments_MAR_n150

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	100.00	100.00	100	100.00	96.55	96.55	100.00	100.00	93.10
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	96.43	96.43	100	96.43	100.00	100.00	100.00	96.43	89.29
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	100.00	100.00	100	100.00	96.55	89.66	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	100.00	96.43	96.43	100	96.43	92.86	96.43	100.00	100.00	85.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
20%	96.55	100.00	96.55	100.00	100	100.00	96.55	93.10	96.55	96.55	86.21
30%	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00
40%	96.43	96.43	100.00	96.43	100	96.43	96.43	96.43	100.00	100.00	89.29

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.449	1.445	1.453	1.462	1.485	1.493	1.480	1.489	1.504	1.515	1.992
Imputation using linear regression											
10%	1.502 (3.68)	1.498 (3.69)	1.497 (2.99)	1.523 (4.21)	1.520 (2.35)	1.534 (2.74)	1.537 (3.86)	1.544 (3.69)	1.545 (2.72)	1.557 (2.80)	2.074 (4.11)
20%	1.594 (10.08)	1.584 (9.65)	1.571 (8.12)	1.589 (8.69)	1.609 (8.31)	1.626 (8.87)	1.626 (9.87)	1.619 (8.71)	1.617 (7.50)	1.646 (8.64)	2.201 (10.53)
30%	1.707 (17.84)	1.705 (18.03)	1.721 (18.45)	1.740 (19.04)	1.759 (18.42)	1.729 (15.77)	1.768 (19.50)	1.787 (20.00)	1.753 (16.60)	1.807 (19.26)	2.358 (18.40)
40%	1.852 (27.89)	1.876 (29.86)	1.875 (29.01)	1.867 (27.71)	1.893 (27.43)	1.909 (27.84)	1.909 (29.05)	1.934 (29.89)	1.927 (28.14)	1.971 (30.08)	2.557 (28.39)
Imputation using predictive mean matching method											
10%	1.493 (3.09)	1.483 (2.67)	1.491 (2.63)	1.524 (4.26)	1.521 (2.39)	1.527 (2.24)	1.537 (3.91)	1.544 (3.71)	1.534 (2.02)	1.559 (2.93)	2.061 (3.46)
20%	1.555 (7.34)	1.552 (7.45)	1.543 (6.20)	1.575 (7.74)	1.591 (7.12)	1.601 (7.20)	1.610 (8.79)	1.594 (7.02)	1.590 (5.74)	1.632 (7.71)	2.147 (7.79)
30%	1.628 (12.41)	1.619 (12.05)	1.663 (14.44)	1.677 (14.73)	1.664 (12.05)	1.671 (11.92)	1.712 (15.74)	1.705 (14.53)	1.704 (13.31)	1.743 (15.03)	2.250 (12.94)
40%	1.809 (24.86)	1.797 (24.35)	1.827 (25.73)	1.836 (25.62)	1.835 (23.55)	1.822 (22.04)	1.858 (25.57)	1.830 (22.87)	1.849 (22.98)	1.904 (25.66)	2.489 (24.99)
Imputation using propensity−score method											
10%	1.490 (2.87)	1.487 (2.92)	1.490 (2.52)	1.520 (3.99)	1.516 (2.07)	1.530 (2.46)	1.531 (3.51)	1.539 (3.33)	1.532 (1.88)	1.558 (2.83)	2.058 (3.31)
20%	1.553 (7.25)	1.557 (7.75)	1.534 (5.58)	1.577 (7.92)	1.585 (6.69)	1.619 (8.39)	1.614 (9.11)	1.601 (7.53)	1.592 (5.87)	1.633 (7.81)	2.147 (7.82)
30%	1.661 (14.64)	1.633 (13.05)	1.691 (16.38)	1.690 (15.63)	1.690 (13.77)	1.706 (14.23)	1.728 (16.77)	1.726 (15.91)	1.731 (15.14)	1.737 (14.65)	2.297 (15.33)
40%	1.823 (25.83)	1.840 (27.36)	1.863 (28.18)	1.850 (26.55)	1.841 (23.95)	1.876 (25.64)	1.909 (29.03)	1.919 (28.86)	1.912 (27.17)	1.910 (26.05)	2.515 (26.27)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.421	1.426	1.446	1.446	1.452	1.454	1.473	1.468	1.473	1.466	1.957
Imputation using linear regression											

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.498 (5.41)	1.524 (6.82)	1.512 (4.57)	1.521 (5.21)	1.519 (4.63)	1.534 (5.47)	1.526 (3.62)	1.550 (5.56)	1.563 (6.13)	1.564 (6.66)	2.066 (5.56)
20%	1.570 (10.45)	1.582 (10.88)	1.601 (10.75)	1.604 (10.93)	1.615 (11.23)	1.623 (11.59)	1.635 (11.02)	1.630 (11.05)	1.648 (11.93)	1.646 (12.25)	2.164 (10.58)
30%	1.687 (18.76)	1.693 (18.68)	1.673 (15.71)	1.681 (16.27)	1.707 (17.59)	1.727 (18.75)	1.707 (15.93)	1.749 (19.16)	1.753 (19.03)	1.750 (19.37)	2.335 (19.31)
40%	1.876 (32.02)	1.860 (30.42)	1.904 (31.71)	1.845 (27.62)	1.904 (31.16)	1.875 (28.92)	1.895 (28.64)	1.931 (31.51)	1.973 (34.00)	1.954 (33.25)	2.588 (32.20)
Imputation using predictive mean matching method											
10%	1.484 (4.41)	1.513 (6.10)	1.502 (3.92)	1.508 (4.27)	1.505 (3.67)	1.529 (5.12)	1.518 (3.07)	1.541 (4.96)	1.547 (5.05)	1.555 (6.06)	2.047 (4.56)
20%	1.544 (8.67)	1.555 (8.99)	1.565 (8.23)	1.579 (9.22)	1.593 (9.73)	1.596 (9.73)	1.607 (9.12)	1.615 (10.03)	1.620 (10.01)	1.617 (10.27)	2.129 (8.75)
30%	1.658 (16.68)	1.649 (15.61)	1.642 (13.55)	1.639 (13.36)	1.656 (14.10)	1.698 (16.75)	1.674 (13.65)	1.713 (16.68)	1.738 (17.99)	1.702 (16.05)	2.291 (17.07)
40%	1.755 (23.50)	1.777 (24.57)	1.774 (22.70)	1.761 (21.76)	1.796 (23.74)	1.774 (22.01)	1.783 (21.07)	1.800 (22.63)	1.847 (25.45)	1.824 (24.38)	2.425 (23.89)
Imputation using propensity−score method											
10%	1.479 (4.11)	1.512 (5.97)	1.504 (4.01)	1.511 (4.49)	1.504 (3.64)	1.528 (5.09)	1.520 (3.22)	1.541 (4.98)	1.549 (5.16)	1.554 (5.98)	2.041 (4.28)
20%	1.554 (9.34)	1.562 (9.49)	1.582 (9.39)	1.596 (10.36)	1.605 (10.60)	1.616 (11.09)	1.629 (10.62)	1.613 (9.89)	1.629 (10.61)	1.628 (10.99)	2.142 (9.43)
30%	1.647 (15.94)	1.674 (17.37)	1.660 (14.84)	1.649 (14.02)	1.690 (16.41)	1.701 (16.98)	1.692 (14.87)	1.739 (18.44)	1.723 (17.02)	1.731 (18.07)	2.278 (16.38)
40%	1.846 (29.88)	1.849 (29.65)	1.867 (29.12)	1.819 (25.82)	1.859 (28.08)	1.850 (27.22)	1.875 (27.31)	1.874 (27.68)	1.900 (29.00)	1.910 (30.24)	2.546 (30.07)

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.987	1.998	2.010	2.042	2.053	2.034	2.047	2.068	2.083
Imputation using linear regression									
10%	2.068 (4.11)	2.067 (3.41)	2.103 (4.64)	2.099 (2.77)	2.118 (3.16)	2.122 (4.29)	2.132 (4.13)	2.133 (3.14)	2.150 (3.22)
20%	2.187 (10.09)	2.170 (8.57)	2.194 (9.14)	2.221 (8.75)	2.245 (9.31)	2.245 (10.33)	2.235 (9.15)	2.232 (7.93)	2.272 (9.09)
30%	2.356 (18.57)	2.378 (19.03)	2.404 (19.60)	2.429 (18.95)	2.389 (16.36)	2.443 (20.07)	2.468 (20.56)	2.423 (17.17)	2.495 (19.78)
40%	2.589 (30.32)	2.588 (29.52)	2.577 (28.23)	2.611 (27.88)	2.633 (28.25)	2.635 (29.51)	2.669 (30.33)	2.660 (28.62)	2.720 (30.56)
Imputation using predictive mean matching method									
10%	2.047 (3.03)	2.058 (3.00)	2.103 (4.63)	2.099 (2.76)	2.107 (2.60)	2.122 (4.28)	2.131 (4.09)	2.117 (2.38)	2.152 (3.29)
20%	2.143 (7.89)	2.131 (6.64)	2.174 (8.16)	2.196 (7.53)	2.210 (7.64)	2.223 (9.25)	2.200 (7.46)	2.196 (6.17)	2.253 (8.14)
30%	2.237 (12.59)	2.298 (14.98)	2.316 (15.24)	2.299 (12.59)	2.309 (12.44)	2.365 (16.25)	2.356 (15.07)	2.354 (13.82)	2.406 (15.51)
40%	2.474 (24.50)	2.515 (25.87)	2.528 (25.77)	2.526 (23.67)	2.508 (22.15)	2.557 (25.71)	2.519 (23.04)	2.547 (23.16)	2.620 (25.78)
Imputation using propensity−score method									
10%	2.053 (3.35)	2.057 (2.96)	2.099 (4.44)	2.093 (2.51)	2.113 (2.90)	2.115 (3.96)	2.125 (3.78)	2.116 (2.32)	2.151 (3.27)
20%	2.152 (8.31)	2.121 (6.14)	2.180 (8.46)	2.190 (7.24)	2.237 (8.93)	2.231 (9.67)	2.213 (8.08)	2.200 (6.41)	2.257 (8.37)
30%	2.260 (13.75)	2.339 (17.07)	2.338 (16.32)	2.337 (14.46)	2.359 (14.90)	2.390 (17.48)	2.388 (16.62)	2.395 (15.81)	2.402 (15.33)
40%	2.538 (27.77)	2.569 (28.58)	2.552 (26.99)	2.539 (24.34)	2.587 (26.00)	2.632 (29.37)	2.646 (29.21)	2.638 (27.58)	2.635 (26.51)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.965	1.992	1.992	2.000	2.003	2.029	2.022	2.029	2.020
Imputation using linear regression									

SampleSize_n150_10Treatments/table3_nonshrink_10_treatments_MAR_n150

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.102 (6.97)	2.086 (4.72)	2.099 (5.37)	2.095 (4.78)	2.116 (5.63)	2.105 (3.76)	2.138 (5.71)	2.156 (6.27)	2.158 (6.82)
20%	2.181 (11.00)	2.208 (10.87)	2.212 (11.06)	2.227 (11.37)	2.238 (11.70)	2.255 (11.15)	2.248 (11.16)	2.273 (12.04)	2.269 (12.36)
30%	2.343 (19.23)	2.315 (16.25)	2.327 (16.84)	2.362 (18.14)	2.390 (19.30)	2.363 (16.48)	2.421 (19.72)	2.426 (19.58)	2.422 (19.93)
40%	2.567 (30.66)	2.626 (31.85)	2.546 (27.83)	2.626 (31.32)	2.587 (29.15)	2.613 (28.80)	2.663 (31.67)	2.722 (34.16)	2.695 (33.44)
Imputation using predictive mean matching method									
10%	2.088 (6.25)	2.073 (4.07)	2.080 (4.43)	2.076 (3.82)	2.109 (5.27)	2.094 (3.21)	2.125 (5.10)	2.134 (5.20)	2.145 (6.22)
20%	2.143 (9.07)	2.157 (8.32)	2.177 (9.32)	2.196 (9.82)	2.200 (9.82)	2.216 (9.21)	2.226 (10.09)	2.233 (10.08)	2.229 (10.35)
30%	2.279 (15.99)	2.269 (13.91)	2.266 (13.74)	2.290 (14.50)	2.346 (17.12)	2.313 (14.02)	2.367 (17.05)	2.401 (18.34)	2.352 (16.45)
40%	2.455 (24.95)	2.451 (23.07)	2.433 (22.14)	2.481 (24.10)	2.453 (22.42)	2.464 (21.43)	2.488 (23.01)	2.552 (25.81)	2.521 (24.81)
Imputation using propensity−score method									
10%	2.086 (6.15)	2.075 (4.19)	2.085 (4.67)	2.076 (3.82)	2.109 (5.27)	2.097 (3.39)	2.126 (5.15)	2.137 (5.33)	2.144 (6.17)
20%	2.153 (9.57)	2.180 (9.47)	2.200 (10.43)	2.213 (10.68)	2.227 (11.16)	2.246 (10.69)	2.224 (9.97)	2.245 (10.68)	2.243 (11.07)
30%	2.314 (17.76)	2.295 (15.22)	2.279 (14.43)	2.336 (16.81)	2.352 (17.38)	2.339 (15.27)	2.403 (18.82)	2.383 (17.44)	2.393 (18.46)
40%	2.551 (29.85)	2.575 (29.29)	2.511 (26.05)	2.565 (28.28)	2.553 (27.44)	2.586 (27.46)	2.586 (27.87)	2.621 (29.22)	2.635 (30.46)

SampleSize_n150_10Treatments/table3_shrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.449	1.445	1.453	1.462	1.485	1.493	1.480	1.489	1.504	1.515	1.993
Imputation using linear regression											
10%	1.502 (3.68)	1.498 (3.69)	1.497 (2.99)	1.523 (4.21)	1.520 (2.35)	1.534 (2.74)	1.537 (3.86)	1.544 (3.69)	1.545 (2.72)	1.557 (2.80)	2.079 (4.31)
20%	1.594 (10.08)	1.584 (9.65)	1.571 (8.12)	1.589 (8.69)	1.609 (8.31)	1.626 (8.87)	1.626 (9.87)	1.619 (8.71)	1.617 (7.50)	1.646 (8.64)	2.195 (10.16)
30%	1.707 (17.84)	1.705 (18.03)	1.721 (18.45)	1.740 (19.04)	1.759 (18.42)	1.729 (15.77)	1.768 (19.50)	1.787 (20.00)	1.753 (16.60)	1.807 (19.26)	2.367 (18.76)
40%	1.852 (27.89)	1.876 (29.86)	1.875 (29.01)	1.867 (27.71)	1.893 (27.43)	1.909 (27.84)	1.909 (29.05)	1.934 (29.89)	1.927 (28.14)	1.971 (30.08)	2.556 (28.26)
Imputation using predictive mean matching method											
10%	1.493 (3.09)	1.483 (2.67)	1.491 (2.63)	1.524 (4.26)	1.521 (2.39)	1.527 (2.24)	1.537 (3.91)	1.544 (3.71)	1.534 (2.02)	1.559 (2.93)	2.068 (3.78)
20%	1.555 (7.34)	1.552 (7.45)	1.543 (6.20)	1.575 (7.74)	1.591 (7.12)	1.601 (7.20)	1.610 (8.79)	1.594 (7.02)	1.590 (5.74)	1.632 (7.71)	2.140 (7.38)
30%	1.628 (12.41)	1.619 (12.05)	1.663 (14.44)	1.677 (14.73)	1.664 (12.05)	1.671 (11.92)	1.712 (15.74)	1.705 (14.53)	1.704 (13.31)	1.743 (15.03)	2.259 (13.35)
40%	1.809 (24.86)	1.797 (24.35)	1.827 (25.73)	1.836 (25.62)	1.835 (23.55)	1.822 (22.04)	1.858 (25.57)	1.830 (22.87)	1.849 (22.98)	1.904 (25.66)	2.489 (24.92)
Imputation using propensity−score method											
10%	1.490 (2.87)	1.487 (2.92)	1.490 (2.52)	1.520 (3.99)	1.516 (2.07)	1.530 (2.46)	1.531 (3.51)	1.539 (3.33)	1.532 (1.88)	1.558 (2.83)	2.064 (3.57)
20%	1.553 (7.25)	1.557 (7.75)	1.534 (5.58)	1.577 (7.92)	1.585 (6.69)	1.619 (8.39)	1.614 (9.11)	1.601 (7.53)	1.592 (5.87)	1.633 (7.81)	2.141 (7.46)
30%	1.661 (14.64)	1.633 (13.05)	1.691 (16.38)	1.690 (15.63)	1.690 (13.77)	1.706 (14.23)	1.728 (16.77)	1.726 (15.91)	1.731 (15.14)	1.737 (14.65)	2.307 (15.79)
40%	1.823 (25.83)	1.840 (27.36)	1.863 (28.18)	1.850 (26.55)	1.841 (23.95)	1.876 (25.64)	1.909 (29.03)	1.919 (28.86)	1.912 (27.17)	1.910 (26.05)	2.515 (26.19)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.421	1.426	1.446	1.446	1.452	1.454	1.473	1.468	1.473	1.466	1.976
Imputation using linear regression											

SampleSize_n150_10Treatments/table3_shrink_10_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.498 (5.41)	1.524 (6.82)	1.512 (4.57)	1.521 (5.21)	1.519 (4.63)	1.534 (5.47)	1.526 (3.62)	1.550 (5.56)	1.563 (6.13)	1.564 (6.66)	2.077 (5.13)
20%	1.570 (10.45)	1.582 (10.88)	1.601 (10.75)	1.604 (10.93)	1.615 (11.23)	1.623 (11.59)	1.635 (11.02)	1.630 (11.05)	1.648 (11.93)	1.646 (12.25)	2.172 (9.94)
30%	1.687 (18.76)	1.693 (18.68)	1.673 (15.71)	1.681 (16.27)	1.707 (17.59)	1.727 (18.75)	1.707 (15.93)	1.749 (19.16)	1.753 (19.03)	1.750 (19.37)	2.344 (18.68)
40%	1.876 (32.02)	1.860 (30.42)	1.904 (31.71)	1.845 (27.62)	1.904 (31.16)	1.875 (28.92)	1.895 (28.64)	1.931 (31.51)	1.973 (34.00)	1.954 (33.25)	2.585 (30.85)
Imputation using predictive mean matching method											
10%	1.484 (4.41)	1.513 (6.10)	1.502 (3.92)	1.508 (4.27)	1.505 (3.67)	1.529 (5.12)	1.518 (3.07)	1.541 (4.96)	1.547 (5.05)	1.555 (6.06)	2.060 (4.26)
20%	1.544 (8.67)	1.555 (8.99)	1.565 (8.23)	1.579 (9.22)	1.593 (9.73)	1.596 (9.73)	1.607 (9.12)	1.615 (10.03)	1.620 (10.01)	1.617 (10.27)	2.139 (8.28)
30%	1.658 (16.68)	1.649 (15.61)	1.642 (13.55)	1.639 (13.36)	1.656 (14.10)	1.698 (16.75)	1.674 (13.65)	1.713 (16.68)	1.738 (17.99)	1.702 (16.05)	2.296 (16.22)
40%	1.755 (23.50)	1.777 (24.57)	1.774 (22.70)	1.761 (21.76)	1.796 (23.74)	1.774 (22.01)	1.783 (21.07)	1.800 (22.63)	1.847 (25.45)	1.824 (24.38)	2.425 (22.76)
Imputation using propensity−score method											
10%	1.479 (4.11)	1.512 (5.97)	1.504 (4.01)	1.511 (4.49)	1.504 (3.64)	1.528 (5.09)	1.520 (3.22)	1.541 (4.98)	1.549 (5.16)	1.554 (5.98)	2.054 (3.97)
20%	1.554 (9.34)	1.562 (9.49)	1.582 (9.39)	1.596 (10.36)	1.605 (10.60)	1.616 (11.09)	1.629 (10.62)	1.613 (9.89)	1.629 (10.61)	1.628 (10.99)	2.149 (8.77)
30%	1.647 (15.94)	1.674 (17.37)	1.660 (14.84)	1.649 (14.02)	1.690 (16.41)	1.701 (16.98)	1.692 (14.87)	1.739 (18.44)	1.723 (17.02)	1.731 (18.07)	2.283 (15.58)
40%	1.846 (29.88)	1.849 (29.65)	1.867 (29.12)	1.819 (25.82)	1.859 (28.08)	1.850 (27.22)	1.875 (27.31)	1.874 (27.68)	1.900 (29.00)	1.910 (30.24)	2.544 (28.79)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.992	1.994	1.999	2.003	2.009	2.002	2.004	2.007	2.020
Imputation using linear regression									
10%	2.084 (4.64)	2.069 (3.79)	2.086 (4.36)	2.086 (4.13)	2.099 (4.49)	2.096 (4.66)	2.093 (4.42)	2.103 (4.76)	2.099 (3.94)
20%	2.185 (9.73)	2.176 (9.16)	2.189 (9.53)	2.202 (9.91)	2.214 (10.24)	2.209 (10.31)	2.213 (10.41)	2.209 (10.02)	2.229 (10.35)
30%	2.368 (18.91)	2.373 (19.04)	2.394 (19.81)	2.422 (20.93)	2.373 (18.16)	2.421 (20.92)	2.443 (21.91)	2.400 (19.54)	2.468 (22.22)
40%	2.587 (29.90)	2.580 (29.42)	2.566 (28.41)	2.604 (29.99)	2.627 (30.78)	2.617 (30.69)	2.653 (32.37)	2.635 (31.26)	2.687 (33.05)
Imputation using predictive mean matching method									
10%	2.064 (3.63)	2.061 (3.39)	2.086 (4.40)	2.085 (4.10)	2.088 (3.93)	2.095 (4.65)	2.092 (4.40)	2.089 (4.05)	2.102 (4.09)
20%	2.143 (7.62)	2.138 (7.24)	2.171 (8.61)	2.178 (8.75)	2.182 (8.64)	2.186 (9.19)	2.176 (8.55)	2.171 (8.13)	2.209 (9.40)
30%	2.251 (13.01)	2.295 (15.09)	2.310 (15.60)	2.294 (14.50)	2.296 (14.33)	2.345 (17.10)	2.332 (16.34)	2.329 (16.03)	2.378 (17.76)
40%	2.471 (24.05)	2.506 (25.68)	2.514 (25.80)	2.518 (25.69)	2.500 (24.46)	2.536 (26.66)	2.502 (24.82)	2.517 (25.38)	2.584 (27.93)
Imputation using propensity−score method									
10%	2.070 (3.94)	2.060 (3.35)	2.083 (4.20)	2.081 (3.87)	2.093 (4.22)	2.091 (4.42)	2.087 (4.13)	2.088 (4.03)	2.101 (4.05)
20%	2.151 (8.02)	2.128 (6.74)	2.178 (8.96)	2.173 (8.47)	2.208 (9.94)	2.197 (9.73)	2.190 (9.28)	2.179 (8.52)	2.214 (9.61)
30%	2.274 (14.18)	2.338 (17.29)	2.333 (16.74)	2.333 (16.48)	2.348 (16.90)	2.371 (18.40)	2.366 (18.04)	2.374 (18.26)	2.377 (17.71)
40%	2.536 (27.34)	2.561 (28.45)	2.540 (27.07)	2.533 (26.44)	2.580 (28.45)	2.616 (30.65)	2.630 (31.20)	2.613 (30.14)	2.601 (28.79)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.974	1.988	1.985	1.984	1.983	1.991	1.988	1.990	1.992
Imputation using linear regression									

SampleSize_n150_10Treatments/table3_shrink_10_treatments_MAR_n150

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	2.088 (5.77)	2.077 (4.45)	2.083 (4.96)	2.090 (5.34)	2.090 (5.38)	2.088 (4.90)	2.101 (5.65)	2.110 (6.04)	2.105 (5.66)
20%	2.185 (10.67)	2.198 (10.52)	2.199 (10.80)	2.202 (11.01)	2.218 (11.83)	2.222 (11.65)	2.220 (11.67)	2.242 (12.64)	2.235 (12.19)
30%	2.351 (19.09)	2.327 (17.04)	2.332 (17.49)	2.359 (18.91)	2.382 (20.11)	2.354 (18.26)	2.397 (20.54)	2.400 (20.60)	2.403 (20.61)
40%	2.563 (29.81)	2.621 (31.81)	2.545 (28.22)	2.615 (31.83)	2.577 (29.96)	2.605 (30.87)	2.645 (33.04)	2.694 (35.35)	2.668 (33.89)
Imputation using predictive mean matching method									
10%	2.076 (5.14)	2.063 (3.73)	2.065 (4.05)	2.074 (4.55)	2.082 (5.01)	2.077 (4.35)	2.087 (4.97)	2.091 (5.05)	2.091 (4.97)
20%	2.148 (8.82)	2.148 (8.04)	2.164 (9.04)	2.174 (9.57)	2.178 (9.81)	2.184 (9.73)	2.199 (10.60)	2.203 (10.69)	2.198 (10.32)
30%	2.289 (15.96)	2.281 (14.71)	2.272 (14.48)	2.285 (15.18)	2.337 (17.83)	2.303 (15.72)	2.341 (17.76)	2.373 (19.24)	2.332 (17.07)
40%	2.456 (24.41)	2.449 (23.14)	2.435 (22.66)	2.469 (24.44)	2.445 (23.27)	2.454 (23.31)	2.471 (24.26)	2.526 (26.94)	2.493 (25.12)
Imputation using propensity−score method									
10%	2.074 (5.04)	2.066 (3.88)	2.069 (4.26)	2.072 (4.46)	2.083 (5.03)	2.079 (4.45)	2.088 (5.04)	2.092 (5.13)	2.091 (4.96)
20%	2.156 (9.24)	2.169 (9.07)	2.185 (10.11)	2.191 (10.43)	2.205 (11.17)	2.214 (11.21)	2.196 (10.45)	2.213 (11.18)	2.209 (10.86)
30%	2.320 (17.50)	2.305 (15.93)	2.283 (15.04)	2.332 (17.55)	2.342 (18.12)	2.328 (16.95)	2.378 (19.61)	2.355 (18.33)	2.373 (19.10)
40%	2.548 (29.07)	2.571 (29.31)	2.512 (26.55)	2.555 (28.80)	2.545 (28.32)	2.579 (29.55)	2.572 (29.37)	2.595 (30.39)	2.610 (30.99)

SampleSize_n20_10Treatments/table2_nonshrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	93.10	93.10	93.10	100.00	96.55	93.10	100.00	100.00	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	93.10	100.00	96.55	72.41
20%	92.86	92.86	92.86	89.29	100.00	89.29	96.43	100.00	89.29	96.43	71.43
30%	96.43	92.86	96.43	96.43	100.00	100.00	92.86	92.86	96.43	96.43	75.00
40%	93.10	89.66	100.00	100.00	96.55	89.66	96.55	100.00	93.10	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	86.21	93.10	96.55	93.10	89.66	93.10	89.66	100.00	96.55	68.97
20%	85.71	92.86	89.29	85.71	100.00	89.29	96.43	96.43	85.71	96.43	64.29
30%	92.86	92.86	85.71	85.71	82.14	92.86	89.29	92.86	85.71	92.86	39.29
40%	86.21	75.86	89.66	82.76	75.86	86.21	89.66	89.66	75.86	82.76	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	89.66	100.00	93.10	72.41
20%	82.14	92.86	89.29	85.71	100.00	89.29	96.43	96.43	92.86	96.43	64.29
30%	92.86	92.86	92.86	92.86	78.57	92.86	85.71	85.71	82.14	92.86	39.29
40%	89.66	79.31	86.21	89.66	79.31	93.10	89.66	93.10	82.76	79.31	41.38
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	96.55	96.55	93.10	96.55	96.55	96.55	89.66	93.10	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_nonshrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	96.55	89.66	96.55	100.00	100.00	100.00	100.00	82.76	65.52
20%	96.43	100.00	96.43	96.43	89.29	100.00	92.86	100.00	96.43	85.71	75.00
30%	93.10	96.55	96.55	96.55	96.55	93.10	89.66	89.66	96.55	96.55	75.86
40%	96.55	100.00	100.00	93.10	96.55	96.55	96.55	100.00	96.55	96.55	75.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	86.21	93.10	100.00	96.55	96.55	100.00	82.76	62.07
20%	85.71	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	57.14
30%	86.21	93.10	86.21	93.10	86.21	86.21	89.66	86.21	93.10	89.66	58.62
40%	89.66	86.21	93.10	93.10	86.21	86.21	89.66	79.31	93.10	93.10	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	89.66	89.66	89.66	89.66	100.00	96.55	96.55	100.00	82.76	62.07
20%	92.86	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	60.71
30%	93.10	89.66	93.10	93.10	86.21	86.21	89.66	82.76	89.66	89.66	55.17
40%	86.21	93.10	93.10	86.21	86.21	93.10	79.31	82.76	96.55	89.66	51.72

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	92.86
30%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	96.43	96.43	85.71
30%	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43	96.43	96.43	89.29
40%	96.55	93.10	96.55	100.00	100.00	96.55	96.55	100.00	100.00	96.55	79.31
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	100.00	100.00	100.00	96.43	96.43	89.29
30%	96.43	96.43	100.00	96.43	100.00	96.43	96.43	100.00	92.86	92.86	82.14
40%	100.00	96.55	96.55	100.00	100.00	96.55	100.00	100.00	100.00	93.10	86.21
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_nonshrink_10_treatments_MAR_n20

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	93.1	96.55	100.00	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.0	100.00	100.00	100.00	100.00	100.00	96.43	92.86
30%	96.55	96.55	100	100.0	100.00	100.00	96.55	100.00	100.00	100.00	96.55
40%	100.00	100.00	100	100.0	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.1	96.55	100.00	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.0	96.43	100.00	96.43	100.00	100.00	96.43	89.29
30%	93.10	96.55	100	100.0	100.00	100.00	96.55	100.00	96.55	100.00	89.66
40%	96.55	96.55	100	93.1	96.55	96.55	96.55	96.55	96.55	96.55	79.31
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.1	96.55	100.00	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.0	96.43	100.00	92.86	100.00	100.00	96.43	85.71
30%	93.10	96.55	100	100.0	100.00	100.00	96.55	96.55	100.00	96.55	86.21
40%	96.55	100.00	100	93.1	100.00	100.00	93.10	96.55	96.55	93.10	82.76

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	93.10	93.10	93.10	93.10	100.00	96.55	93.10	100.00	100.00	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	93.10	100.00	96.55	72.41
20%	92.86	92.86	92.86	89.29	100.00	89.29	96.43	100.00	89.29	96.43	71.43
30%	96.43	92.86	96.43	96.43	100.00	100.00	92.86	92.86	96.43	96.43	75.00
40%	93.10	89.66	100.00	100.00	96.55	89.66	96.55	100.00	93.10	93.10	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	86.21	93.10	96.55	93.10	89.66	93.10	89.66	100.00	96.55	68.97
20%	85.71	92.86	89.29	85.71	100.00	89.29	96.43	96.43	85.71	96.43	64.29
30%	92.86	92.86	85.71	85.71	82.14	92.86	89.29	92.86	85.71	92.86	39.29
40%	86.21	75.86	89.66	82.76	75.86	86.21	89.66	89.66	75.86	82.76	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	93.10	93.10	96.55	93.10	89.66	93.10	89.66	100.00	93.10	72.41
20%	82.14	92.86	89.29	85.71	100.00	89.29	96.43	96.43	92.86	96.43	64.29
30%	92.86	92.86	92.86	92.86	78.57	92.86	85.71	85.71	82.14	92.86	39.29
40%	89.66	79.31	86.21	89.66	79.31	93.10	89.66	93.10	82.76	79.31	41.38
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	96.55	96.55	93.10	96.55	96.55	96.55	89.66	93.10	68.97
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	96.55	89.66	96.55	100.00	100.00	100.00	100.00	82.76	65.52
20%	96.43	100.00	96.43	96.43	89.29	100.00	92.86	100.00	96.43	85.71	75.00
30%	93.10	96.55	96.55	96.55	96.55	93.10	89.66	89.66	96.55	96.55	75.86
40%	96.55	100.00	100.00	93.10	96.55	96.55	96.55	100.00	96.55	96.55	75.86
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	86.21	93.10	100.00	96.55	96.55	100.00	82.76	62.07
20%	85.71	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	57.14
30%	86.21	93.10	86.21	93.10	86.21	86.21	89.66	86.21	93.10	89.66	58.62
40%	89.66	86.21	93.10	93.10	86.21	86.21	89.66	79.31	93.10	93.10	48.28
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	89.66	89.66	89.66	89.66	100.00	96.55	96.55	100.00	82.76	62.07
20%	92.86	100.00	89.29	92.86	92.86	96.43	92.86	92.86	92.86	82.14	60.71
30%	93.10	89.66	93.10	93.10	86.21	86.21	89.66	82.76	89.66	89.66	55.17
40%	86.21	93.10	93.10	86.21	86.21	93.10	79.31	82.76	96.55	89.66	51.72

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.43	92.86
30%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	96.43	96.43	85.71
30%	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43	100.00	92.86	89.29
40%	96.55	100.00	96.55	100.00	96.55	96.55	96.55	100.00	100.00	96.55	86.21
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.55	100.00	100.00	96.55	96.55	96.55	100.00	100.00	100.00	93.10
20%	100.00	96.43	96.43	96.43	100.00	100.00	100.00	100.00	96.43	100.00	92.86
30%	96.43	96.43	100.00	96.43	100.00	96.43	96.43	100.00	92.86	92.86	82.14
40%	100.00	100.00	96.55	100.00	100.00	96.55	96.55	100.00	100.00	93.10	86.21
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	96.55	100.00	100.00	100.00	100.00	96.55	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_10Treatments/table2_shrink_10_treatments_MAR_n20

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	100.00	96.55	100	100.00	100.00	100.00	96.55	96.55
20%	96.43	100.00	100	100.00	100.00	100	100.00	100.00	100.00	96.43	92.86
30%	96.55	96.55	100	100.00	100.00	100	96.55	100.00	100.00	100.00	96.55
40%	100.00	100.00	100	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.10	96.55	100	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.00	96.43	100	96.43	100.00	100.00	96.43	89.29
30%	93.10	96.55	100	96.55	100.00	100	96.55	96.55	96.55	100.00	89.66
40%	96.55	96.55	100	93.10	96.55	100	96.55	93.10	96.55	96.55	79.31
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	93.10	96.55	100	100.00	100.00	100.00	96.55	89.66
20%	96.43	100.00	100	100.00	96.43	100	100.00	100.00	100.00	96.43	89.29
30%	93.10	96.55	100	100.00	100.00	100	96.55	96.55	100.00	96.55	86.21
40%	96.55	100.00	100	93.10	100.00	100	96.55	100.00	96.55	89.66	79.31

SampleSize_n20_10Treatments/table3_nonshrink_10_treatments_MAR_n20

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	3.911	4.030	3.924	4.010	4.047	4.112	4.267	4.109	4.036	4.241	5.284
Imputation using linear regression											
10%	4.195 (7.24)	4.237 (5.14)	4.316 (10.00)	4.345 (8.34)	4.433 (9.56)	4.412 (7.30)	4.335 (1.60)	4.408 (7.28)	4.384 (8.62)	4.568 (7.71)	5.755 (8.91)
20%	4.661 (19.17)	4.759 (18.09)	4.664 (18.87)	4.727 (17.87)	4.644 (14.77)	4.678 (13.76)	4.822 (13.02)	4.828 (17.50)	4.822 (19.48)	4.756 (12.15)	6.423 (21.55)
30%	4.857 (24.17)	5.396 (33.90)	5.013 (27.77)	5.155 (28.54)	5.017 (23.98)	5.057 (22.98)	5.054 (18.45)	4.987 (21.36)	5.143 (27.44)	5.143 (21.28)	6.678 (26.37)
40%	5.807 (48.46)	5.781 (43.45)	5.890 (50.12)	5.890 (46.87)	5.623 (38.95)	5.932 (44.27)	5.896 (38.19)	6.024 (46.60)	6.142 (52.19)	6.316 (48.94)	7.975 (50.92)
Imputation using predictive mean matching method											
10%	4.028 (2.97)	4.095 (1.60)	4.127 (5.18)	4.147 (3.42)	4.222 (4.33)	4.216 (2.54)	4.194 (−1.70)	4.233 (3.02)	4.194 (3.92)	4.388 (3.46)	5.521 (4.47)
20%	4.175 (6.75)	4.312 (6.99)	4.193 (6.86)	4.283 (6.79)	4.280 (5.76)	4.277 (4.02)	4.416 (3.49)	4.344 (5.73)	4.365 (8.16)	4.295 (1.27)	5.752 (8.85)
30%	4.109 (5.04)	4.338 (7.63)	4.213 (7.37)	4.211 (4.99)	4.241 (4.80)	4.208 (2.34)	4.239 (−0.65)	4.184 (1.82)	4.241 (5.10)	4.341 (2.35)	5.653 (6.98)
40%	4.396 (12.39)	4.383 (8.76)	4.430 (12.90)	4.424 (10.31)	4.308 (6.46)	4.548 (10.60)	4.645 (8.87)	4.532 (10.29)	4.658 (15.41)	4.636 (9.31)	6.036 (14.24)
Imputation using propensity−score method											
10%	4.024 (2.87)	4.086 (1.39)	4.091 (4.27)	4.167 (3.90)	4.246 (4.93)	4.236 (3.02)	4.175 (−2.15)	4.227 (2.88)	4.170 (3.33)	4.376 (3.19)	5.516 (4.39)
20%	4.193 (7.20)	4.386 (8.83)	4.140 (5.52)	4.263 (6.30)	4.246 (4.92)	4.284 (4.17)	4.428 (3.78)	4.443 (8.13)	4.374 (8.40)	4.348 (2.52)	5.774 (9.27)
30%	4.064 (3.90)	4.201 (4.23)	4.199 (7.02)	4.371 (8.98)	4.248 (4.98)	4.159 (1.15)	4.263 (−0.10)	4.193 (2.05)	4.273 (5.89)	4.413 (4.05)	5.587 (5.74)
40%	4.251 (8.69)	4.286 (6.35)	4.200 (7.05)	4.333 (8.05)	4.196 (3.70)	4.423 (7.57)	4.529 (6.14)	4.343 (5.69)	4.503 (11.59)	4.484 (5.74)	5.856 (10.82)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	3.949	3.853	3.952	3.981	4.063	4.035	4.010	3.895	4.095	4.041	5.320
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	4.130 (4.59)	4.230 (9.78)	4.319 (9.28)	4.271 (7.27)	4.282 (5.40)	4.376 (8.44)	4.403 (9.81)	4.353 (11.75)	4.465 (9.04)	4.372 (8.21)	5.690 (6.96)
20%	4.557 (15.38)	4.476 (16.17)	4.361 (10.35)	4.709 (18.29)	4.479 (10.24)	4.615 (14.37)	4.575 (14.09)	4.582 (17.64)	4.621 (12.85)	4.627 (14.50)	6.280 (18.05)
30%	5.278 (33.66)	5.225 (35.58)	5.282 (33.64)	5.570 (39.90)	5.301 (30.46)	5.460 (35.32)	5.249 (30.91)	5.510 (41.45)	5.509 (34.53)	5.445 (34.76)	7.196 (35.27)
40%	5.737 (45.27)	5.712 (48.23)	5.891 (49.04)	5.528 (38.84)	5.638 (38.77)	6.005 (48.83)	6.169 (53.85)	6.552 (68.23)	5.826 (42.27)	5.847 (44.70)	7.861 (47.77)
Imputation using predictive mean matching method											
10%	3.893 (−1.42)	4.054 (5.22)	4.066 (2.89)	4.045 (1.60)	4.049 (−0.35)	4.162 (3.14)	4.190 (4.51)	4.160 (6.81)	4.241 (3.56)	4.141 (2.48)	5.371 (0.96)
20%	4.156 (5.24)	4.003 (3.87)	3.948 (−0.11)	4.214 (5.84)	4.063 (0.01)	4.179 (3.57)	4.191 (4.53)	4.153 (6.62)	4.224 (3.16)	4.248 (5.13)	5.726 (7.64)
30%	4.245 (7.49)	4.287 (11.25)	4.400 (11.32)	4.418 (10.96)	4.271 (5.11)	4.501 (11.55)	4.325 (7.86)	4.531 (16.33)	4.449 (8.64)	4.512 (11.67)	5.826 (9.52)
40%	4.207 (6.53)	4.278 (11.01)	4.345 (9.94)	4.301 (8.04)	4.301 (5.85)	4.361 (8.09)	4.426 (10.39)	4.458 (14.44)	4.478 (9.35)	4.463 (10.44)	5.804 (9.10)
Imputation using propensity−score method											
10%	3.912 (−0.93)	4.071 (5.64)	4.126 (4.39)	4.106 (3.13)	4.064 (0.02)	4.191 (3.86)	4.204 (4.84)	4.203 (7.90)	4.275 (4.39)	4.197 (3.88)	5.391 (1.33)
20%	4.234 (7.21)	4.074 (5.72)	4.004 (1.32)	4.209 (5.72)	4.175 (2.75)	4.291 (6.35)	4.224 (5.34)	4.239 (8.83)	4.242 (3.59)	4.337 (7.33)	5.827 (9.54)
30%	4.372 (10.71)	4.393 (14.01)	4.518 (14.32)	4.544 (14.14)	4.355 (7.19)	4.561 (13.03)	4.427 (10.40)	4.786 (22.88)	4.559 (11.34)	4.545 (12.47)	5.972 (12.26)
40%	4.163 (5.41)	4.230 (9.77)	4.195 (6.13)	4.230 (6.24)	4.282 (5.39)	4.243 (5.15)	4.331 (8.02)	4.437 (13.91)	4.492 (9.70)	4.444 (9.99)	5.750 (8.08)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	5.444	5.299	5.418	5.468	5.553	5.764	5.550	5.453	5.729
Imputation using linear regression									
10%	5.817 (6.84)	5.923 (11.76)	5.961 (10.03)	6.086 (11.30)	6.055 (9.03)	5.949 (3.22)	6.049 (8.99)	6.017 (10.35)	6.273 (9.50)
20%	6.561 (20.51)	6.423 (21.20)	6.514 (20.23)	6.400 (17.04)	6.447 (16.08)	6.645 (15.28)	6.657 (19.96)	6.643 (21.83)	6.556 (14.44)
30%	7.414 (36.17)	6.891 (30.03)	7.089 (30.86)	6.902 (26.22)	6.951 (25.16)	6.949 (20.56)	6.859 (23.59)	7.064 (29.54)	7.075 (23.50)
40%	7.936 (45.77)	8.087 (52.60)	8.085 (49.24)	7.719 (41.16)	8.146 (46.68)	8.098 (40.51)	8.263 (48.89)	8.433 (54.64)	8.668 (51.31)
Imputation using predictive mean matching method									
10%	5.615 (3.13)	5.658 (6.77)	5.687 (4.96)	5.790 (5.89)	5.781 (4.09)	5.750 (−0.23)	5.804 (4.57)	5.751 (5.46)	6.019 (5.07)
20%	5.942 (9.14)	5.773 (8.94)	5.899 (8.89)	5.894 (7.78)	5.891 (6.09)	6.083 (5.54)	5.987 (7.88)	6.012 (10.26)	5.918 (3.31)
30%	5.963 (9.53)	5.792 (9.30)	5.794 (6.95)	5.836 (6.73)	5.788 (4.23)	5.835 (1.23)	5.754 (3.68)	5.835 (7.00)	5.971 (4.23)
40%	6.014 (10.47)	6.081 (14.76)	6.072 (12.08)	5.916 (8.19)	6.246 (12.47)	6.379 (10.68)	6.216 (12.01)	6.398 (17.32)	6.365 (11.10)
Imputation using propensity−score method									
10%	5.604 (2.94)	5.611 (5.88)	5.714 (5.46)	5.825 (6.53)	5.809 (4.59)	5.726 (−0.66)	5.796 (4.44)	5.720 (4.90)	6.005 (4.83)
20%	6.042 (10.97)	5.700 (7.57)	5.872 (8.38)	5.846 (6.90)	5.898 (6.21)	6.099 (5.82)	6.121 (10.29)	6.024 (10.47)	5.987 (4.51)
30%	5.772 (6.02)	5.771 (8.89)	6.009 (10.91)	5.843 (6.86)	5.719 (2.97)	5.863 (1.73)	5.764 (3.87)	5.875 (7.74)	6.064 (5.86)
40%	5.899 (8.35)	5.784 (9.14)	5.965 (10.09)	5.785 (5.79)	6.100 (9.85)	6.244 (8.34)	5.979 (7.73)	6.207 (13.83)	6.178 (7.84)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	5.193	5.325	5.365	5.475	5.438	5.405	5.248	5.520	5.444
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	5.831 (12.29)	5.952 (11.77)	5.884 (9.68)	5.900 (7.76)	6.033 (10.94)	6.068 (12.27)	5.998 (14.29)	6.154 (11.48)	6.025 (10.67)
20%	6.168 (18.77)	6.011 (12.88)	6.492 (21.01)	6.176 (12.80)	6.361 (16.98)	6.310 (16.74)	6.316 (20.35)	6.369 (15.38)	6.378 (17.14)
30%	7.119 (37.10)	7.204 (35.28)	7.589 (41.46)	7.220 (31.87)	7.445 (36.91)	7.150 (32.28)	7.511 (43.12)	7.503 (35.92)	7.423 (36.35)
40%	7.826 (50.71)	8.076 (51.66)	7.576 (41.23)	7.725 (41.10)	8.217 (51.11)	8.453 (56.40)	8.993 (71.35)	7.988 (44.72)	8.016 (47.23)
Imputation using predictive mean matching method									
10%	5.595 (7.76)	5.612 (5.39)	5.582 (4.04)	5.586 (2.04)	5.745 (5.66)	5.783 (6.98)	5.741 (9.39)	5.851 (5.99)	5.715 (4.97)
20%	5.514 (6.19)	5.440 (2.16)	5.806 (8.23)	5.602 (2.32)	5.757 (5.87)	5.779 (6.91)	5.722 (9.03)	5.822 (5.47)	5.854 (7.52)
30%	5.881 (13.26)	6.040 (13.42)	6.062 (12.99)	5.860 (7.03)	6.176 (13.58)	5.934 (9.79)	6.218 (18.48)	6.104 (10.58)	6.192 (13.73)
40%	5.902 (13.67)	5.995 (12.59)	5.935 (10.63)	5.930 (8.32)	6.016 (10.63)	6.105 (12.95)	6.146 (17.10)	6.177 (11.90)	6.156 (13.07)
Imputation using propensity−score method									
10%	5.612 (8.08)	5.688 (6.81)	5.658 (5.47)	5.600 (2.29)	5.778 (6.26)	5.794 (7.20)	5.793 (10.38)	5.891 (6.73)	5.786 (6.28)
20%	5.604 (7.92)	5.511 (3.48)	5.794 (8.00)	5.748 (4.99)	5.904 (8.57)	5.815 (7.59)	5.832 (11.12)	5.838 (5.77)	5.968 (9.62)
30%	5.996 (15.48)	6.173 (15.92)	6.209 (15.74)	5.947 (8.63)	6.233 (14.62)	6.047 (11.87)	6.533 (24.47)	6.227 (12.82)	6.211 (14.09)
40%	5.847 (12.60)	5.798 (8.89)	5.850 (9.05)	5.921 (8.15)	5.865 (7.86)	5.990 (10.83)	6.133 (16.86)	6.211 (12.52)	6.145 (12.87)

SampleSize_n20_10Treatments/table3_shrink_10_treatments_MAR_n20

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	3.911	4.030	3.924	4.010	4.047	4.112	4.267	4.109	4.036	4.241	5.553
Imputation using linear regression											
10%	4.195 (7.24)	4.237 (5.14)	4.316 (10.00)	4.345 (8.34)	4.433 (9.56)	4.412 (7.30)	4.335 (1.60)	4.408 (7.28)	4.384 (8.62)	4.568 (7.71)	5.827 (4.95)
20%	4.661 (19.17)	4.759 (18.09)	4.664 (18.87)	4.727 (17.87)	4.644 (14.77)	4.678 (13.76)	4.822 (13.02)	4.828 (17.50)	4.822 (19.48)	4.756 (12.15)	6.494 (16.96)
30%	4.857 (24.17)	5.396 (33.90)	5.013 (27.77)	5.155 (28.54)	5.017 (23.98)	5.057 (22.98)	5.054 (18.45)	4.987 (21.36)	5.143 (27.44)	5.143 (21.28)	6.773 (21.99)
40%	5.807 (48.46)	5.781 (43.45)	5.890 (50.12)	5.890 (46.87)	5.623 (38.95)	5.932 (44.27)	5.896 (38.19)	6.024 (46.60)	6.142 (52.19)	6.316 (48.94)	7.996 (44.01)
Imputation using predictive mean matching method											
10%	4.028 (2.97)	4.095 (1.60)	4.127 (5.18)	4.147 (3.42)	4.222 (4.33)	4.216 (2.54)	4.194 (−1.70)	4.233 (3.02)	4.194 (3.92)	4.388 (3.46)	5.579 (0.47)
20%	4.175 (6.75)	4.312 (6.99)	4.193 (6.86)	4.283 (6.79)	4.280 (5.76)	4.277 (4.02)	4.416 (3.49)	4.344 (5.73)	4.365 (8.16)	4.295 (1.27)	5.821 (4.83)
30%	4.109 (5.04)	4.338 (7.63)	4.213 (7.37)	4.211 (4.99)	4.241 (4.80)	4.208 (2.34)	4.239 (−0.65)	4.184 (1.82)	4.241 (5.10)	4.341 (2.35)	5.747 (3.50)
40%	4.396 (12.39)	4.383 (8.76)	4.430 (12.90)	4.424 (10.31)	4.308 (6.46)	4.548 (10.60)	4.645 (8.87)	4.532 (10.29)	4.658 (15.41)	4.636 (9.31)	6.049 (8.94)
Imputation using propensity−score method											
10%	4.024 (2.87)	4.086 (1.39)	4.091 (4.27)	4.167 (3.90)	4.246 (4.93)	4.236 (3.02)	4.175 (−2.15)	4.227 (2.88)	4.170 (3.33)	4.376 (3.19)	5.573 (0.37)
20%	4.193 (7.20)	4.386 (8.83)	4.140 (5.52)	4.263 (6.30)	4.246 (4.92)	4.284 (4.17)	4.428 (3.78)	4.443 (8.13)	4.374 (8.40)	4.348 (2.52)	5.838 (5.14)
30%	4.064 (3.90)	4.201 (4.23)	4.199 (7.02)	4.371 (8.98)	4.248 (4.98)	4.159 (1.15)	4.263 (−0.10)	4.193 (2.05)	4.273 (5.89)	4.413 (4.05)	5.710 (2.84)
40%	4.251 (8.69)	4.286 (6.35)	4.200 (7.05)	4.333 (8.05)	4.196 (3.70)	4.423 (7.57)	4.529 (6.14)	4.343 (5.69)	4.503 (11.59)	4.484 (5.74)	5.901 (6.27)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	3.949	3.853	3.952	3.981	4.063	4.035	4.010	3.895	4.095	4.041	5.396
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	4.130 (4.59)	4.230 (9.78)	4.319 (9.28)	4.271 (7.27)	4.282 (5.40)	4.376 (8.44)	4.403 (9.81)	4.353 (11.75)	4.465 (9.04)	4.372 (8.21)	5.880 (8.96)
20%	4.557 (15.38)	4.476 (16.17)	4.361 (10.35)	4.709 (18.29)	4.479 (10.24)	4.615 (14.37)	4.575 (14.09)	4.582 (17.64)	4.621 (12.85)	4.627 (14.50)	6.273 (16.25)
30%	5.278 (33.66)	5.225 (35.58)	5.282 (33.64)	5.570 (39.90)	5.301 (30.46)	5.460 (35.32)	5.249 (30.91)	5.510 (41.45)	5.509 (34.53)	5.445 (34.76)	7.212 (33.65)
40%	5.737 (45.27)	5.712 (48.23)	5.891 (49.04)	5.528 (38.84)	5.638 (38.77)	6.005 (48.83)	6.169 (53.85)	6.552 (68.23)	5.826 (42.27)	5.847 (44.70)	7.920 (46.77)
Imputation using predictive mean matching method											
10%	3.893 (−1.42)	4.054 (5.22)	4.066 (2.89)	4.045 (1.60)	4.049 (−0.35)	4.162 (3.14)	4.190 (4.51)	4.160 (6.81)	4.241 (3.56)	4.141 (2.48)	5.598 (3.73)
20%	4.156 (5.24)	4.003 (3.87)	3.948 (−0.11)	4.214 (5.84)	4.063 (0.01)	4.179 (3.57)	4.191 (4.53)	4.153 (6.62)	4.224 (3.16)	4.248 (5.13)	5.726 (6.11)
30%	4.245 (7.49)	4.287 (11.25)	4.400 (11.32)	4.418 (10.96)	4.271 (5.11)	4.501 (11.55)	4.325 (7.86)	4.531 (16.33)	4.449 (8.64)	4.512 (11.67)	5.889 (9.14)
40%	4.207 (6.53)	4.278 (11.01)	4.345 (9.94)	4.301 (8.04)	4.301 (5.85)	4.361 (8.09)	4.426 (10.39)	4.458 (14.44)	4.478 (9.35)	4.463 (10.44)	5.910 (9.52)
Imputation using propensity−score method											
10%	3.912 (−0.93)	4.071 (5.64)	4.126 (4.39)	4.106 (3.13)	4.064 (0.02)	4.191 (3.86)	4.204 (4.84)	4.203 (7.90)	4.275 (4.39)	4.197 (3.88)	5.628 (4.29)
20%	4.234 (7.21)	4.074 (5.72)	4.004 (1.32)	4.209 (5.72)	4.175 (2.75)	4.291 (6.35)	4.224 (5.34)	4.239 (8.83)	4.242 (3.59)	4.337 (7.33)	5.788 (7.25)
30%	4.372 (10.71)	4.393 (14.01)	4.518 (14.32)	4.544 (14.14)	4.355 (7.19)	4.561 (13.03)	4.427 (10.40)	4.786 (22.88)	4.559 (11.34)	4.545 (12.47)	6.026 (11.67)
40%	4.163 (5.41)	4.230 (9.77)	4.195 (6.13)	4.230 (6.24)	4.282 (5.39)	4.243 (5.15)	4.331 (8.02)	4.437 (13.91)	4.492 (9.70)	4.444 (9.99)	5.877 (8.91)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	5.554	5.565	5.563	5.584	5.578	5.595	5.582	5.569	5.608
Imputation using linear regression									
10%	5.829 (4.95)	5.907 (6.16)	5.908 (6.20)	5.986 (7.20)	5.926 (6.24)	5.879 (5.07)	5.962 (6.81)	5.889 (5.75)	6.051 (7.90)
20%	6.479 (16.64)	6.491 (16.64)	6.501 (16.86)	6.433 (15.21)	6.428 (15.24)	6.573 (17.47)	6.524 (16.89)	6.565 (17.89)	6.515 (16.17)
30%	7.394 (33.13)	6.899 (23.98)	7.080 (27.27)	6.855 (22.77)	6.974 (25.03)	6.946 (24.14)	6.854 (22.79)	7.049 (26.58)	7.021 (25.19)
40%	7.980 (43.67)	8.115 (45.83)	8.101 (45.62)	7.781 (39.33)	8.121 (45.60)	8.053 (43.93)	8.274 (48.23)	8.355 (50.03)	8.579 (52.97)
Imputation using predictive mean matching method									
10%	5.611 (1.02)	5.630 (1.17)	5.623 (1.08)	5.675 (1.64)	5.653 (1.34)	5.657 (1.11)	5.695 (2.03)	5.615 (0.82)	5.758 (2.68)
20%	5.848 (5.29)	5.860 (5.31)	5.872 (5.55)	5.887 (5.43)	5.869 (5.22)	5.950 (6.35)	5.861 (5.00)	5.880 (5.59)	5.865 (4.58)
30%	5.942 (6.98)	5.848 (5.10)	5.766 (3.64)	5.800 (3.88)	5.804 (4.05)	5.840 (4.37)	5.743 (2.89)	5.804 (4.22)	5.905 (5.28)
40%	6.049 (8.91)	6.079 (9.25)	6.101 (9.66)	5.984 (7.16)	6.186 (10.89)	6.264 (11.96)	6.227 (11.56)	6.251 (12.25)	6.258 (11.58)
Imputation using propensity−score method									
10%	5.606 (0.94)	5.587 (0.40)	5.645 (1.47)	5.706 (2.19)	5.678 (1.80)	5.631 (0.63)	5.693 (1.99)	5.588 (0.33)	5.762 (2.74)
20%	5.918 (6.54)	5.802 (4.27)	5.850 (5.15)	5.850 (4.77)	5.881 (5.44)	5.942 (6.20)	5.976 (7.05)	5.885 (5.67)	5.943 (5.98)
30%	5.792 (4.27)	5.836 (4.88)	5.966 (7.25)	5.804 (3.95)	5.751 (3.09)	5.853 (4.61)	5.759 (3.17)	5.858 (5.19)	6.018 (7.31)
40%	5.961 (7.32)	5.837 (4.89)	6.023 (8.26)	5.836 (4.52)	6.072 (8.86)	6.132 (9.60)	6.017 (7.80)	6.100 (9.53)	6.123 (9.18)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	5.358	5.378	5.365	5.399	5.392	5.382	5.380	5.376	5.398
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	5.941 (10.90)	5.979 (11.16)	5.934 (10.61)	5.961 (10.40)	5.973 (10.76)	5.988 (11.25)	5.944 (10.49)	6.009 (11.76)	6.025 (11.60)
20%	6.252 (16.70)	6.132 (14.01)	6.443 (20.09)	6.221 (15.22)	6.328 (17.34)	6.263 (16.36)	6.237 (15.93)	6.281 (16.83)	6.277 (16.28)
30%	7.137 (33.21)	7.137 (32.70)	7.511 (40.01)	7.236 (34.03)	7.340 (36.11)	7.112 (32.14)	7.356 (36.74)	7.392 (37.50)	7.320 (35.59)
40%	7.891 (47.30)	8.098 (50.56)	7.638 (42.37)	7.796 (44.39)	8.271 (53.38)	8.426 (56.56)	8.955 (66.47)	7.994 (48.70)	7.998 (48.15)
Imputation using predictive mean matching method									
10%	5.697 (6.34)	5.656 (5.15)	5.643 (5.19)	5.648 (4.61)	5.678 (5.30)	5.704 (5.99)	5.670 (5.39)	5.709 (6.19)	5.703 (5.64)
20%	5.636 (5.19)	5.570 (3.57)	5.764 (7.45)	5.646 (4.57)	5.761 (6.83)	5.717 (6.22)	5.645 (4.92)	5.705 (6.11)	5.751 (6.53)
30%	5.909 (10.28)	5.951 (10.65)	6.014 (12.09)	5.935 (9.93)	6.088 (12.91)	5.906 (9.73)	6.056 (12.57)	5.998 (11.56)	6.096 (12.91)
40%	6.002 (12.03)	6.042 (12.34)	6.002 (11.87)	6.011 (11.33)	6.103 (13.18)	6.106 (13.46)	6.149 (14.29)	6.154 (14.47)	6.137 (13.68)
Imputation using propensity−score method									
10%	5.707 (6.52)	5.719 (6.34)	5.706 (6.36)	5.666 (4.95)	5.711 (5.91)	5.718 (6.25)	5.718 (6.29)	5.751 (6.96)	5.761 (6.72)
20%	5.721 (6.78)	5.643 (4.93)	5.776 (7.66)	5.789 (7.21)	5.880 (9.03)	5.758 (6.99)	5.762 (7.11)	5.746 (6.89)	5.871 (8.75)
30%	6.028 (12.51)	6.094 (13.31)	6.144 (14.52)	6.031 (11.70)	6.145 (13.96)	6.020 (11.85)	6.387 (18.72)	6.125 (13.93)	6.113 (13.24)
40%	5.960 (11.25)	5.886 (9.44)	5.924 (10.42)	5.997 (11.08)	5.991 (11.10)	5.990 (11.29)	6.123 (13.81)	6.196 (15.25)	6.117 (13.30)

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	100.00	85.71	100.00	100.00	82.14	96.43	96.43	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	89.66	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	86.21	93.10	93.10	96.55	89.66	96.55	96.55	93.10	86.21	100.00	62.07
40%	96.43	96.43	96.43	96.43	96.43	92.86	92.86	92.86	96.43	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	86.21	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	62.07
30%	89.66	93.10	86.21	89.66	89.66	96.55	96.55	89.66	93.10	96.55	62.07
40%	96.43	92.86	85.71	100.00	92.86	85.71	85.71	96.43	96.43	96.43	60.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	100.00	96.43	96.43	89.29	100.00	100.00	89.29	67.86
20%	89.66	79.31	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	89.66	93.10	89.66	89.66	89.66	96.55	96.55	93.10	86.21	96.55	62.07
40%	92.86	96.43	96.43	100.00	96.43	92.86	92.86	96.43	100.00	96.43	67.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.43	96.43	96.43	92.86	92.86	100.00	96.43	100.00	82.14
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	79.31	100.00	93.10	96.55	93.10	68.97
30%	96.43	100.00	96.43	100.00	96.43	92.86	100.00	92.86	96.43	96.43	78.57
40%	92.86	82.14	96.43	100.00	92.86	92.86	100.00	89.29	92.86	100.00	60.71
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	96.55	96.55	100.00	82.76
20%	93.10	96.55	96.55	86.21	93.10	82.76	96.55	93.10	96.55	93.10	68.97
30%	96.43	100.00	100.00	100.00	100.00	92.86	96.43	96.43	92.86	96.43	82.14
40%	89.29	82.14	96.43	100.00	92.86	92.86	100.00	96.43	92.86	96.43	71.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	93.10	100.00	100.00	100.00	96.55	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	86.21	100.00	93.10	96.55	93.10	72.41
30%	96.43	100.00	100.00	100.00	96.43	92.86	96.43	92.86	92.86	96.43	78.57
40%	92.86	85.71	96.43	100.00	89.29	96.43	96.43	96.43	96.43	96.43	75.00

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	96.43	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	96.55	100	96.55	96.55	100.00	100.00	100.00	93.10	100	82.76
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	93.10	100	89.66
40%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	96.43	100.00	100.00	100	92.86
20%	96.55	100.00	100	100.00	96.55	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	100.00	96.55	100.00	100.00	96.55	93.10	100	86.21
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_nonshrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	96.55	100.00	100	93.10	100.00	96.55	100	96.55	100.00	100	93.10
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100	96.43
40%	100.00	96.43	100	100.00	100.00	100.00	100	100.00	96.43	100	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	96.55	100.00	100	93.10	100.00	96.55	100	100.00	100.00	100	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100	96.43
40%	100.00	96.43	100	100.00	96.43	100.00	100	100.00	96.43	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	96.55	100.00	100	89.66	100.00	96.55	100	96.55	100.00	100	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100	96.43
40%	100.00	100.00	100	100.00	96.43	100.00	100	100.00	96.43	100	96.43

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	100.00	85.71	100.00	100.00	82.14	96.43	96.43	92.86	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	89.66	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	86.21	93.10	93.10	96.55	89.66	96.55	96.55	93.10	86.21	100.00	62.07
40%	96.43	96.43	96.43	96.43	96.43	92.86	92.86	92.86	96.43	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	96.43	96.43	96.43	89.29	100.00	100.00	89.29	64.29
20%	86.21	82.76	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	62.07
30%	89.66	93.10	86.21	89.66	89.66	96.55	96.55	89.66	93.10	96.55	62.07
40%	96.43	92.86	85.71	100.00	92.86	85.71	85.71	96.43	96.43	96.43	60.71
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	92.86	100.00	96.43	96.43	89.29	100.00	100.00	89.29	67.86
20%	89.66	79.31	96.55	96.55	93.10	100.00	93.10	100.00	96.55	93.10	65.52
30%	89.66	93.10	89.66	89.66	89.66	96.55	96.55	93.10	86.21	96.55	62.07
40%	92.86	96.43	96.43	100.00	96.43	92.86	92.86	96.43	100.00	96.43	67.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	96.43	96.43	96.43	92.86	92.86	100.00	96.43	100.00	82.14
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	79.31	100.00	93.10	96.55	93.10	68.97
30%	96.43	100.00	96.43	100.00	96.43	92.86	100.00	92.86	96.43	96.43	78.57
40%	92.86	82.14	96.43	100.00	92.86	92.86	100.00	89.29	92.86	100.00	60.71
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	96.55	100.00	100.00	100.00	93.10	96.55	96.55	100.00	82.76
20%	93.10	96.55	96.55	86.21	93.10	82.76	96.55	93.10	96.55	93.10	68.97
30%	96.43	100.00	100.00	100.00	100.00	92.86	96.43	96.43	92.86	96.43	82.14
40%	89.29	82.14	96.43	100.00	92.86	92.86	100.00	96.43	92.86	96.43	71.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	93.10	100.00	100.00	100.00	96.55	93.10	100.00	100.00	82.76
20%	93.10	93.10	93.10	86.21	96.55	86.21	100.00	93.10	96.55	93.10	72.41
30%	96.43	100.00	100.00	100.00	96.43	92.86	96.43	92.86	92.86	96.43	78.57
40%	92.86	85.71	96.43	100.00	89.29	96.43	96.43	96.43	96.43	96.43	75.00

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	96.43	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	96.43	100.00	100.00	100	92.86
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	96.55	100	96.55	96.55	100.00	100.00	100.00	93.10	100	82.76
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	96.43	100.00	100.00	100	89.29
20%	96.55	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	93.10	100	89.66
40%	100.00	96.43	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	100	100.00	100.00	96.43	92.86	100.00	100.00	100	89.29
20%	96.55	100.00	100	100.00	96.55	100.00	100.00	100.00	100.00	100	96.55
30%	100.00	100.00	100	100.00	96.55	100.00	100.00	96.55	93.10	100	86.21
40%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_10Treatments/table2_shrink_10_treatments_MAR_n200

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.55	100.00	100	93.10	100.00	96.55	100	93.10	100.00	100.00	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100.00	96.43
40%	100.00	96.43	100	100.00	100.00	100.00	100	100.00	96.43	100.00	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.55	100.00	100	96.55	100.00	96.55	100	100.00	100.00	96.55	93.10
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100.00	96.43
40%	100.00	96.43	100	100.00	96.43	100.00	100	100.00	96.43	100.00	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.55	100.00	100	89.66	100.00	96.55	100	96.55	100.00	96.55	89.66
30%	100.00	100.00	100	100.00	100.00	100.00	100	96.43	100.00	100.00	96.43
40%	100.00	100.00	100	100.00	96.43	100.00	100	100.00	96.43	100.00	96.43

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.255	1.248	1.256	1.267	1.263	1.271	1.279	1.294	1.294	1.301	1.728
Imputation using linear regression											
10%	1.310 (4.38)	1.308 (4.76)	1.322 (5.31)	1.326 (4.61)	1.324 (4.89)	1.333 (4.92)	1.336 (4.46)	1.333 (2.96)	1.358 (4.96)	1.367 (5.09)	1.805 (4.47)
20%	1.361 (8.48)	1.375 (10.12)	1.364 (8.61)	1.381 (8.96)	1.383 (9.52)	1.394 (9.69)	1.402 (9.58)	1.395 (7.75)	1.411 (9.08)	1.421 (9.22)	1.882 (8.92)
30%	1.463 (16.58)	1.472 (17.92)	1.483 (18.09)	1.484 (17.08)	1.501 (18.86)	1.517 (19.36)	1.505 (17.69)	1.504 (16.19)	1.524 (17.81)	1.524 (17.10)	2.021 (16.95)
40%	1.601 (27.61)	1.628 (30.37)	1.622 (29.21)	1.665 (31.35)	1.649 (30.58)	1.662 (30.77)	1.659 (29.74)	1.658 (28.10)	1.645 (27.14)	1.690 (29.89)	2.208 (27.76)
Imputation using predictive mean matching method											
10%	1.302 (3.79)	1.301 (4.19)	1.313 (4.60)	1.324 (4.47)	1.320 (4.54)	1.328 (4.50)	1.329 (3.89)	1.327 (2.53)	1.350 (4.33)	1.355 (4.12)	1.795 (3.88)
20%	1.346 (7.24)	1.351 (8.23)	1.343 (6.99)	1.355 (6.96)	1.372 (8.64)	1.369 (7.72)	1.387 (8.48)	1.386 (7.10)	1.391 (7.52)	1.401 (7.66)	1.860 (7.62)
30%	1.445 (15.16)	1.465 (17.35)	1.457 (16.01)	1.429 (12.74)	1.474 (16.77)	1.489 (17.16)	1.485 (16.12)	1.480 (14.34)	1.519 (17.41)	1.493 (14.71)	1.992 (15.24)
40%	1.541 (22.80)	1.570 (25.78)	1.573 (25.26)	1.604 (26.55)	1.614 (27.81)	1.594 (25.39)	1.592 (24.45)	1.591 (22.95)	1.579 (22.04)	1.593 (22.45)	2.123 (22.84)
Imputation using propensity−score method											
10%	1.307 (4.20)	1.308 (4.80)	1.319 (5.06)	1.325 (4.59)	1.323 (4.79)	1.332 (4.82)	1.333 (4.24)	1.339 (3.48)	1.361 (5.16)	1.361 (4.60)	1.802 (4.26)
20%	1.346 (7.24)	1.353 (8.39)	1.356 (7.99)	1.355 (6.95)	1.378 (9.16)	1.379 (8.50)	1.395 (9.07)	1.383 (6.88)	1.394 (7.71)	1.411 (8.43)	1.862 (7.72)
30%	1.458 (16.17)	1.444 (15.67)	1.470 (17.05)	1.438 (13.44)	1.461 (15.71)	1.471 (15.75)	1.471 (15.03)	1.479 (14.29)	1.500 (15.92)	1.493 (14.71)	2.014 (16.53)
40%	1.564 (24.68)	1.616 (29.44)	1.592 (26.76)	1.626 (28.28)	1.630 (29.12)	1.620 (27.42)	1.626 (27.11)	1.632 (26.07)	1.620 (25.24)	1.645 (26.41)	2.159 (24.95)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.244	1.257	1.256	1.263	1.277	1.279	1.292	1.277	1.299	1.295	1.712
Imputation using linear regression											

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.291 (3.81)	1.294 (2.90)	1.307 (4.06)	1.305 (3.36)	1.309 (2.50)	1.319 (3.06)	1.330 (2.96)	1.338 (4.75)	1.342 (3.32)	1.348 (4.14)	1.783 (4.16)
20%	1.372 (10.25)	1.382 (9.91)	1.382 (10.00)	1.388 (9.88)	1.395 (9.17)	1.406 (9.87)	1.407 (8.89)	1.419 (11.09)	1.409 (8.51)	1.430 (10.49)	1.894 (10.63)
30%	1.445 (16.16)	1.478 (17.54)	1.475 (17.47)	1.481 (17.27)	1.481 (15.93)	1.489 (16.34)	1.505 (16.51)	1.484 (16.19)	1.524 (17.32)	1.499 (15.80)	1.999 (16.75)
40%	1.595 (28.23)	1.603 (27.50)	1.615 (28.60)	1.604 (27.03)	1.629 (27.51)	1.611 (25.93)	1.614 (24.90)	1.608 (25.94)	1.668 (28.44)	1.678 (29.64)	2.201 (28.60)
Imputation using predictive mean matching method											
10%	1.279 (2.79)	1.285 (2.21)	1.296 (3.15)	1.298 (2.76)	1.303 (2.01)	1.309 (2.34)	1.313 (1.62)	1.325 (3.74)	1.327 (2.22)	1.336 (3.17)	1.766 (3.17)
20%	1.350 (8.50)	1.366 (8.65)	1.374 (9.36)	1.370 (8.49)	1.386 (8.52)	1.389 (8.57)	1.392 (7.75)	1.404 (9.96)	1.386 (6.75)	1.408 (8.77)	1.862 (8.79)
30%	1.431 (15.06)	1.434 (14.07)	1.443 (14.88)	1.424 (12.77)	1.430 (11.95)	1.445 (12.91)	1.462 (13.17)	1.452 (13.70)	1.472 (13.31)	1.475 (13.97)	1.980 (15.69)
40%	1.546 (24.29)	1.562 (24.22)	1.560 (24.17)	1.570 (24.32)	1.567 (22.66)	1.592 (24.45)	1.609 (24.59)	1.596 (24.98)	1.605 (23.61)	1.602 (23.75)	2.124 (24.07)
Imputation using propensity−score method											
10%	1.286 (3.33)	1.286 (2.29)	1.301 (3.60)	1.301 (2.99)	1.306 (2.22)	1.308 (2.20)	1.321 (2.25)	1.331 (4.21)	1.332 (2.60)	1.339 (3.45)	1.775 (3.68)
20%	1.360 (9.33)	1.373 (9.20)	1.365 (8.68)	1.374 (8.84)	1.389 (8.76)	1.394 (8.94)	1.410 (9.14)	1.409 (10.35)	1.398 (7.67)	1.419 (9.60)	1.878 (9.70)
30%	1.443 (15.96)	1.455 (15.72)	1.441 (14.74)	1.458 (15.46)	1.466 (14.81)	1.458 (13.99)	1.479 (14.47)	1.477 (15.63)	1.491 (14.79)	1.486 (14.79)	1.996 (16.58)
40%	1.539 (23.72)	1.556 (23.80)	1.603 (27.60)	1.593 (26.16)	1.571 (23.01)	1.607 (25.63)	1.603 (24.06)	1.597 (25.05)	1.625 (25.11)	1.634 (26.21)	2.125 (24.15)

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.719	1.729	1.745	1.739	1.751	1.762	1.783	1.782	1.792
Imputation using linear regression									
10%	1.803 (4.86)	1.823 (5.41)	1.828 (4.71)	1.826 (4.99)	1.838 (5.01)	1.842 (4.56)	1.837 (3.05)	1.872 (5.06)	1.885 (5.18)
20%	1.901 (10.55)	1.886 (9.05)	1.909 (9.37)	1.912 (9.96)	1.928 (10.12)	1.938 (10.02)	1.929 (8.19)	1.951 (9.51)	1.965 (9.64)
30%	2.034 (18.30)	2.049 (18.45)	2.050 (17.44)	2.073 (19.23)	2.096 (19.71)	2.080 (18.06)	2.078 (16.58)	2.106 (18.18)	2.105 (17.45)
40%	2.245 (30.54)	2.238 (29.41)	2.295 (31.47)	2.274 (30.76)	2.292 (30.90)	2.288 (29.90)	2.287 (28.27)	2.269 (27.31)	2.330 (30.01)
Imputation using predictive mean matching method									
10%	1.793 (4.28)	1.811 (4.70)	1.825 (4.55)	1.820 (4.63)	1.831 (4.59)	1.832 (3.99)	1.829 (2.62)	1.861 (4.42)	1.868 (4.21)
20%	1.867 (8.61)	1.857 (7.39)	1.873 (7.33)	1.896 (9.04)	1.892 (8.10)	1.918 (8.86)	1.916 (7.47)	1.923 (7.90)	1.937 (8.04)
30%	2.019 (17.42)	2.008 (16.11)	1.970 (12.87)	2.032 (16.86)	2.052 (17.24)	2.047 (16.21)	2.040 (14.44)	2.094 (17.48)	2.058 (14.81)
40%	2.163 (25.80)	2.167 (25.32)	2.208 (26.52)	2.222 (27.78)	2.196 (25.44)	2.193 (24.50)	2.193 (23.00)	2.177 (22.14)	2.196 (22.50)
Imputation using propensity−score method									
10%	1.803 (4.86)	1.818 (5.12)	1.827 (4.65)	1.823 (4.86)	1.836 (4.88)	1.837 (4.30)	1.846 (3.53)	1.875 (5.22)	1.876 (4.66)
20%	1.872 (8.87)	1.876 (8.47)	1.875 (7.41)	1.907 (9.65)	1.907 (8.96)	1.930 (9.54)	1.914 (7.35)	1.928 (8.18)	1.952 (8.89)
30%	1.996 (16.08)	2.031 (17.42)	1.987 (13.83)	2.020 (16.13)	2.033 (16.15)	2.034 (15.43)	2.045 (14.68)	2.073 (16.30)	2.063 (15.09)
40%	2.229 (29.66)	2.197 (27.04)	2.243 (28.49)	2.249 (29.35)	2.235 (27.70)	2.244 (27.36)	2.252 (26.32)	2.236 (25.50)	2.270 (26.65)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.730	1.728	1.738	1.758	1.761	1.778	1.757	1.787	1.782
Imputation using linear regression									

SampleSize_n200_10Treatments/table3_nonshrink_10_treatments_MAR_n200

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.786 (3.24)	1.804 (4.40)	1.802 (3.70)	1.808 (2.84)	1.820 (3.40)	1.836 (3.29)	1.847 (5.10)	1.852 (3.65)	1.861 (4.48)
20%	1.908 (10.28)	1.908 (10.38)	1.916 (10.26)	1.925 (9.53)	1.941 (10.24)	1.942 (9.26)	1.959 (11.47)	1.946 (8.89)	1.975 (10.86)
30%	2.043 (18.12)	2.041 (18.06)	2.048 (17.87)	2.048 (16.51)	2.059 (16.93)	2.081 (17.09)	2.053 (16.81)	2.107 (17.91)	2.074 (16.41)
40%	2.212 (27.89)	2.229 (28.97)	2.215 (27.47)	2.247 (27.85)	2.224 (26.33)	2.228 (25.34)	2.220 (26.36)	2.302 (28.81)	2.316 (30.00)
Imputation using predictive mean matching method									
10%	1.774 (2.58)	1.789 (3.53)	1.792 (3.13)	1.800 (2.38)	1.808 (2.71)	1.813 (1.99)	1.830 (4.13)	1.833 (2.59)	1.845 (3.54)
20%	1.884 (8.93)	1.895 (9.65)	1.891 (8.79)	1.912 (8.79)	1.917 (8.86)	1.921 (8.04)	1.937 (10.25)	1.913 (7.04)	1.943 (9.06)
30%	1.985 (14.73)	1.997 (15.53)	1.972 (13.45)	1.980 (12.62)	1.999 (13.57)	2.023 (13.81)	2.010 (14.36)	2.037 (13.98)	2.042 (14.63)
40%	2.145 (24.01)	2.142 (23.96)	2.157 (24.15)	2.152 (22.45)	2.186 (24.16)	2.211 (24.36)	2.192 (24.77)	2.206 (23.43)	2.201 (23.56)
Imputation using propensity−score method									
10%	1.775 (2.63)	1.797 (3.94)	1.796 (3.33)	1.803 (2.56)	1.805 (2.55)	1.824 (2.59)	1.837 (4.56)	1.839 (2.94)	1.849 (3.79)
20%	1.895 (9.57)	1.885 (9.07)	1.898 (9.22)	1.918 (9.13)	1.925 (9.31)	1.947 (9.50)	1.946 (10.74)	1.931 (8.04)	1.959 (9.97)
30%	2.012 (16.33)	1.994 (15.39)	2.017 (16.09)	2.029 (15.44)	2.018 (14.62)	2.046 (15.10)	2.043 (16.25)	2.063 (15.44)	2.057 (15.44)
40%	2.149 (24.23)	2.212 (27.96)	2.200 (26.58)	2.169 (23.40)	2.218 (26.00)	2.213 (24.49)	2.204 (25.45)	2.243 (25.52)	2.256 (26.61)

SampleSize_n200_10Treatments/table3_shrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.255	1.248	1.256	1.267	1.263	1.271	1.279	1.294	1.294	1.301	1.724
Imputation using linear regression											
10%	1.310 (4.38)	1.308 (4.76)	1.322 (5.31)	1.326 (4.61)	1.324 (4.89)	1.333 (4.92)	1.336 (4.46)	1.333 (2.96)	1.358 (4.96)	1.367 (5.09)	1.804 (4.64)
20%	1.361 (8.48)	1.375 (10.12)	1.364 (8.61)	1.381 (8.96)	1.383 (9.52)	1.394 (9.69)	1.402 (9.58)	1.395 (7.75)	1.411 (9.08)	1.421 (9.22)	1.887 (9.45)
30%	1.463 (16.58)	1.472 (17.92)	1.483 (18.09)	1.484 (17.08)	1.501 (18.86)	1.517 (19.36)	1.505 (17.69)	1.504 (16.19)	1.524 (17.81)	1.524 (17.10)	2.023 (17.33)
40%	1.601 (27.61)	1.628 (30.37)	1.622 (29.21)	1.665 (31.35)	1.649 (30.58)	1.662 (30.77)	1.659 (29.74)	1.658 (28.10)	1.645 (27.14)	1.690 (29.89)	2.209 (28.10)
Imputation using predictive mean matching method											
10%	1.302 (3.79)	1.301 (4.19)	1.313 (4.60)	1.324 (4.47)	1.320 (4.54)	1.328 (4.50)	1.329 (3.89)	1.327 (2.53)	1.350 (4.33)	1.355 (4.12)	1.794 (4.04)
20%	1.346 (7.24)	1.351 (8.23)	1.343 (6.99)	1.355 (6.96)	1.372 (8.64)	1.369 (7.72)	1.387 (8.48)	1.386 (7.10)	1.391 (7.52)	1.401 (7.66)	1.864 (8.09)
30%	1.445 (15.16)	1.465 (17.35)	1.457 (16.01)	1.429 (12.74)	1.474 (16.77)	1.489 (17.16)	1.485 (16.12)	1.480 (14.34)	1.519 (17.41)	1.493 (14.71)	1.995 (15.67)
40%	1.541 (22.80)	1.570 (25.78)	1.573 (25.26)	1.604 (26.55)	1.614 (27.81)	1.594 (25.39)	1.592 (24.45)	1.591 (22.95)	1.579 (22.04)	1.593 (22.45)	2.124 (23.17)
Imputation using propensity−score method											
10%	1.307 (4.20)	1.308 (4.80)	1.319 (5.06)	1.325 (4.59)	1.323 (4.79)	1.332 (4.82)	1.333 (4.24)	1.339 (3.48)	1.361 (5.16)	1.361 (4.60)	1.801 (4.42)
20%	1.346 (7.24)	1.353 (8.39)	1.356 (7.99)	1.355 (6.95)	1.378 (9.16)	1.379 (8.50)	1.395 (9.07)	1.383 (6.88)	1.394 (7.71)	1.411 (8.43)	1.866 (8.19)
30%	1.458 (16.17)	1.444 (15.67)	1.470 (17.05)	1.438 (13.44)	1.461 (15.71)	1.471 (15.75)	1.471 (15.03)	1.479 (14.29)	1.500 (15.92)	1.493 (14.71)	2.018 (17.01)
40%	1.564 (24.68)	1.616 (29.44)	1.592 (26.76)	1.626 (28.28)	1.630 (29.12)	1.620 (27.42)	1.626 (27.11)	1.632 (26.07)	1.620 (25.24)	1.645 (26.41)	2.160 (25.28)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.244	1.257	1.256	1.263	1.277	1.279	1.292	1.277	1.299	1.295	1.717
Imputation using linear regression											

SampleSize_n200_10Treatments/table3_shrink_10_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.291 (3.81)	1.294 (2.90)	1.307 (4.06)	1.305 (3.36)	1.309 (2.50)	1.319 (3.06)	1.330 (2.96)	1.338 (4.75)	1.342 (3.32)	1.348 (4.14)	1.788 (4.13)
20%	1.372 (10.25)	1.382 (9.91)	1.382 (10.00)	1.388 (9.88)	1.395 (9.17)	1.406 (9.87)	1.407 (8.89)	1.419 (11.09)	1.409 (8.51)	1.430 (10.49)	1.901 (10.72)
30%	1.445 (16.16)	1.478 (17.54)	1.475 (17.47)	1.481 (17.27)	1.481 (15.93)	1.489 (16.34)	1.505 (16.51)	1.484 (16.19)	1.524 (17.32)	1.499 (15.80)	2.010 (17.12)
40%	1.595 (28.23)	1.603 (27.50)	1.615 (28.60)	1.604 (27.03)	1.629 (27.51)	1.611 (25.93)	1.614 (24.90)	1.608 (25.94)	1.668 (28.44)	1.678 (29.64)	2.202 (28.29)
Imputation using predictive mean matching method											
10%	1.279 (2.79)	1.285 (2.21)	1.296 (3.15)	1.298 (2.76)	1.303 (2.01)	1.309 (2.34)	1.313 (1.62)	1.325 (3.74)	1.327 (2.22)	1.336 (3.17)	1.772 (3.22)
20%	1.350 (8.50)	1.366 (8.65)	1.374 (9.36)	1.370 (8.49)	1.386 (8.52)	1.389 (8.57)	1.392 (7.75)	1.404 (9.96)	1.386 (6.75)	1.408 (8.77)	1.871 (9.00)
30%	1.431 (15.06)	1.434 (14.07)	1.443 (14.88)	1.424 (12.77)	1.430 (11.95)	1.445 (12.91)	1.462 (13.17)	1.452 (13.70)	1.472 (13.31)	1.475 (13.97)	1.991 (15.98)
40%	1.546 (24.29)	1.562 (24.22)	1.560 (24.17)	1.570 (24.32)	1.567 (22.66)	1.592 (24.45)	1.609 (24.59)	1.596 (24.98)	1.605 (23.61)	1.602 (23.75)	2.124 (23.75)
Imputation using propensity−score method											
10%	1.286 (3.33)	1.286 (2.29)	1.301 (3.60)	1.301 (2.99)	1.306 (2.22)	1.308 (2.20)	1.321 (2.25)	1.331 (4.21)	1.332 (2.60)	1.339 (3.45)	1.780 (3.68)
20%	1.360 (9.33)	1.373 (9.20)	1.365 (8.68)	1.374 (8.84)	1.389 (8.76)	1.394 (8.94)	1.410 (9.14)	1.409 (10.35)	1.398 (7.67)	1.419 (9.60)	1.884 (9.76)
30%	1.443 (15.96)	1.455 (15.72)	1.441 (14.74)	1.458 (15.46)	1.466 (14.81)	1.458 (13.99)	1.479 (14.47)	1.477 (15.63)	1.491 (14.79)	1.486 (14.79)	2.006 (16.85)
40%	1.539 (23.72)	1.556 (23.80)	1.603 (27.60)	1.593 (26.16)	1.571 (23.01)	1.607 (25.63)	1.603 (24.06)	1.597 (25.05)	1.625 (25.11)	1.634 (26.21)	2.127 (23.89)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.722	1.727	1.731	1.728	1.731	1.738	1.743	1.746	1.748
Imputation using linear regression									
10%	1.799 (4.51)	1.809 (4.77)	1.811 (4.62)	1.811 (4.75)	1.817 (4.94)	1.816 (4.48)	1.810 (3.83)	1.832 (4.91)	1.839 (5.23)
20%	1.899 (10.32)	1.887 (9.27)	1.911 (10.36)	1.901 (9.96)	1.914 (10.53)	1.916 (10.25)	1.914 (9.81)	1.930 (10.51)	1.940 (10.98)
30%	2.034 (18.15)	2.046 (18.44)	2.049 (18.38)	2.064 (19.45)	2.083 (20.33)	2.067 (18.92)	2.062 (18.34)	2.086 (19.46)	2.086 (19.37)
40%	2.235 (29.82)	2.227 (28.94)	2.283 (31.85)	2.261 (30.80)	2.275 (31.38)	2.275 (30.89)	2.267 (30.07)	2.253 (28.99)	2.310 (32.17)
Imputation using predictive mean matching method									
10%	1.789 (3.94)	1.797 (4.06)	1.807 (4.37)	1.804 (4.36)	1.808 (4.43)	1.805 (3.87)	1.801 (3.36)	1.821 (4.26)	1.824 (4.35)
20%	1.866 (8.36)	1.859 (7.64)	1.876 (8.35)	1.883 (8.93)	1.879 (8.54)	1.895 (9.02)	1.901 (9.06)	1.899 (8.76)	1.909 (9.24)
30%	2.020 (17.30)	2.005 (16.10)	1.971 (13.82)	2.025 (17.17)	2.041 (17.91)	2.033 (16.96)	2.026 (16.24)	2.072 (18.64)	2.038 (16.63)
40%	2.154 (25.12)	2.156 (24.82)	2.195 (26.80)	2.207 (27.70)	2.176 (25.69)	2.175 (25.12)	2.172 (24.62)	2.159 (23.64)	2.173 (24.36)
Imputation using propensity−score method									
10%	1.800 (4.57)	1.805 (4.51)	1.810 (4.52)	1.806 (4.52)	1.813 (4.73)	1.811 (4.22)	1.818 (4.32)	1.834 (5.04)	1.831 (4.75)
20%	1.870 (8.59)	1.877 (8.68)	1.878 (8.49)	1.894 (9.58)	1.895 (9.48)	1.908 (9.76)	1.898 (8.90)	1.904 (9.06)	1.925 (10.15)
30%	1.997 (16.02)	2.029 (17.50)	1.991 (14.97)	2.014 (16.53)	2.023 (16.87)	2.022 (16.32)	2.032 (16.58)	2.055 (17.66)	2.046 (17.07)
40%	2.221 (28.99)	2.186 (26.57)	2.231 (28.88)	2.236 (29.37)	2.219 (28.13)	2.229 (28.25)	2.232 (28.04)	2.222 (27.24)	2.251 (28.81)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.723	1.722	1.725	1.729	1.725	1.735	1.728	1.740	1.737
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.789 (3.80)	1.797 (4.35)	1.797 (4.16)	1.798 (4.00)	1.802 (4.51)	1.814 (4.51)	1.816 (5.07)	1.821 (4.67)	1.827 (5.22)
20%	1.906 (10.63)	1.905 (10.59)	1.906 (10.49)	1.919 (11.01)	1.922 (11.46)	1.922 (10.78)	1.932 (11.80)	1.926 (10.70)	1.946 (12.05)
30%	2.042 (18.54)	2.036 (18.21)	2.044 (18.49)	2.043 (18.20)	2.050 (18.85)	2.067 (19.13)	2.040 (18.05)	2.083 (19.73)	2.054 (18.26)
40%	2.207 (28.08)	2.223 (29.08)	2.204 (27.77)	2.241 (29.63)	2.216 (28.46)	2.211 (27.43)	2.207 (27.73)	2.282 (31.17)	2.294 (32.08)
Imputation using predictive mean matching method									
10%	1.779 (3.27)	1.783 (3.52)	1.786 (3.57)	1.789 (3.49)	1.791 (3.86)	1.791 (3.24)	1.798 (4.06)	1.803 (3.60)	1.810 (4.24)
20%	1.885 (9.38)	1.892 (9.86)	1.881 (9.04)	1.905 (10.18)	1.897 (9.96)	1.899 (9.44)	1.910 (10.52)	1.894 (8.84)	1.913 (10.17)
30%	1.987 (15.31)	1.995 (15.85)	1.967 (14.06)	1.973 (14.14)	1.991 (15.44)	2.009 (15.77)	1.997 (15.59)	2.010 (15.53)	2.021 (16.39)
40%	2.141 (24.26)	2.136 (24.03)	2.145 (24.34)	2.147 (24.20)	2.177 (26.20)	2.192 (26.35)	2.177 (26.00)	2.185 (25.61)	2.177 (25.38)
Imputation using propensity−score method									
10%	1.779 (3.23)	1.790 (3.93)	1.790 (3.78)	1.792 (3.68)	1.788 (3.66)	1.800 (3.76)	1.805 (4.46)	1.808 (3.94)	1.814 (4.49)
20%	1.895 (10.01)	1.883 (9.33)	1.889 (9.53)	1.911 (10.55)	1.907 (10.54)	1.927 (11.05)	1.918 (11.00)	1.911 (9.83)	1.931 (11.19)
30%	2.013 (16.85)	1.991 (15.63)	2.014 (16.76)	2.020 (16.86)	2.012 (16.63)	2.031 (17.06)	2.031 (17.56)	2.038 (17.14)	2.035 (17.18)
40%	2.145 (24.50)	2.205 (28.05)	2.189 (26.89)	2.165 (25.22)	2.210 (28.12)	2.197 (26.59)	2.192 (26.85)	2.225 (27.86)	2.234 (28.63)

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	100.00	100.00	100.00	96.43	96.43	100.00	89.29	96.43	96.43	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	93.10	96.55	96.55	100.00	96.55	96.55	100.00	96.55	93.10	93.10	68.97
30%	100.00	89.29	89.29	92.86	96.43	96.43	100.00	89.29	89.29	92.86	67.86
40%	96.43	100.00	89.29	100.00	100.00	92.86	96.43	100.00	85.71	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	89.66	89.66	96.55	100.00	96.55	96.55	96.55	96.55	93.10	89.66	58.62
30%	96.43	89.29	89.29	89.29	92.86	89.29	96.43	89.29	89.29	92.86	64.29
40%	96.43	96.43	96.43	100.00	100.00	89.29	96.43	100.00	85.71	96.43	78.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	100.00	96.55	100.00	96.55	93.10	82.76
20%	89.66	89.66	93.10	100.00	93.10	96.55	100.00	96.55	93.10	89.66	62.07
30%	100.00	85.71	89.29	89.29	96.43	96.43	100.00	96.43	92.86	96.43	67.86
40%	96.43	96.43	92.86	100.00	100.00	89.29	100.00	100.00	85.71	96.43	75.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	92.86	92.86	96.43	100.00	96.43	96.43	96.43	96.43	100.00	71.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	89.66	93.10	96.55	86.21	96.55	100.00	96.55	96.55	65.52
20%	96.55	86.21	100.00	89.66	86.21	86.21	93.10	89.66	100.00	86.21	65.52
30%	92.86	92.86	92.86	85.71	82.14	92.86	82.14	89.29	85.71	92.86	46.43
40%	96.55	89.66	93.10	96.55	96.55	93.10	96.55	100.00	93.10	89.66	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	93.10	93.10	93.10	93.10	89.66	100.00	96.55	100.00	96.55	65.52
20%	96.55	96.55	93.10	86.21	89.66	86.21	96.55	89.66	100.00	89.66	65.52
30%	92.86	89.29	89.29	85.71	82.14	89.29	82.14	85.71	92.86	89.29	42.86
40%	96.55	89.66	89.66	93.10	96.55	93.10	96.55	96.55	93.10	89.66	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	89.66	89.66	93.10	86.21	100.00	96.55	96.55	96.55	58.62
20%	89.66	93.10	96.55	86.21	93.10	86.21	93.10	93.10	100.00	89.66	68.97
30%	89.29	92.86	89.29	85.71	85.71	82.14	85.71	85.71	82.14	92.86	46.43
40%	96.55	89.66	96.55	96.55	89.66	96.55	93.10	100.00	93.10	93.10	65.52

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	96.43	100	100	100.00	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	96.55	100	96.55
30%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	96.55	100	100	100.00	96.55	100	93.10
30%	100	100	96.43	100	96.43	100	100	100.00	92.86	100	85.71
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
30%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
40%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	100.00	100	100	96.43	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_nonshrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	96.55	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
30%	100.00	100.00	100.00	96.43	96.43	96.43	96.43	100.00	100.00	96.43	82.14
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	100.00	92.86	92.86	100.00	85.71
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	96.43	89.29	96.43	96.43	82.14
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	93.10	93.10

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	100.00	100.00	100.00	96.43	96.43	100.00	89.29	96.43	96.43	67.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	93.10	96.55	96.55	100.00	96.55	96.55	100.00	96.55	93.10	93.10	68.97
30%	100.00	89.29	89.29	92.86	96.43	96.43	100.00	89.29	89.29	92.86	67.86
40%	96.43	100.00	89.29	100.00	100.00	92.86	96.43	100.00	85.71	96.43	75.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	96.55	93.10	100.00	93.10	93.10	79.31
20%	89.66	89.66	96.55	100.00	96.55	96.55	96.55	96.55	93.10	89.66	58.62
30%	96.43	89.29	89.29	89.29	92.86	89.29	96.43	89.29	89.29	92.86	64.29
40%	96.43	96.43	96.43	100.00	100.00	89.29	96.43	100.00	85.71	96.43	78.57
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	93.10	100.00	96.55	100.00	96.55	100.00	96.55	93.10	82.76
20%	89.66	89.66	93.10	100.00	93.10	96.55	100.00	96.55	93.10	89.66	62.07
30%	100.00	85.71	89.29	89.29	96.43	96.43	100.00	96.43	92.86	96.43	67.86
40%	96.43	96.43	92.86	100.00	100.00	89.29	100.00	100.00	85.71	96.43	75.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	92.86	92.86	96.43	100.00	96.43	96.43	96.43	96.43	100.00	71.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	93.10	89.66	93.10	96.55	86.21	96.55	100.00	96.55	96.55	65.52
20%	96.55	86.21	100.00	89.66	86.21	86.21	93.10	89.66	100.00	86.21	65.52
30%	92.86	92.86	92.86	85.71	82.14	92.86	82.14	89.29	85.71	92.86	46.43
40%	96.55	89.66	93.10	96.55	96.55	93.10	96.55	100.00	93.10	89.66	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	93.10	93.10	93.10	93.10	89.66	100.00	96.55	100.00	96.55	65.52
20%	96.55	96.55	93.10	86.21	89.66	86.21	96.55	89.66	100.00	89.66	65.52
30%	92.86	89.29	89.29	85.71	82.14	89.29	82.14	85.71	92.86	89.29	42.86
40%	96.55	89.66	89.66	93.10	96.55	93.10	96.55	96.55	93.10	89.66	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	89.66	89.66	93.10	86.21	100.00	96.55	96.55	96.55	58.62
20%	89.66	93.10	96.55	86.21	93.10	86.21	93.10	93.10	100.00	89.66	68.97
30%	89.29	92.86	89.29	85.71	85.71	82.14	85.71	85.71	82.14	92.86	46.43
40%	96.55	89.66	96.55	96.55	89.66	96.55	93.10	100.00	93.10	93.10	65.52

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	96.43	100	100	100.00	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	96.55	100	96.55
30%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	96.55	96.55	100	93.10
30%	100	100	96.43	100	96.43	100	100	100.00	92.86	100	85.71
40%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
20%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
30%	100	100	100.00	100	100.00	100	100	100.00	96.43	100	96.43
40%	100	100	100.00	100	100.00	100	100	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100	100	100.00	100	100.00	100	100	96.43	100.00	100	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_10Treatments/table2_shrink_10_treatments_MAR_n250

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	96.55	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
30%	100.00	100.00	100.00	96.43	96.43	100.00	96.43	100.00	100.00	96.43	85.71
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	96.55
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	100.00	92.86	92.86	100.00	85.71
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	93.10
20%	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
30%	100.00	100.00	100.00	100.00	96.43	96.43	96.43	92.86	96.43	96.43	82.14
40%	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100.00	96.55	96.55	96.55

SampleSize_n250_10Treatments/table3_nonshrink_10_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.124	1.111	1.126	1.131	1.131	1.135	1.140	1.147	1.162	1.154	1.549
Imputation using linear regression											
10%	1.177 (4.68)	1.179 (6.14)	1.179 (4.72)	1.176 (3.96)	1.199 (6.03)	1.195 (5.29)	1.208 (6.03)	1.209 (5.33)	1.216 (4.65)	1.221 (5.83)	1.622 (4.71)
20%	1.216 (8.19)	1.220 (9.85)	1.244 (10.50)	1.233 (8.95)	1.233 (9.02)	1.252 (10.34)	1.247 (9.42)	1.260 (9.79)	1.265 (8.87)	1.265 (9.61)	1.681 (8.51)
30%	1.328 (18.11)	1.341 (20.71)	1.343 (19.28)	1.351 (19.37)	1.357 (20.02)	1.363 (20.09)	1.366 (19.89)	1.372 (19.54)	1.371 (18.00)	1.389 (20.40)	1.831 (18.17)
40%	1.417 (26.08)	1.414 (27.26)	1.439 (27.79)	1.417 (25.22)	1.465 (29.53)	1.432 (26.17)	1.440 (26.39)	1.473 (28.37)	1.454 (25.16)	1.483 (28.53)	1.961 (26.55)
Imputation using predictive mean matching method											
10%	1.172 (4.24)	1.173 (5.65)	1.174 (4.23)	1.174 (3.74)	1.193 (5.49)	1.194 (5.18)	1.204 (5.62)	1.201 (4.66)	1.211 (4.23)	1.213 (5.09)	1.615 (4.25)
20%	1.199 (6.65)	1.211 (9.07)	1.230 (9.19)	1.226 (8.34)	1.209 (6.90)	1.240 (9.21)	1.236 (8.48)	1.239 (7.94)	1.245 (7.21)	1.250 (8.36)	1.657 (6.94)
30%	1.289 (14.67)	1.293 (16.37)	1.309 (16.24)	1.294 (14.39)	1.295 (14.46)	1.313 (15.66)	1.314 (15.29)	1.333 (16.16)	1.334 (14.84)	1.333 (15.56)	1.781 (14.95)
40%	1.373 (22.10)	1.378 (24.06)	1.406 (24.90)	1.421 (25.57)	1.410 (24.69)	1.397 (23.07)	1.409 (23.61)	1.439 (25.42)	1.429 (22.97)	1.431 (24.00)	1.894 (22.28)
Imputation using propensity−score method											
10%	1.172 (4.21)	1.174 (5.65)	1.178 (4.58)	1.174 (3.77)	1.197 (5.87)	1.196 (5.40)	1.202 (5.47)	1.205 (4.99)	1.212 (4.30)	1.216 (5.37)	1.614 (4.20)
20%	1.212 (7.78)	1.217 (9.52)	1.228 (9.03)	1.240 (9.61)	1.228 (8.60)	1.245 (9.67)	1.252 (9.87)	1.253 (9.22)	1.262 (8.59)	1.261 (9.30)	1.674 (8.07)
30%	1.297 (15.41)	1.313 (18.22)	1.314 (16.67)	1.320 (16.69)	1.330 (17.61)	1.331 (17.26)	1.349 (18.40)	1.359 (18.41)	1.351 (16.28)	1.351 (17.08)	1.792 (15.64)
40%	1.397 (24.30)	1.431 (28.80)	1.436 (27.55)	1.420 (25.47)	1.452 (28.39)	1.430 (26.01)	1.449 (27.16)	1.482 (29.14)	1.461 (25.77)	1.489 (29.02)	1.931 (24.61)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.097	1.109	1.116	1.114	1.123	1.131	1.134	1.146	1.149	1.155	1.513
Imputation using linear regression											

SampleSize_n250_10Treatments/table3_nonshrink_10_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.165 (6.18)	1.161 (4.66)	1.170 (4.89)	1.174 (5.31)	1.180 (5.16)	1.184 (4.70)	1.197 (5.60)	1.206 (5.25)	1.207 (5.02)	1.209 (4.71)	1.607 (6.21)
20%	1.224 (11.61)	1.237 (11.47)	1.231 (10.34)	1.246 (11.77)	1.247 (11.10)	1.248 (10.42)	1.276 (12.57)	1.260 (9.95)	1.275 (10.95)	1.277 (10.58)	1.691 (11.71)
30%	1.321 (20.41)	1.305 (17.61)	1.326 (18.89)	1.326 (18.96)	1.335 (18.91)	1.330 (17.62)	1.350 (19.09)	1.350 (17.87)	1.362 (18.55)	1.356 (17.44)	1.824 (20.50)
40%	1.431 (30.41)	1.425 (28.46)	1.432 (28.36)	1.454 (30.46)	1.427 (27.09)	1.447 (28.00)	1.449 (27.84)	1.454 (26.94)	1.480 (28.77)	1.486 (28.68)	1.977 (30.61)
Imputation using predictive mean matching method											
10%	1.158 (5.58)	1.153 (3.91)	1.160 (3.99)	1.166 (4.66)	1.170 (4.20)	1.172 (3.71)	1.184 (4.47)	1.195 (4.28)	1.198 (4.26)	1.196 (3.55)	1.599 (5.65)
20%	1.213 (10.58)	1.219 (9.89)	1.225 (9.78)	1.234 (10.71)	1.224 (9.02)	1.243 (9.93)	1.263 (11.39)	1.253 (9.34)	1.253 (9.03)	1.260 (9.08)	1.674 (10.63)
30%	1.285 (17.11)	1.282 (15.55)	1.290 (15.60)	1.298 (16.44)	1.299 (15.73)	1.305 (15.41)	1.315 (15.99)	1.334 (16.41)	1.330 (15.71)	1.318 (14.09)	1.775 (17.26)
40%	1.385 (26.24)	1.356 (22.26)	1.368 (22.63)	1.413 (26.84)	1.386 (23.47)	1.395 (23.36)	1.404 (23.84)	1.396 (21.89)	1.431 (24.51)	1.423 (23.18)	1.914 (26.45)
Imputation using propensity−score method											
10%	1.159 (5.65)	1.157 (4.28)	1.164 (4.34)	1.171 (5.05)	1.175 (4.68)	1.175 (3.90)	1.187 (4.69)	1.197 (4.46)	1.199 (4.37)	1.204 (4.20)	1.599 (5.67)
20%	1.208 (10.12)	1.225 (10.41)	1.229 (10.18)	1.231 (10.50)	1.228 (9.42)	1.246 (10.22)	1.259 (11.07)	1.245 (8.69)	1.259 (9.55)	1.266 (9.64)	1.668 (10.20)
30%	1.301 (18.59)	1.305 (17.62)	1.319 (18.19)	1.315 (17.97)	1.319 (17.54)	1.301 (15.10)	1.329 (17.24)	1.336 (16.61)	1.353 (17.71)	1.332 (15.37)	1.799 (18.85)
40%	1.409 (28.43)	1.403 (26.50)	1.399 (25.37)	1.432 (28.54)	1.427 (27.09)	1.449 (28.14)	1.457 (28.48)	1.438 (25.53)	1.485 (29.28)	1.476 (27.81)	1.946 (28.58)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.531	1.552	1.559	1.559	1.564	1.571	1.581	1.601	1.590
Imputation using linear regression									
10%	1.625 (6.18)	1.626 (4.76)	1.621 (4.00)	1.653 (6.07)	1.648 (5.33)	1.666 (6.06)	1.666 (5.36)	1.676 (4.69)	1.683 (5.87)
20%	1.687 (10.19)	1.719 (10.80)	1.704 (9.28)	1.704 (9.34)	1.731 (10.67)	1.724 (9.75)	1.741 (10.11)	1.748 (9.19)	1.748 (9.94)
30%	1.848 (20.75)	1.852 (19.33)	1.862 (19.41)	1.871 (20.06)	1.879 (20.13)	1.884 (19.96)	1.892 (19.62)	1.890 (18.08)	1.915 (20.45)
40%	1.956 (27.78)	1.991 (28.31)	1.961 (25.75)	2.026 (29.97)	1.981 (26.64)	1.993 (26.89)	2.038 (28.88)	2.012 (25.67)	2.051 (29.00)
Imputation using predictive mean matching method									
10%	1.617 (5.67)	1.618 (4.25)	1.618 (3.76)	1.645 (5.52)	1.646 (5.20)	1.659 (5.64)	1.655 (4.69)	1.669 (4.25)	1.671 (5.11)
20%	1.674 (9.36)	1.699 (9.46)	1.694 (8.63)	1.671 (7.19)	1.713 (9.49)	1.708 (8.76)	1.712 (8.24)	1.721 (7.50)	1.728 (8.66)
30%	1.786 (16.67)	1.808 (16.52)	1.788 (14.68)	1.789 (14.78)	1.814 (15.95)	1.816 (15.61)	1.842 (16.46)	1.843 (15.15)	1.842 (15.86)
40%	1.902 (24.27)	1.941 (25.08)	1.960 (25.70)	1.946 (24.86)	1.927 (23.23)	1.944 (23.77)	1.986 (25.62)	1.971 (23.12)	1.975 (24.20)
Imputation using propensity−score method									
10%	1.617 (5.66)	1.623 (4.57)	1.618 (3.77)	1.650 (5.87)	1.649 (5.40)	1.656 (5.46)	1.660 (4.99)	1.670 (4.30)	1.675 (5.36)
20%	1.681 (9.83)	1.696 (9.32)	1.713 (9.89)	1.697 (8.89)	1.720 (9.96)	1.730 (10.15)	1.732 (9.51)	1.743 (8.88)	1.743 (9.61)
30%	1.813 (18.44)	1.814 (16.91)	1.823 (16.90)	1.836 (17.82)	1.837 (17.47)	1.863 (18.62)	1.876 (18.63)	1.865 (16.51)	1.866 (17.32)
40%	1.976 (29.09)	1.984 (27.86)	1.961 (25.78)	2.005 (28.65)	1.975 (26.30)	2.001 (27.42)	2.046 (29.39)	2.018 (26.07)	2.056 (29.29)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.530	1.539	1.537	1.549	1.560	1.564	1.580	1.585	1.593
Imputation using linear regression									

SampleSize_n250_10Treatments/table3_nonshrink_10_treatments_MAR_n250

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.602 (4.70)	1.615 (4.93)	1.619 (5.35)	1.629 (5.19)	1.633 (4.74)	1.652 (5.64)	1.664 (5.28)	1.665 (5.06)	1.669 (4.74)
20%	1.707 (11.57)	1.700 (10.45)	1.720 (11.87)	1.722 (11.19)	1.724 (10.53)	1.762 (12.66)	1.739 (10.05)	1.760 (11.05)	1.763 (10.67)
30%	1.802 (17.72)	1.832 (19.00)	1.830 (19.06)	1.843 (19.00)	1.836 (17.72)	1.864 (19.18)	1.864 (17.96)	1.880 (18.63)	1.873 (17.54)
40%	1.969 (28.65)	1.979 (28.58)	2.008 (30.64)	1.972 (27.34)	2.000 (28.23)	2.004 (28.10)	2.010 (27.18)	2.044 (28.98)	2.053 (28.86)
Imputation using predictive mean matching method									
10%	1.591 (3.98)	1.602 (4.06)	1.610 (4.73)	1.615 (4.26)	1.618 (3.77)	1.635 (4.53)	1.649 (4.35)	1.654 (4.32)	1.651 (3.62)
20%	1.683 (9.95)	1.690 (9.83)	1.703 (10.76)	1.689 (9.08)	1.715 (9.99)	1.743 (11.43)	1.729 (9.39)	1.729 (9.10)	1.739 (9.15)
30%	1.771 (15.70)	1.782 (15.76)	1.792 (16.58)	1.795 (15.89)	1.802 (15.56)	1.816 (16.15)	1.841 (16.51)	1.836 (15.86)	1.820 (14.23)
40%	1.874 (22.48)	1.891 (22.88)	1.953 (27.02)	1.915 (23.69)	1.928 (23.61)	1.941 (24.08)	1.931 (22.16)	1.977 (24.73)	1.966 (23.39)
Imputation using propensity−score method									
10%	1.596 (4.30)	1.606 (4.37)	1.615 (5.07)	1.621 (4.70)	1.621 (3.93)	1.638 (4.72)	1.651 (4.49)	1.655 (4.40)	1.661 (4.22)
20%	1.691 (10.48)	1.697 (10.24)	1.700 (10.57)	1.696 (9.50)	1.720 (10.28)	1.738 (11.13)	1.719 (8.76)	1.738 (9.62)	1.748 (9.71)
30%	1.804 (17.86)	1.823 (18.45)	1.817 (18.22)	1.824 (17.80)	1.800 (15.40)	1.838 (17.50)	1.847 (16.85)	1.870 (17.95)	1.842 (15.62)
40%	1.938 (26.65)	1.933 (25.58)	1.978 (28.66)	1.970 (27.23)	2.001 (28.29)	2.012 (28.62)	1.987 (25.71)	2.051 (29.37)	2.038 (27.91)

SampleSize_n250_10Treatments/table3_shrink_10_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.124	1.111	1.126	1.131	1.131	1.135	1.140	1.147	1.162	1.154	1.543
Imputation using linear regression											
10%	1.177 (4.68)	1.179 (6.14)	1.179 (4.72)	1.176 (3.96)	1.199 (6.03)	1.195 (5.29)	1.208 (6.03)	1.209 (5.33)	1.216 (4.65)	1.221 (5.83)	1.625 (5.29)
20%	1.216 (8.19)	1.220 (9.85)	1.244 (10.50)	1.233 (8.95)	1.233 (9.02)	1.252 (10.34)	1.247 (9.42)	1.260 (9.79)	1.265 (8.87)	1.265 (9.61)	1.689 (9.45)
30%	1.328 (18.11)	1.341 (20.71)	1.343 (19.28)	1.351 (19.37)	1.357 (20.02)	1.363 (20.09)	1.366 (19.89)	1.372 (19.54)	1.371 (18.00)	1.389 (20.40)	1.830 (18.61)
40%	1.417 (26.08)	1.414 (27.26)	1.439 (27.79)	1.417 (25.22)	1.465 (29.53)	1.432 (26.17)	1.440 (26.39)	1.473 (28.37)	1.454 (25.16)	1.483 (28.53)	1.967 (27.51)
Imputation using predictive mean matching method											
10%	1.172 (4.24)	1.173 (5.65)	1.174 (4.23)	1.174 (3.74)	1.193 (5.49)	1.194 (5.18)	1.204 (5.62)	1.201 (4.66)	1.211 (4.23)	1.213 (5.09)	1.616 (4.77)
20%	1.199 (6.65)	1.211 (9.07)	1.230 (9.19)	1.226 (8.34)	1.209 (6.90)	1.240 (9.21)	1.236 (8.48)	1.239 (7.94)	1.245 (7.21)	1.250 (8.36)	1.663 (7.81)
30%	1.289 (14.67)	1.293 (16.37)	1.309 (16.24)	1.294 (14.39)	1.295 (14.46)	1.313 (15.66)	1.314 (15.29)	1.333 (16.16)	1.334 (14.84)	1.333 (15.56)	1.780 (15.39)
40%	1.373 (22.10)	1.378 (24.06)	1.406 (24.90)	1.421 (25.57)	1.410 (24.69)	1.397 (23.07)	1.409 (23.61)	1.439 (25.42)	1.429 (22.97)	1.431 (24.00)	1.901 (23.18)
Imputation using propensity−score method											
10%	1.172 (4.21)	1.174 (5.65)	1.178 (4.58)	1.174 (3.77)	1.197 (5.87)	1.196 (5.40)	1.202 (5.47)	1.205 (4.99)	1.212 (4.30)	1.216 (5.37)	1.616 (4.73)
20%	1.212 (7.78)	1.217 (9.52)	1.228 (9.03)	1.240 (9.61)	1.228 (8.60)	1.245 (9.67)	1.252 (9.87)	1.253 (9.22)	1.262 (8.59)	1.261 (9.30)	1.680 (8.91)
30%	1.297 (15.41)	1.313 (18.22)	1.314 (16.67)	1.320 (16.69)	1.330 (17.61)	1.331 (17.26)	1.349 (18.40)	1.359 (18.41)	1.351 (16.28)	1.351 (17.08)	1.791 (16.07)
40%	1.397 (24.30)	1.431 (28.80)	1.436 (27.55)	1.420 (25.47)	1.452 (28.39)	1.430 (26.01)	1.449 (27.16)	1.482 (29.14)	1.461 (25.77)	1.489 (29.02)	1.936 (25.49)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.097	1.109	1.116	1.114	1.123	1.131	1.134	1.146	1.149	1.155	1.530
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.165 (6.18)	1.161 (4.66)	1.170 (4.89)	1.174 (5.31)	1.180 (5.16)	1.184 (4.70)	1.197 (5.60)	1.206 (5.25)	1.207 (5.02)	1.209 (4.71)	1.607 (5.00)
20%	1.224 (11.61)	1.237 (11.47)	1.231 (10.34)	1.246 (11.77)	1.247 (11.10)	1.248 (10.42)	1.276 (12.57)	1.260 (9.95)	1.275 (10.95)	1.277 (10.58)	1.696 (10.84)
30%	1.321 (20.41)	1.305 (17.61)	1.326 (18.89)	1.326 (18.96)	1.335 (18.91)	1.330 (17.62)	1.350 (19.09)	1.350 (17.87)	1.362 (18.55)	1.356 (17.44)	1.821 (19.02)
40%	1.431 (30.41)	1.425 (28.46)	1.432 (28.36)	1.454 (30.46)	1.427 (27.09)	1.447 (28.00)	1.449 (27.84)	1.454 (26.94)	1.480 (28.77)	1.486 (28.68)	1.978 (29.27)
Imputation using predictive mean matching method											
10%	1.158 (5.58)	1.153 (3.91)	1.160 (3.99)	1.166 (4.66)	1.170 (4.20)	1.172 (3.71)	1.184 (4.47)	1.195 (4.28)	1.198 (4.26)	1.196 (3.55)	1.598 (4.43)
20%	1.213 (10.58)	1.219 (9.89)	1.225 (9.78)	1.234 (10.71)	1.224 (9.02)	1.243 (9.93)	1.263 (11.39)	1.253 (9.34)	1.253 (9.03)	1.260 (9.08)	1.679 (9.74)
30%	1.285 (17.11)	1.282 (15.55)	1.290 (15.60)	1.298 (16.44)	1.299 (15.73)	1.305 (15.41)	1.315 (15.99)	1.334 (16.41)	1.330 (15.71)	1.318 (14.09)	1.773 (15.88)
40%	1.385 (26.24)	1.356 (22.26)	1.368 (22.63)	1.413 (26.84)	1.386 (23.47)	1.395 (23.36)	1.404 (23.84)	1.396 (21.89)	1.431 (24.51)	1.423 (23.18)	1.915 (25.11)
Imputation using propensity−score method											
10%	1.159 (5.65)	1.157 (4.28)	1.164 (4.34)	1.171 (5.05)	1.175 (4.68)	1.175 (3.90)	1.187 (4.69)	1.197 (4.46)	1.199 (4.37)	1.204 (4.20)	1.599 (4.46)
20%	1.208 (10.12)	1.225 (10.41)	1.229 (10.18)	1.231 (10.50)	1.228 (9.42)	1.246 (10.22)	1.259 (11.07)	1.245 (8.69)	1.259 (9.55)	1.266 (9.64)	1.673 (9.32)
30%	1.301 (18.59)	1.305 (17.62)	1.319 (18.19)	1.315 (17.97)	1.319 (17.54)	1.301 (15.10)	1.329 (17.24)	1.336 (16.61)	1.353 (17.71)	1.332 (15.37)	1.797 (17.44)
40%	1.409 (28.43)	1.403 (26.50)	1.399 (25.37)	1.432 (28.54)	1.427 (27.09)	1.449 (28.14)	1.457 (28.48)	1.438 (25.53)	1.485 (29.28)	1.476 (27.81)	1.949 (27.33)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.540	1.548	1.547	1.548	1.550	1.547	1.553	1.558	1.557
Imputation using linear regression									
10%	1.621 (5.27)	1.622 (4.82)	1.615 (4.42)	1.637 (5.78)	1.632 (5.28)	1.643 (6.20)	1.642 (5.75)	1.646 (5.65)	1.654 (6.21)
20%	1.687 (9.56)	1.716 (10.88)	1.695 (9.53)	1.702 (9.96)	1.715 (10.67)	1.709 (10.51)	1.724 (10.99)	1.725 (10.69)	1.722 (10.62)
30%	1.846 (19.85)	1.844 (19.11)	1.855 (19.92)	1.863 (20.33)	1.868 (20.55)	1.866 (20.65)	1.869 (20.38)	1.867 (19.82)	1.895 (21.71)
40%	1.958 (27.14)	1.987 (28.39)	1.958 (26.56)	2.022 (30.63)	1.981 (27.80)	1.988 (28.54)	2.024 (30.31)	2.002 (28.46)	2.037 (30.84)
Imputation using predictive mean matching method									
10%	1.613 (4.73)	1.613 (4.25)	1.611 (4.15)	1.628 (5.16)	1.628 (5.04)	1.635 (5.70)	1.631 (5.01)	1.638 (5.12)	1.640 (5.37)
20%	1.674 (8.69)	1.695 (9.51)	1.684 (8.84)	1.670 (7.87)	1.697 (9.53)	1.695 (9.59)	1.692 (8.99)	1.698 (8.95)	1.701 (9.26)
30%	1.783 (15.75)	1.801 (16.39)	1.782 (15.19)	1.780 (15.01)	1.802 (16.24)	1.798 (16.26)	1.819 (17.16)	1.820 (16.79)	1.823 (17.09)
40%	1.906 (23.75)	1.936 (25.08)	1.956 (26.43)	1.941 (25.41)	1.926 (24.26)	1.939 (25.33)	1.969 (26.81)	1.960 (25.77)	1.960 (25.87)
Imputation using propensity−score method									
10%	1.613 (4.70)	1.619 (4.63)	1.611 (4.13)	1.634 (5.55)	1.632 (5.27)	1.634 (5.60)	1.637 (5.40)	1.639 (5.18)	1.644 (5.63)
20%	1.680 (9.10)	1.693 (9.36)	1.703 (10.09)	1.696 (9.55)	1.704 (9.96)	1.715 (10.87)	1.714 (10.36)	1.720 (10.39)	1.716 (10.24)
30%	1.810 (17.55)	1.808 (16.80)	1.819 (17.56)	1.829 (18.14)	1.828 (17.95)	1.847 (19.38)	1.855 (19.44)	1.843 (18.27)	1.846 (18.59)
40%	1.977 (28.37)	1.979 (27.86)	1.959 (26.59)	2.000 (29.21)	1.974 (27.36)	1.997 (29.11)	2.032 (30.82)	2.007 (28.77)	2.041 (31.08)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.536	1.539	1.539	1.540	1.545	1.547	1.551	1.553	1.555
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.604 (4.43)	1.609 (4.57)	1.613 (4.81)	1.618 (5.06)	1.618 (4.74)	1.631 (5.41)	1.634 (5.37)	1.635 (5.23)	1.638 (5.33)
20%	1.706 (11.08)	1.698 (10.30)	1.712 (11.25)	1.715 (11.35)	1.710 (10.67)	1.744 (12.72)	1.723 (11.09)	1.736 (11.77)	1.740 (11.90)
30%	1.802 (17.31)	1.822 (18.40)	1.823 (18.47)	1.832 (18.96)	1.826 (18.19)	1.848 (19.43)	1.847 (19.13)	1.858 (19.62)	1.856 (19.36)
40%	1.971 (28.32)	1.976 (28.38)	2.004 (30.21)	1.967 (27.70)	1.991 (28.90)	1.989 (28.56)	1.998 (28.83)	2.028 (30.55)	2.038 (31.09)
Imputation using predictive mean matching method									
10%	1.593 (3.75)	1.597 (3.78)	1.604 (4.20)	1.604 (4.17)	1.603 (3.73)	1.614 (4.32)	1.619 (4.38)	1.622 (4.44)	1.619 (4.14)
20%	1.683 (9.59)	1.689 (9.76)	1.696 (10.24)	1.684 (9.35)	1.702 (10.14)	1.726 (11.53)	1.712 (10.39)	1.706 (9.83)	1.714 (10.24)
30%	1.771 (15.32)	1.775 (15.33)	1.786 (16.09)	1.783 (15.77)	1.793 (16.05)	1.800 (16.35)	1.824 (17.61)	1.814 (16.76)	1.805 (16.06)
40%	1.879 (22.32)	1.889 (22.77)	1.948 (26.59)	1.912 (24.14)	1.920 (24.25)	1.928 (24.60)	1.920 (23.79)	1.959 (26.12)	1.952 (25.57)
Imputation using propensity−score method									
10%	1.598 (4.07)	1.602 (4.10)	1.609 (4.56)	1.610 (4.56)	1.606 (3.95)	1.618 (4.58)	1.622 (4.62)	1.624 (4.57)	1.630 (4.80)
20%	1.689 (10.00)	1.694 (10.10)	1.692 (9.97)	1.689 (9.70)	1.706 (10.45)	1.720 (11.18)	1.702 (9.76)	1.714 (10.36)	1.724 (10.86)
30%	1.803 (17.40)	1.815 (17.95)	1.811 (17.69)	1.814 (17.78)	1.790 (15.87)	1.822 (17.75)	1.830 (17.99)	1.848 (18.99)	1.827 (17.48)
40%	1.941 (26.39)	1.931 (25.48)	1.974 (28.31)	1.966 (27.66)	1.993 (28.99)	2.000 (29.25)	1.975 (27.38)	2.035 (31.01)	2.025 (30.26)

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	89.29	89.29	92.86	96.43	89.29	96.43	89.29	100.00	96.43	64.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	96.55	89.66	100.00	100.00	96.55	93.10	93.10	72.41
20%	96.55	100.00	93.10	93.10	96.55	96.55	96.55	93.10	93.10	96.55	82.76
30%	92.86	100.00	92.86	96.43	89.29	92.86	96.43	89.29	92.86	96.43	64.29
40%	89.66	96.55	100.00	93.10	96.55	96.55	96.55	96.55	89.66	100.00	72.41
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	100.00	86.21	100.00	100.00	96.55	89.66	89.66	65.52
20%	100.00	96.55	96.55	96.55	96.55	93.10	96.55	96.55	93.10	100.00	86.21
30%	96.43	100.00	92.86	96.43	89.29	96.43	96.43	92.86	89.29	96.43	71.43
40%	89.66	93.10	96.55	93.10	96.55	89.66	93.10	96.55	96.55	96.55	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	96.55	100.00	89.66	100.00	100.00	96.55	93.10	93.10	68.97
20%	96.55	100.00	93.10	89.66	96.55	96.55	100.00	96.55	93.10	100.00	79.31
30%	92.86	100.00	92.86	92.86	96.43	100.00	100.00	85.71	92.86	96.43	67.86
40%	89.66	96.55	93.10	93.10	96.55	93.10	96.55	96.55	89.66	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	93.10	93.10	96.55	100.00	93.10	93.10	100.00	75.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	86.21
20%	92.86	100.00	78.57	96.43	85.71	89.29	85.71	89.29	96.43	92.86	64.29
30%	100.00	96.43	82.14	96.43	85.71	92.86	96.43	96.43	96.43	92.86	75.00
40%	89.66	93.10	96.55	100.00	93.10	100.00	93.10	93.10	100.00	100.00	62.07
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	96.55	82.76
20%	92.86	100.00	78.57	96.43	89.29	92.86	85.71	89.29	96.43	89.29	64.29
30%	96.43	96.43	85.71	96.43	85.71	89.29	92.86	92.86	96.43	92.86	71.43
40%	96.55	82.76	96.55	100.00	93.10	96.55	93.10	86.21	100.00	100.00	55.17
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	89.66
20%	92.86	100.00	85.71	96.43	89.29	89.29	82.14	89.29	89.29	85.71	67.86
30%	96.43	96.43	85.71	96.43	89.29	92.86	100.00	100.00	96.43	92.86	75.00
40%	93.10	89.66	96.55	100.00	93.10	100.00	93.10	96.55	100.00	100.00	68.97

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	100	100	100.00	100.00	96.43	100	96.43	100.00	100	89.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	96.55	100	100.00	100.00	100	96.55
30%	96.43	100	100	100.00	100.00	100.00	100	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	96.55	100	96.55
20%	100.00	100	100	100.00	100.00	96.55	100	96.55	100.00	100	93.10
30%	96.43	100	100	100.00	100.00	100.00	100	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
30%	96.43	100	100	100.00	100.00	100.00	100	96.43	96.43	100	89.29
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100	100	96.55	96.55	100.00	100	100.00	100.00	100	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_nonshrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100.00	100	100.00	96.43
30%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	100	96.43	96.43
40%	100.00	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	96.43	100.00	96.43	100.00	96.43	96.43	96.43	96.43	100	100.00	92.86
30%	100.00	96.43	92.86	100.00	96.43	96.43	100.00	100.00	100	96.43	89.29
40%	100.00	100.00	96.55	100.00	93.10	100.00	100.00	100.00	100	100.00	89.66
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	96.43	96.43	100.00	96.43	100	100.00	92.86
30%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	100	96.43	96.43
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	89.29	89.29	92.86	96.43	89.29	96.43	89.29	100.00	96.43	64.29
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	96.55	89.66	100.00	100.00	96.55	93.10	93.10	72.41
20%	96.55	100.00	93.10	93.10	96.55	96.55	96.55	93.10	93.10	96.55	82.76
30%	92.86	100.00	92.86	96.43	89.29	92.86	96.43	89.29	92.86	96.43	64.29
40%	89.66	96.55	100.00	93.10	96.55	96.55	96.55	96.55	89.66	100.00	72.41
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	100.00	86.21	100.00	100.00	96.55	89.66	89.66	65.52
20%	100.00	96.55	96.55	96.55	96.55	93.10	96.55	96.55	93.10	100.00	86.21
30%	96.43	100.00	92.86	96.43	89.29	96.43	96.43	92.86	89.29	96.43	71.43
40%	89.66	93.10	96.55	93.10	96.55	89.66	93.10	96.55	96.55	96.55	65.52
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	96.55	100.00	89.66	100.00	100.00	96.55	93.10	93.10	68.97
20%	96.55	100.00	93.10	89.66	96.55	96.55	100.00	96.55	93.10	100.00	79.31
30%	92.86	100.00	92.86	92.86	96.43	100.00	100.00	85.71	92.86	96.43	67.86
40%	89.66	96.55	93.10	93.10	96.55	93.10	96.55	96.55	89.66	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	93.10	93.10	96.55	100.00	93.10	93.10	100.00	75.86
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	86.21
20%	92.86	100.00	78.57	96.43	85.71	89.29	85.71	89.29	96.43	92.86	64.29
30%	100.00	96.43	82.14	96.43	85.71	92.86	96.43	96.43	96.43	92.86	75.00
40%	89.66	93.10	96.55	100.00	93.10	100.00	93.10	93.10	100.00	100.00	62.07
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	96.55	82.76
20%	92.86	100.00	78.57	96.43	89.29	92.86	85.71	89.29	96.43	89.29	64.29
30%	96.43	96.43	85.71	96.43	85.71	89.29	92.86	92.86	96.43	92.86	71.43
40%	96.55	82.76	96.55	100.00	93.10	96.55	93.10	86.21	100.00	100.00	55.17
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	96.55	96.55	100.00	96.55	100.00	100.00	100.00	89.66
20%	92.86	100.00	85.71	96.43	89.29	89.29	82.14	89.29	89.29	85.71	67.86
30%	96.43	96.43	85.71	96.43	89.29	92.86	100.00	100.00	96.43	92.86	75.00
40%	93.10	89.66	96.55	100.00	93.10	100.00	93.10	96.55	100.00	100.00	68.97

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	100	100	100.00	100.00	96.43	96.43	96.43	100.00	100	85.71
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	96.55	100.00	96.55	100.00	100	93.10
30%	96.43	100	100	100.00	100.00	100.00	100.00	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	96.55	100	96.55
20%	100.00	100	100	100.00	100.00	96.55	100.00	96.55	100.00	100	93.10
30%	96.43	100	100	100.00	100.00	100.00	100.00	96.43	100.00	100	92.86
40%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
20%	100.00	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00
30%	96.43	100	100	100.00	100.00	100.00	100.00	96.43	96.43	100	89.29
40%	96.55	100	100	100.00	100.00	100.00	100.00	100.00	100.00	100	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100	100	96.55	96.55	100.00	100.00	100.00	100.00	100	93.10
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_10Treatments/table2_shrink_10_treatments_MAR_n300

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	100.00	96.43	100.00	100.00	100	100.00	96.43
30%	100.00	96.43	96.43	96.43	96.43	96.43	100.00	100.00	100	96.43	92.86
40%	100.00	100.00	100.00	100.00	96.55	100.00	100.00	100.00	100	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	96.43	100.00	96.43	100.00	96.43	96.43	96.43	96.43	100	100.00	92.86
30%	100.00	96.43	92.86	100.00	96.43	96.43	100.00	100.00	100	96.43	89.29
40%	100.00	100.00	96.55	100.00	93.10	100.00	100.00	100.00	100	100.00	89.66
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00
20%	100.00	100.00	96.43	100.00	96.43	96.43	100.00	96.43	100	100.00	92.86
30%	100.00	96.43	96.43	96.43	100.00	96.43	100.00	100.00	100	96.43	96.43
40%	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100	100.00	100.00

SampleSize_n300_10Treatments/table3_nonshrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.016	1.022	1.020	1.028	1.033	1.043	1.040	1.043	1.061	1.060	1.400
Imputation using linear regression											
10%	1.061 (4.43)	1.061 (3.80)	1.077 (5.62)	1.077 (4.72)	1.081 (4.60)	1.093 (4.89)	1.098 (5.52)	1.100 (5.45)	1.106 (4.28)	1.109 (4.67)	1.463 (4.52)
20%	1.124 (10.65)	1.124 (9.91)	1.130 (10.84)	1.138 (10.64)	1.145 (10.78)	1.149 (10.23)	1.149 (10.42)	1.161 (11.36)	1.164 (9.70)	1.169 (10.36)	1.551 (10.80)
30%	1.190 (17.10)	1.220 (19.30)	1.214 (19.03)	1.238 (20.36)	1.232 (19.20)	1.241 (19.09)	1.242 (19.42)	1.241 (18.98)	1.247 (17.56)	1.266 (19.44)	1.642 (17.32)
40%	1.300 (27.94)	1.304 (27.53)	1.308 (28.27)	1.329 (29.23)	1.321 (27.86)	1.324 (27.03)	1.350 (29.76)	1.360 (30.38)	1.341 (26.41)	1.365 (28.78)	1.795 (28.23)
Imputation using predictive mean matching method											
10%	1.054 (3.76)	1.062 (3.90)	1.072 (5.06)	1.071 (4.11)	1.075 (4.02)	1.084 (3.99)	1.091 (4.88)	1.092 (4.67)	1.099 (3.59)	1.100 (3.83)	1.454 (3.86)
20%	1.120 (10.21)	1.112 (8.77)	1.123 (10.11)	1.124 (9.28)	1.130 (9.38)	1.133 (8.70)	1.143 (9.91)	1.148 (10.05)	1.148 (8.19)	1.149 (8.42)	1.545 (10.39)
30%	1.177 (15.89)	1.183 (15.73)	1.189 (16.59)	1.188 (15.49)	1.184 (14.57)	1.215 (16.57)	1.212 (16.50)	1.218 (16.81)	1.219 (14.90)	1.227 (15.75)	1.624 (16.00)
40%	1.261 (24.17)	1.252 (22.48)	1.256 (23.11)	1.300 (26.47)	1.294 (25.26)	1.296 (24.32)	1.295 (24.50)	1.320 (26.54)	1.295 (22.08)	1.323 (24.81)	1.744 (24.57)
Imputation using propensity−score method											
10%	1.060 (4.35)	1.066 (4.30)	1.079 (5.76)	1.076 (4.67)	1.079 (4.42)	1.093 (4.87)	1.098 (5.60)	1.094 (4.88)	1.101 (3.77)	1.106 (4.36)	1.462 (4.41)
20%	1.125 (10.69)	1.129 (10.45)	1.133 (11.10)	1.140 (10.90)	1.143 (10.64)	1.154 (10.69)	1.151 (10.64)	1.161 (11.34)	1.166 (9.89)	1.172 (10.63)	1.551 (10.79)
30%	1.176 (15.74)	1.184 (15.83)	1.190 (16.68)	1.201 (16.76)	1.207 (16.78)	1.232 (18.15)	1.227 (17.93)	1.222 (17.23)	1.227 (15.68)	1.246 (17.56)	1.624 (16.00)
40%	1.301 (28.06)	1.301 (27.24)	1.300 (27.43)	1.336 (29.89)	1.318 (27.59)	1.319 (26.51)	1.333 (28.14)	1.351 (29.52)	1.323 (24.72)	1.363 (28.62)	1.796 (28.28)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.014	1.021	1.025	1.022	1.036	1.035	1.045	1.045	1.050	1.057	1.398
Imputation using linear regression											

SampleSize_n300_10Treatments/table3_nonshrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.055 (4.05)	1.062 (4.04)	1.062 (3.56)	1.074 (5.14)	1.080 (4.30)	1.086 (4.97)	1.084 (3.70)	1.095 (4.75)	1.101 (4.90)	1.095 (3.60)	1.458 (4.27)
20%	1.112 (9.64)	1.117 (9.38)	1.126 (9.78)	1.132 (10.78)	1.130 (9.13)	1.141 (10.30)	1.150 (10.07)	1.164 (11.38)	1.149 (9.42)	1.160 (9.84)	1.535 (9.79)
30%	1.198 (18.13)	1.194 (17.01)	1.207 (17.73)	1.205 (17.92)	1.198 (15.72)	1.222 (18.04)	1.224 (17.18)	1.238 (18.52)	1.251 (19.20)	1.253 (18.56)	1.655 (18.37)
40%	1.294 (27.59)	1.292 (26.60)	1.292 (26.01)	1.289 (26.13)	1.320 (27.47)	1.328 (28.29)	1.329 (27.16)	1.356 (29.76)	1.361 (29.63)	1.352 (27.93)	1.790 (28.02)
Imputation using predictive mean matching method											
10%	1.053 (3.83)	1.058 (3.62)	1.060 (3.37)	1.069 (4.63)	1.070 (3.34)	1.080 (4.34)	1.077 (3.09)	1.090 (4.33)	1.095 (4.35)	1.091 (3.27)	1.455 (4.04)
20%	1.115 (9.89)	1.106 (8.32)	1.119 (9.15)	1.122 (9.84)	1.136 (9.71)	1.140 (10.16)	1.139 (9.04)	1.154 (10.44)	1.141 (8.72)	1.150 (8.85)	1.537 (9.95)
30%	1.194 (17.72)	1.169 (14.52)	1.198 (16.89)	1.204 (17.83)	1.199 (15.74)	1.220 (17.90)	1.212 (16.00)	1.224 (17.18)	1.239 (18.05)	1.236 (16.95)	1.646 (17.76)
40%	1.256 (23.81)	1.247 (22.13)	1.261 (22.98)	1.244 (21.78)	1.253 (21.01)	1.274 (23.11)	1.284 (22.92)	1.293 (23.77)	1.303 (24.14)	1.305 (23.49)	1.738 (24.28)
Imputation using propensity−score method											
10%	1.054 (3.87)	1.059 (3.71)	1.059 (3.27)	1.069 (4.62)	1.072 (3.54)	1.081 (4.41)	1.082 (3.50)	1.094 (4.67)	1.098 (4.62)	1.094 (3.53)	1.455 (4.08)
20%	1.111 (9.51)	1.110 (8.72)	1.129 (10.09)	1.125 (10.08)	1.130 (9.15)	1.137 (9.88)	1.145 (9.59)	1.150 (10.03)	1.154 (9.96)	1.161 (9.92)	1.532 (9.58)
30%	1.182 (16.51)	1.177 (15.28)	1.202 (17.26)	1.207 (18.14)	1.210 (16.84)	1.207 (16.63)	1.221 (16.88)	1.226 (17.29)	1.229 (17.08)	1.232 (16.61)	1.633 (16.78)
40%	1.276 (25.82)	1.276 (24.99)	1.281 (24.95)	1.292 (26.50)	1.309 (26.44)	1.327 (28.27)	1.317 (26.02)	1.334 (27.65)	1.361 (29.67)	1.337 (26.54)	1.764 (26.15)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.409	1.406	1.417	1.424	1.437	1.433	1.437	1.462	1.460
Imputation using linear regression									
10%	1.464 (3.90)	1.486 (5.72)	1.485 (4.81)	1.491 (4.69)	1.508 (4.98)	1.514 (5.61)	1.517 (5.54)	1.526 (4.36)	1.530 (4.76)
20%	1.551 (10.05)	1.560 (10.99)	1.570 (10.79)	1.579 (10.92)	1.586 (10.37)	1.585 (10.57)	1.602 (11.50)	1.606 (9.84)	1.614 (10.49)
30%	1.683 (19.46)	1.676 (19.22)	1.708 (20.51)	1.700 (19.38)	1.714 (19.28)	1.715 (19.62)	1.712 (19.15)	1.722 (17.74)	1.747 (19.64)
40%	1.801 (27.83)	1.807 (28.59)	1.835 (29.51)	1.825 (28.16)	1.829 (27.32)	1.864 (30.02)	1.877 (30.64)	1.853 (26.71)	1.885 (29.07)
Imputation using predictive mean matching method									
10%	1.465 (4.01)	1.478 (5.17)	1.477 (4.22)	1.483 (4.13)	1.496 (4.10)	1.505 (4.99)	1.506 (4.78)	1.516 (3.69)	1.518 (3.94)
20%	1.535 (8.95)	1.550 (10.30)	1.551 (9.47)	1.560 (9.56)	1.564 (8.89)	1.578 (10.08)	1.584 (10.22)	1.585 (8.37)	1.586 (8.61)
30%	1.632 (15.84)	1.641 (16.71)	1.639 (15.63)	1.634 (14.73)	1.677 (16.69)	1.672 (16.63)	1.680 (16.90)	1.682 (15.02)	1.692 (15.89)
40%	1.732 (22.92)	1.737 (23.57)	1.797 (26.84)	1.789 (25.65)	1.792 (24.70)	1.791 (24.92)	1.824 (26.92)	1.791 (22.50)	1.829 (25.22)
Imputation using propensity−score method									
10%	1.470 (4.36)	1.487 (5.81)	1.484 (4.73)	1.488 (4.48)	1.507 (4.92)	1.514 (5.65)	1.508 (4.93)	1.518 (3.82)	1.525 (4.42)
20%	1.557 (10.54)	1.563 (11.19)	1.573 (10.99)	1.577 (10.73)	1.592 (10.77)	1.587 (10.73)	1.601 (11.42)	1.608 (9.98)	1.617 (10.72)
30%	1.636 (16.10)	1.644 (16.95)	1.658 (17.03)	1.667 (17.05)	1.701 (18.39)	1.694 (18.19)	1.688 (17.47)	1.695 (15.93)	1.721 (17.82)
40%	1.796 (27.49)	1.795 (27.70)	1.844 (30.09)	1.820 (27.83)	1.821 (26.75)	1.840 (28.39)	1.865 (29.76)	1.828 (25.00)	1.882 (28.85)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.407	1.413	1.408	1.427	1.426	1.440	1.440	1.447	1.456
Imputation using linear regression									

SampleSize_n300_10Treatments/table3_nonshrink_10_treatments_MAR_n300

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.467 (4.26)	1.467 (3.78)	1.484 (5.36)	1.492 (4.51)	1.500 (5.18)	1.497 (3.91)	1.512 (4.96)	1.521 (5.11)	1.512 (3.82)
20%	1.541 (9.53)	1.554 (9.94)	1.562 (10.93)	1.560 (9.28)	1.575 (10.44)	1.587 (10.22)	1.606 (11.53)	1.585 (9.58)	1.602 (9.98)
30%	1.650 (17.25)	1.667 (17.98)	1.664 (18.18)	1.656 (15.99)	1.687 (18.30)	1.691 (17.41)	1.711 (18.78)	1.728 (19.44)	1.730 (18.80)
40%	1.788 (27.07)	1.787 (26.45)	1.783 (26.59)	1.826 (27.90)	1.836 (28.72)	1.838 (27.59)	1.874 (30.14)	1.882 (30.06)	1.869 (28.34)
Imputation using predictive mean matching method									
10%	1.461 (3.83)	1.464 (3.58)	1.476 (4.85)	1.478 (3.55)	1.491 (4.55)	1.488 (3.29)	1.506 (4.53)	1.513 (4.56)	1.507 (3.48)
20%	1.525 (8.41)	1.544 (9.24)	1.548 (9.92)	1.567 (9.77)	1.572 (10.22)	1.572 (9.11)	1.592 (10.51)	1.574 (8.80)	1.586 (8.92)
30%	1.613 (14.61)	1.653 (16.96)	1.660 (17.88)	1.653 (15.82)	1.683 (17.96)	1.672 (16.07)	1.689 (17.27)	1.709 (18.12)	1.704 (17.03)
40%	1.726 (22.65)	1.745 (23.45)	1.722 (22.31)	1.735 (21.55)	1.763 (23.61)	1.777 (23.39)	1.790 (24.26)	1.803 (24.64)	1.805 (23.97)
Imputation using propensity−score method									
10%	1.462 (3.91)	1.462 (3.48)	1.476 (4.83)	1.481 (3.75)	1.492 (4.62)	1.494 (3.70)	1.511 (4.88)	1.517 (4.83)	1.511 (3.73)
20%	1.531 (8.80)	1.557 (10.16)	1.551 (10.16)	1.559 (9.22)	1.568 (9.96)	1.579 (9.66)	1.586 (10.11)	1.592 (10.03)	1.602 (9.99)
30%	1.626 (15.55)	1.661 (17.51)	1.667 (18.39)	1.671 (17.08)	1.668 (16.91)	1.687 (17.12)	1.693 (17.56)	1.698 (17.36)	1.702 (16.88)
40%	1.763 (25.33)	1.770 (25.27)	1.786 (26.80)	1.809 (26.75)	1.834 (28.54)	1.819 (26.31)	1.843 (27.94)	1.880 (29.95)	1.847 (26.85)

SampleSize_n300_10Treatments/table3_shrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	1.016	1.022	1.020	1.028	1.033	1.043	1.040	1.043	1.061	1.060	1.404
Imputation using linear regression											
10%	1.061 (4.43)	1.061 (3.80)	1.077 (5.62)	1.077 (4.72)	1.081 (4.60)	1.093 (4.89)	1.098 (5.52)	1.100 (5.45)	1.106 (4.28)	1.109 (4.67)	1.467 (4.44)
20%	1.124 (10.65)	1.124 (9.91)	1.130 (10.84)	1.138 (10.64)	1.145 (10.78)	1.149 (10.23)	1.149 (10.42)	1.161 (11.36)	1.164 (9.70)	1.169 (10.36)	1.550 (10.33)
30%	1.190 (17.10)	1.220 (19.30)	1.214 (19.03)	1.238 (20.36)	1.232 (19.20)	1.241 (19.09)	1.242 (19.42)	1.241 (18.98)	1.247 (17.56)	1.266 (19.44)	1.647 (17.29)
40%	1.300 (27.94)	1.304 (27.53)	1.308 (28.27)	1.329 (29.23)	1.321 (27.86)	1.324 (27.03)	1.350 (29.76)	1.360 (30.38)	1.341 (26.41)	1.365 (28.78)	1.796 (27.89)
Imputation using predictive mean matching method											
10%	1.054 (3.76)	1.062 (3.90)	1.072 (5.06)	1.071 (4.11)	1.075 (4.02)	1.084 (3.99)	1.091 (4.88)	1.092 (4.67)	1.099 (3.59)	1.100 (3.83)	1.459 (3.85)
20%	1.120 (10.21)	1.112 (8.77)	1.123 (10.11)	1.124 (9.28)	1.130 (9.38)	1.133 (8.70)	1.143 (9.91)	1.148 (10.05)	1.148 (8.19)	1.149 (8.42)	1.542 (9.82)
30%	1.177 (15.89)	1.183 (15.73)	1.189 (16.59)	1.188 (15.49)	1.184 (14.57)	1.215 (16.57)	1.212 (16.50)	1.218 (16.81)	1.219 (14.90)	1.227 (15.75)	1.629 (15.98)
40%	1.261 (24.17)	1.252 (22.48)	1.256 (23.11)	1.300 (26.47)	1.294 (25.26)	1.296 (24.32)	1.295 (24.50)	1.320 (26.54)	1.295 (22.08)	1.323 (24.81)	1.745 (24.23)
Imputation using propensity−score method											
10%	1.060 (4.35)	1.066 (4.30)	1.079 (5.76)	1.076 (4.67)	1.079 (4.42)	1.093 (4.87)	1.098 (5.60)	1.094 (4.88)	1.101 (3.77)	1.106 (4.36)	1.465 (4.33)
20%	1.125 (10.69)	1.129 (10.45)	1.133 (11.10)	1.140 (10.90)	1.143 (10.64)	1.154 (10.69)	1.151 (10.64)	1.161 (11.34)	1.166 (9.89)	1.172 (10.63)	1.549 (10.30)
30%	1.176 (15.74)	1.184 (15.83)	1.190 (16.68)	1.201 (16.76)	1.207 (16.78)	1.232 (18.15)	1.227 (17.93)	1.222 (17.23)	1.227 (15.68)	1.246 (17.56)	1.629 (15.99)
40%	1.301 (28.06)	1.301 (27.24)	1.300 (27.43)	1.336 (29.89)	1.318 (27.59)	1.319 (26.51)	1.333 (28.14)	1.351 (29.52)	1.323 (24.72)	1.363 (28.62)	1.797 (27.93)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	1.014	1.021	1.025	1.022	1.036	1.035	1.045	1.045	1.050	1.057	1.399
Imputation using linear regression											

SampleSize_n300_10Treatments/table3_shrink_10_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	1.055 (4.05)	1.062 (4.04)	1.062 (3.56)	1.074 (5.14)	1.080 (4.30)	1.086 (4.97)	1.084 (3.70)	1.095 (4.75)	1.101 (4.90)	1.095 (3.60)	1.464 (4.65)
20%	1.112 (9.64)	1.117 (9.38)	1.126 (9.78)	1.132 (10.78)	1.130 (9.13)	1.141 (10.30)	1.150 (10.07)	1.164 (11.38)	1.149 (9.42)	1.160 (9.84)	1.537 (9.85)
30%	1.198 (18.13)	1.194 (17.01)	1.207 (17.73)	1.205 (17.92)	1.198 (15.72)	1.222 (18.04)	1.224 (17.18)	1.238 (18.52)	1.251 (19.20)	1.253 (18.56)	1.657 (18.39)
40%	1.294 (27.59)	1.292 (26.60)	1.292 (26.01)	1.289 (26.13)	1.320 (27.47)	1.328 (28.29)	1.329 (27.16)	1.356 (29.76)	1.361 (29.63)	1.352 (27.93)	1.791 (28.01)
Imputation using predictive mean matching method											
10%	1.053 (3.83)	1.058 (3.62)	1.060 (3.37)	1.069 (4.63)	1.070 (3.34)	1.080 (4.34)	1.077 (3.09)	1.090 (4.33)	1.095 (4.35)	1.091 (3.27)	1.461 (4.39)
20%	1.115 (9.89)	1.106 (8.32)	1.119 (9.15)	1.122 (9.84)	1.136 (9.71)	1.140 (10.16)	1.139 (9.04)	1.154 (10.44)	1.141 (8.72)	1.150 (8.85)	1.539 (9.98)
30%	1.194 (17.72)	1.169 (14.52)	1.198 (16.89)	1.204 (17.83)	1.199 (15.74)	1.220 (17.90)	1.212 (16.00)	1.224 (17.18)	1.239 (18.05)	1.236 (16.95)	1.647 (17.72)
40%	1.256 (23.81)	1.247 (22.13)	1.261 (22.98)	1.244 (21.78)	1.253 (21.01)	1.274 (23.11)	1.284 (22.92)	1.293 (23.77)	1.303 (24.14)	1.305 (23.49)	1.740 (24.33)
Imputation using propensity−score method											
10%	1.054 (3.87)	1.059 (3.71)	1.059 (3.27)	1.069 (4.62)	1.072 (3.54)	1.081 (4.41)	1.082 (3.50)	1.094 (4.67)	1.098 (4.62)	1.094 (3.53)	1.461 (4.41)
20%	1.111 (9.51)	1.110 (8.72)	1.129 (10.09)	1.125 (10.08)	1.130 (9.15)	1.137 (9.88)	1.145 (9.59)	1.150 (10.03)	1.154 (9.96)	1.161 (9.92)	1.534 (9.64)
30%	1.182 (16.51)	1.177 (15.28)	1.202 (17.26)	1.207 (18.14)	1.210 (16.84)	1.207 (16.63)	1.221 (16.88)	1.226 (17.29)	1.229 (17.08)	1.232 (16.61)	1.634 (16.78)
40%	1.276 (25.82)	1.276 (24.99)	1.281 (24.95)	1.292 (26.50)	1.309 (26.44)	1.327 (28.27)	1.317 (26.02)	1.334 (27.65)	1.361 (29.67)	1.337 (26.54)	1.766 (26.24)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	1.409	1.406	1.410	1.414	1.418	1.418	1.417	1.429	1.429
Imputation using linear regression									
10%	1.464 (3.93)	1.477 (5.03)	1.477 (4.74)	1.479 (4.58)	1.492 (5.20)	1.491 (5.18)	1.496 (5.62)	1.503 (5.15)	1.501 (5.07)
20%	1.549 (9.95)	1.554 (10.52)	1.559 (10.59)	1.568 (10.86)	1.571 (10.78)	1.569 (10.68)	1.581 (11.63)	1.585 (10.89)	1.592 (11.38)
30%	1.685 (19.60)	1.672 (18.93)	1.703 (20.81)	1.692 (19.64)	1.699 (19.85)	1.700 (19.91)	1.702 (20.13)	1.708 (19.48)	1.727 (20.85)
40%	1.801 (27.82)	1.804 (28.34)	1.829 (29.74)	1.818 (28.58)	1.821 (28.42)	1.854 (30.80)	1.865 (31.61)	1.841 (28.81)	1.869 (30.79)
Imputation using predictive mean matching method									
10%	1.466 (4.06)	1.469 (4.51)	1.468 (4.17)	1.471 (4.02)	1.479 (4.30)	1.483 (4.59)	1.485 (4.84)	1.493 (4.49)	1.489 (4.23)
20%	1.532 (8.73)	1.543 (9.73)	1.541 (9.30)	1.547 (9.41)	1.548 (9.18)	1.560 (10.04)	1.562 (10.29)	1.563 (9.35)	1.563 (9.38)
30%	1.633 (15.92)	1.637 (16.40)	1.634 (15.88)	1.626 (14.95)	1.663 (17.29)	1.656 (16.84)	1.669 (17.78)	1.666 (16.56)	1.672 (17.05)
40%	1.733 (22.96)	1.735 (23.39)	1.791 (27.07)	1.783 (26.07)	1.784 (25.80)	1.781 (25.61)	1.812 (27.89)	1.779 (24.50)	1.813 (26.88)
Imputation using propensity−score method									
10%	1.471 (4.40)	1.478 (5.09)	1.475 (4.61)	1.475 (4.30)	1.491 (5.18)	1.492 (5.24)	1.488 (5.00)	1.495 (4.57)	1.496 (4.72)
20%	1.556 (10.39)	1.556 (10.70)	1.563 (10.87)	1.565 (10.69)	1.577 (11.23)	1.571 (10.84)	1.581 (11.62)	1.588 (11.08)	1.595 (11.60)
30%	1.637 (16.17)	1.640 (16.67)	1.653 (17.26)	1.659 (17.33)	1.688 (19.04)	1.680 (18.49)	1.678 (18.45)	1.681 (17.61)	1.701 (19.07)
40%	1.796 (27.49)	1.793 (27.53)	1.838 (30.36)	1.815 (28.32)	1.814 (27.92)	1.831 (29.19)	1.853 (30.79)	1.817 (27.14)	1.868 (30.70)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	1.403	1.403	1.404	1.407	1.408	1.412	1.414	1.417	1.417
Imputation using linear regression									

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	1.468 (4.62)	1.468 (4.65)	1.476 (5.17)	1.480 (5.24)	1.484 (5.45)	1.483 (5.00)	1.491 (5.43)	1.495 (5.53)	1.489 (5.09)
20%	1.541 (9.81)	1.548 (10.38)	1.557 (10.91)	1.552 (10.36)	1.564 (11.13)	1.573 (11.37)	1.586 (12.11)	1.568 (10.67)	1.581 (11.57)
30%	1.653 (17.77)	1.665 (18.68)	1.659 (18.22)	1.651 (17.37)	1.676 (19.06)	1.682 (19.09)	1.693 (19.69)	1.710 (20.69)	1.709 (20.61)
40%	1.786 (27.26)	1.785 (27.25)	1.781 (26.88)	1.818 (29.26)	1.828 (29.83)	1.829 (29.56)	1.864 (31.78)	1.864 (31.57)	1.853 (30.81)
Imputation using predictive mean matching method									
10%	1.462 (4.21)	1.464 (4.42)	1.469 (4.66)	1.467 (4.26)	1.476 (4.83)	1.475 (4.43)	1.484 (4.96)	1.487 (4.94)	1.483 (4.69)
20%	1.526 (8.74)	1.537 (9.60)	1.542 (9.87)	1.559 (10.83)	1.560 (10.80)	1.557 (10.29)	1.571 (11.08)	1.556 (9.83)	1.565 (10.46)
30%	1.616 (15.13)	1.649 (17.61)	1.655 (17.95)	1.647 (17.07)	1.671 (18.69)	1.661 (17.61)	1.671 (18.15)	1.691 (19.33)	1.682 (18.70)
40%	1.724 (22.88)	1.744 (24.33)	1.721 (22.63)	1.728 (22.85)	1.756 (24.72)	1.769 (25.30)	1.778 (25.69)	1.786 (26.06)	1.789 (26.29)
Imputation using propensity−score method									
10%	1.463 (4.27)	1.464 (4.36)	1.469 (4.65)	1.470 (4.51)	1.477 (4.93)	1.480 (4.78)	1.489 (5.30)	1.490 (5.17)	1.488 (5.01)
20%	1.530 (9.05)	1.551 (10.55)	1.545 (10.09)	1.551 (10.23)	1.556 (10.51)	1.566 (10.87)	1.566 (10.71)	1.574 (11.06)	1.581 (11.57)
30%	1.629 (16.10)	1.658 (18.20)	1.662 (18.42)	1.666 (18.45)	1.656 (17.63)	1.677 (18.74)	1.675 (18.46)	1.680 (18.56)	1.681 (18.63)
40%	1.762 (25.55)	1.770 (26.22)	1.785 (27.18)	1.803 (28.16)	1.825 (29.67)	1.812 (28.30)	1.833 (29.58)	1.864 (31.59)	1.833 (29.36)

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	89.66	89.66	89.66	96.55	93.10	96.55	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	93.10	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	96.43	96.43	92.86	96.43	92.86	100.00	100.00	92.86	100.00	78.57
30%	100.00	89.66	96.55	96.55	93.10	93.10	93.10	89.66	89.66	89.66	62.07
40%	89.66	96.55	96.55	96.55	89.66	96.55	100.00	100.00	96.55	96.55	79.31
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	89.66	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	92.86	92.86	92.86	96.43	92.86	100.00	96.43	89.29	96.43	67.86
30%	96.55	89.66	96.55	96.55	89.66	96.55	89.66	89.66	89.66	89.66	58.62
40%	86.21	93.10	89.66	86.21	86.21	89.66	96.55	96.55	93.10	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	86.21	86.21	96.55	93.10	93.10	93.10	93.10	68.97
20%	92.86	96.43	96.43	92.86	96.43	96.43	100.00	96.43	92.86	96.43	75.00
30%	93.10	79.31	89.66	96.55	93.10	96.55	96.55	93.10	93.10	89.66	55.17
40%	93.10	96.55	96.55	89.66	86.21	89.66	89.66	96.55	93.10	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	100.00	100.00	96.55	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	89.29	92.86	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00	89.29	96.43	75.00
30%	96.55	100.00	96.55	96.55	96.55	100.00	89.66	100.00	100.00	96.55	86.21
40%	93.10	96.55	89.66	93.10	93.10	96.55	100.00	93.10	100.00	96.55	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	89.29	96.43	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	89.29	92.86	85.71	96.43	96.43	100.00	96.43	100.00	89.29	96.43	64.29
30%	93.10	100.00	100.00	89.66	96.55	100.00	93.10	100.00	96.55	96.55	79.31
40%	96.55	100.00	82.76	89.66	89.66	96.55	93.10	89.66	96.55	96.55	51.72
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	92.86	100.00	96.43	96.43	100.00	96.43	96.43	82.14
20%	89.29	96.43	89.29	100.00	96.43	96.43	96.43	100.00	92.86	96.43	71.43
30%	96.55	100.00	96.55	93.10	96.55	100.00	93.10	100.00	100.00	96.55	86.21
40%	93.10	100.00	86.21	86.21	89.66	96.55	96.55	82.76	96.55	93.10	65.52

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	96.55	100.00	96.55	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	100.00	100	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	96.43	100	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43
30%	100.00	93.10	100	100.00	96.55	96.55	96.55	96.55	100.00	96.55	82.76
40%	96.55	100.00	100	96.55	96.55	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100	100.00	96.55	100.00	100.00	96.55	96.55	100.00	93.10
20%	100.00	96.43	100	100.00	96.43	100.00	100.00	100.00	100.00	100.00	92.86
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	96.55	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_nonshrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	100.00	100	96.43	100.00	100.00	100.00	100	100.00	96.43	100.00	92.86
30%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.43	100	92.86	100.00	100.00	100.00	100	100.00	96.43	100.00	85.71
30%	100.00	100	100.00	100.00	96.55	100.00	100	100.00	100.00	100.00	96.55
40%	100.00	100	100.00	96.55	96.55	100.00	100	96.55	100.00	96.55	86.21
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100.00	96.43	100.00	100.00	100	100.00	100.00	100.00	96.43
20%	96.43	100	96.43	100.00	100.00	100.00	100	100.00	96.43	100.00	89.29
30%	100.00	100	100.00	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100.00	100.00	93.10	96.55	100	93.10	100.00	96.55	79.31

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	89.66	89.66	89.66	96.55	93.10	96.55	96.55	93.10	100.00	72.41
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	93.10	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	96.43	96.43	92.86	96.43	92.86	100.00	100.00	92.86	100.00	78.57
30%	100.00	89.66	96.55	96.55	93.10	93.10	93.10	89.66	89.66	89.66	62.07
40%	89.66	96.55	96.55	96.55	89.66	96.55	100.00	100.00	96.55	96.55	79.31
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	89.66	86.21	100.00	93.10	93.10	93.10	96.55	72.41
20%	92.86	92.86	92.86	92.86	96.43	92.86	100.00	96.43	89.29	96.43	67.86
30%	96.55	89.66	96.55	96.55	89.66	96.55	89.66	89.66	89.66	89.66	58.62
40%	86.21	93.10	89.66	86.21	86.21	89.66	96.55	96.55	93.10	96.55	58.62
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	89.66	86.21	86.21	96.55	93.10	93.10	93.10	93.10	68.97
20%	92.86	96.43	96.43	92.86	96.43	96.43	100.00	96.43	92.86	96.43	75.00
30%	93.10	79.31	89.66	96.55	93.10	96.55	96.55	93.10	93.10	89.66	55.17
40%	93.10	96.55	96.55	89.66	86.21	89.66	89.66	96.55	93.10	96.55	65.52
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	100.00	100.00	96.55	100.00	93.10	100.00	82.76
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	T6vs.T0_none	T7vs.T0_none	T8vs.T0_none	T9vs.T0_none	T10vs.T0_none	Joint_none
10%	92.86	100.00	89.29	92.86	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00	89.29	96.43	75.00
30%	96.55	100.00	96.55	96.55	96.55	100.00	89.66	100.00	100.00	96.55	86.21
40%	93.10	96.55	89.66	93.10	93.10	96.55	100.00	93.10	100.00	96.55	65.52
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	89.29	96.43	100.00	96.43	96.43	96.43	96.43	96.43	78.57
20%	89.29	92.86	85.71	96.43	96.43	100.00	96.43	100.00	89.29	96.43	64.29
30%	93.10	100.00	100.00	89.66	96.55	100.00	93.10	100.00	96.55	96.55	79.31
40%	96.55	100.00	82.76	89.66	89.66	96.55	93.10	89.66	96.55	96.55	51.72
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	92.86	100.00	96.43	96.43	100.00	96.43	96.43	82.14
20%	89.29	96.43	89.29	100.00	96.43	96.43	96.43	100.00	92.86	96.43	71.43
30%	96.55	100.00	96.55	93.10	96.55	100.00	93.10	100.00	100.00	96.55	86.21
40%	93.10	100.00	86.21	86.21	89.66	96.55	96.55	82.76	96.55	93.10	65.52

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
Scenario I: all means are 0 for T0,T1,...,T10	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	96.55	100.00	96.55	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	100.00	100	96.43	100.00	100.00	100.00	100.00	100.00	100.00	96.43
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	100.00	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	96.43	100	96.43	96.43	100.00	100.00	100.00	100.00	100.00	92.86
30%	100.00	93.10	100	100.00	100.00	96.55	96.55	96.55	96.55	96.55	82.76
40%	96.55	100.00	100	96.55	96.55	100.00	100.00	96.55	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100.00	100	100.00	96.55	100.00	100.00	100.00	96.55	100.00	93.10
20%	100.00	96.43	100	100.00	96.43	100.00	100.00	100.00	100.00	100.00	92.86
30%	100.00	100.00	100	100.00	100.00	96.55	96.55	100.00	100.00	100.00	93.10
40%	96.55	100.00	100	96.55	100.00	100.00	100.00	100.00	100.00	100.00	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	100	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_10Treatments/table2_shrink_10_treatments_MAR_n50

MR	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many	T6vs.T0_many	T7vs.T0_many	T8vs.T0_many	T9vs.T0_many	T10vs.T0_many	Joint_many
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.43	100	100	100.00	100.00	100.00	100	100.00	96.43	100.00	92.86
30%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	96.55	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
20%	96.43	100	100	100.00	100.00	100.00	100	100.00	96.43	100.00	92.86
30%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100	100.00	96.55	96.55	100	96.55	100.00	96.55	86.21
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	100	100	96.43	100.00	100.00	100	100.00	100.00	100.00	96.43
20%	92.86	100	100	100.00	100.00	100.00	100	100.00	96.43	100.00	89.29
30%	100.00	100	100	100.00	100.00	100.00	100	100.00	100.00	100.00	100.00
40%	100.00	100	100	100.00	93.10	96.55	100	93.10	100.00	96.55	79.31

SampleSize_n50_10Treatments/table3_nonshrink_10_treatments_MAR_n50

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	2.467	2.560	2.584	2.574	2.542	2.621	2.640	2.627	2.618	2.620	3.367
Imputation using linear regression											
10%	2.589 (4.96)	2.604 (1.71)	2.673 (3.45)	2.696 (4.75)	2.664 (4.77)	2.658 (1.42)	2.694 (2.04)	2.734 (4.06)	2.712 (3.59)	2.724 (3.98)	3.572 (6.10)
20%	2.764 (12.04)	2.772 (8.26)	2.798 (8.27)	2.769 (7.57)	2.852 (12.19)	2.803 (6.94)	2.859 (8.29)	2.880 (9.62)	2.855 (9.05)	2.793 (6.63)	3.811 (13.19)
30%	2.980 (20.79)	3.030 (18.36)	3.057 (18.30)	3.066 (19.11)	3.051 (20.00)	3.062 (16.83)	3.044 (15.33)	3.092 (17.68)	3.096 (18.27)	3.073 (17.30)	4.107 (21.97)
40%	3.400 (37.83)	3.318 (29.58)	3.330 (28.87)	3.425 (33.08)	3.413 (34.25)	3.447 (31.53)	3.404 (28.94)	3.430 (30.58)	3.400 (29.87)	3.484 (32.97)	4.684 (39.10)
Imputation using predictive mean matching method											
10%	2.530 (2.56)	2.558 (−0.08)	2.625 (1.59)	2.643 (2.68)	2.620 (3.06)	2.606 (−0.56)	2.647 (0.28)	2.679 (1.98)	2.669 (1.95)	2.676 (2.14)	3.489 (3.62)
20%	2.639 (6.97)	2.667 (4.17)	2.697 (4.36)	2.660 (3.35)	2.735 (7.56)	2.694 (2.79)	2.742 (3.89)	2.778 (5.74)	2.765 (5.63)	2.740 (4.58)	3.637 (8.01)
30%	2.804 (13.65)	2.835 (10.74)	2.838 (9.82)	2.874 (11.65)	2.856 (12.33)	2.830 (8.00)	2.851 (7.99)	2.905 (10.59)	2.922 (11.60)	2.843 (8.51)	3.862 (14.71)
40%	2.969 (20.35)	3.028 (18.26)	3.002 (16.15)	3.030 (17.70)	2.974 (16.98)	3.033 (15.73)	3.010 (14.03)	3.044 (15.87)	3.047 (16.39)	3.123 (19.23)	4.093 (21.57)
Imputation using propensity−score method											
10%	2.542 (3.06)	2.570 (0.39)	2.635 (1.96)	2.639 (2.52)	2.620 (3.03)	2.616 (−0.18)	2.641 (0.05)	2.677 (1.90)	2.668 (1.90)	2.681 (2.35)	3.508 (4.18)
20%	2.653 (7.54)	2.688 (5.00)	2.709 (4.83)	2.685 (4.31)	2.760 (8.57)	2.731 (4.22)	2.755 (4.35)	2.788 (6.13)	2.768 (5.74)	2.753 (5.10)	3.658 (8.66)
30%	2.780 (12.71)	2.812 (9.85)	2.886 (11.69)	2.843 (10.45)	2.901 (14.12)	2.855 (8.95)	2.896 (9.71)	2.934 (11.69)	2.905 (10.98)	2.868 (9.47)	3.844 (14.18)
40%	3.143 (27.39)	3.117 (21.73)	3.101 (20.00)	3.226 (25.31)	3.090 (21.53)	3.143 (19.93)	3.167 (19.98)	3.175 (20.87)	3.166 (20.93)	3.256 (24.27)	4.325 (28.47)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	2.484	2.503	2.536	2.501	2.540	2.562	2.556	2.565	2.601	2.581	3.396
Imputation using linear regression											

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	2.662 (7.17)	2.572 (2.75)	2.639 (4.06)	2.654 (6.13)	2.638 (3.87)	2.643 (3.16)	2.664 (4.22)	2.653 (3.41)	2.702 (3.91)	2.689 (4.20)	3.669 (8.03)
20%	2.687 (8.19)	2.769 (10.60)	2.763 (8.94)	2.739 (9.50)	2.790 (9.85)	2.805 (9.47)	2.807 (9.80)	2.833 (10.41)	2.854 (9.75)	2.853 (10.56)	3.711 (9.26)
30%	2.992 (20.47)	3.060 (22.23)	2.979 (17.44)	3.074 (22.92)	3.050 (20.11)	3.074 (19.99)	3.152 (23.31)	3.171 (23.61)	3.162 (21.58)	3.084 (19.51)	4.121 (21.35)
40%	3.248 (30.78)	3.316 (32.46)	3.289 (29.70)	3.305 (32.14)	3.350 (31.91)	3.399 (32.64)	3.384 (32.38)	3.375 (31.55)	3.469 (33.41)	3.381 (31.00)	4.489 (32.17)
Imputation using predictive mean matching method											
10%	2.601 (4.73)	2.511 (0.30)	2.579 (1.67)	2.594 (3.71)	2.583 (1.69)	2.581 (0.74)	2.606 (1.95)	2.596 (1.17)	2.638 (1.45)	2.637 (2.17)	3.586 (5.60)
20%	2.573 (3.62)	2.640 (5.44)	2.660 (4.89)	2.654 (6.13)	2.678 (5.44)	2.673 (4.32)	2.713 (6.12)	2.720 (6.01)	2.741 (5.39)	2.703 (4.73)	3.557 (4.74)
30%	2.838 (14.27)	2.836 (13.30)	2.790 (9.99)	2.906 (16.17)	2.875 (13.22)	2.901 (13.21)	2.947 (15.30)	3.018 (17.64)	2.946 (13.27)	2.855 (10.65)	3.903 (14.92)
40%	2.971 (19.63)	2.948 (17.75)	2.924 (15.27)	2.968 (18.68)	2.983 (17.46)	3.003 (17.20)	3.038 (18.86)	3.012 (17.39)	3.027 (16.40)	3.023 (17.12)	4.104 (20.85)
Imputation using propensity−score method											
10%	2.615 (5.28)	2.534 (1.24)	2.591 (2.14)	2.607 (4.24)	2.592 (2.07)	2.611 (1.89)	2.634 (3.03)	2.612 (1.83)	2.637 (1.40)	2.658 (3.00)	3.603 (6.10)
20%	2.626 (5.75)	2.683 (7.19)	2.706 (6.70)	2.693 (7.67)	2.731 (7.52)	2.719 (6.13)	2.749 (7.54)	2.765 (7.79)	2.767 (6.40)	2.758 (6.86)	3.623 (6.69)
30%	2.861 (15.18)	2.905 (16.05)	2.854 (12.54)	2.971 (18.78)	2.949 (16.13)	2.947 (15.03)	2.986 (16.81)	3.058 (19.19)	3.005 (15.57)	2.932 (13.63)	3.941 (16.04)
40%	2.997 (20.68)	3.063 (22.38)	2.976 (17.32)	3.032 (21.24)	3.100 (22.07)	3.038 (18.56)	3.123 (22.19)	3.072 (19.73)	3.183 (22.41)	3.115 (20.69)	4.140 (21.89)

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	3.495	3.528	3.514	3.471	3.577	3.604	3.587	3.574	3.576
Imputation using linear regression									
10%	3.593 (2.80)	3.690 (4.58)	3.720 (5.86)	3.676 (5.92)	3.668 (2.52)	3.717 (3.14)	3.773 (5.19)	3.743 (4.71)	3.760 (5.13)
20%	3.823 (9.37)	3.858 (9.36)	3.818 (8.65)	3.934 (13.35)	3.864 (8.01)	3.941 (9.35)	3.972 (10.74)	3.938 (10.16)	3.853 (7.73)
30%	4.177 (19.50)	4.213 (19.42)	4.227 (20.28)	4.203 (21.10)	4.219 (17.93)	4.198 (16.47)	4.262 (18.81)	4.269 (19.42)	4.235 (18.42)
40%	4.575 (30.89)	4.592 (30.16)	4.723 (34.40)	4.702 (35.47)	4.750 (32.78)	4.693 (30.22)	4.729 (31.84)	4.687 (31.14)	4.803 (34.29)
Imputation using predictive mean matching method									
10%	3.528 (0.95)	3.622 (2.66)	3.646 (3.74)	3.614 (4.14)	3.595 (0.48)	3.651 (1.30)	3.696 (3.04)	3.682 (3.00)	3.691 (3.22)
20%	3.676 (5.17)	3.717 (5.35)	3.666 (4.32)	3.769 (8.61)	3.712 (3.76)	3.779 (4.86)	3.829 (6.74)	3.811 (6.62)	3.776 (5.58)
30%	3.907 (11.77)	3.912 (10.89)	3.961 (12.70)	3.935 (13.38)	3.900 (9.01)	3.929 (9.02)	4.004 (11.62)	4.027 (12.66)	3.918 (9.55)
40%	4.173 (19.39)	4.138 (17.30)	4.179 (18.91)	4.101 (18.17)	4.181 (16.86)	4.151 (15.18)	4.197 (17.01)	4.202 (17.55)	4.307 (20.42)
Imputation using propensity−score method									
10%	3.547 (1.47)	3.637 (3.08)	3.642 (3.63)	3.616 (4.17)	3.610 (0.91)	3.645 (1.13)	3.695 (3.01)	3.682 (3.01)	3.701 (3.48)
20%	3.707 (6.07)	3.736 (5.88)	3.702 (5.35)	3.807 (9.69)	3.765 (5.26)	3.799 (5.40)	3.846 (7.21)	3.818 (6.80)	3.797 (6.16)
30%	3.888 (11.24)	3.989 (13.08)	3.931 (11.87)	4.009 (15.52)	3.946 (10.30)	4.004 (11.11)	4.056 (13.08)	4.017 (12.38)	3.965 (10.87)
40%	4.291 (22.77)	4.270 (21.04)	4.442 (26.39)	4.255 (22.59)	4.327 (20.97)	4.362 (21.02)	4.373 (21.92)	4.359 (21.96)	4.482 (25.33)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	3.423	3.468	3.420	3.473	3.504	3.496	3.508	3.557	3.529
Imputation using linear regression									

SampleSize_n50_10Treatments/table3_nonshrink_10_treatments_MAR_n50

MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	3.546 (3.60)	3.639 (4.91)	3.660 (6.99)	3.637 (4.73)	3.644 (4.00)	3.673 (5.06)	3.658 (4.25)	3.725 (4.73)	3.708 (5.06)
20%	3.825 (11.74)	3.817 (10.05)	3.783 (10.60)	3.852 (10.93)	3.874 (10.57)	3.877 (10.90)	3.912 (11.49)	3.943 (10.86)	3.940 (11.65)
30%	4.213 (23.07)	4.105 (18.34)	4.234 (23.79)	4.203 (21.02)	4.233 (20.81)	4.341 (24.18)	4.368 (24.51)	4.354 (22.42)	4.247 (20.35)
40%	4.580 (33.79)	4.542 (30.96)	4.567 (33.52)	4.627 (33.23)	4.693 (33.92)	4.674 (33.71)	4.662 (32.90)	4.792 (34.74)	4.671 (32.35)
Imputation using predictive mean matching method									
10%	3.462 (1.14)	3.556 (2.51)	3.577 (4.57)	3.561 (2.53)	3.559 (1.56)	3.594 (2.80)	3.579 (2.01)	3.637 (2.27)	3.636 (3.02)
20%	3.650 (6.63)	3.678 (6.04)	3.669 (7.27)	3.701 (6.57)	3.696 (5.46)	3.749 (7.25)	3.759 (7.14)	3.789 (6.54)	3.736 (5.87)
30%	3.900 (13.94)	3.838 (10.66)	3.997 (16.85)	3.955 (13.88)	3.988 (13.80)	4.052 (15.92)	4.150 (18.29)	4.052 (13.92)	3.927 (11.28)
40%	4.071 (18.94)	4.037 (16.39)	4.101 (19.89)	4.122 (18.68)	4.150 (18.43)	4.197 (20.05)	4.160 (18.58)	4.183 (17.61)	4.176 (18.33)
Imputation using propensity−score method									
10%	3.493 (2.05)	3.571 (2.95)	3.593 (5.05)	3.573 (2.87)	3.598 (2.68)	3.630 (3.84)	3.601 (2.63)	3.634 (2.19)	3.663 (3.81)
20%	3.703 (8.17)	3.735 (7.67)	3.716 (8.63)	3.767 (8.47)	3.752 (7.07)	3.792 (8.48)	3.815 (8.73)	3.819 (7.38)	3.804 (7.80)
30%	4.000 (16.85)	3.933 (13.40)	4.092 (19.62)	4.063 (16.99)	4.060 (15.85)	4.114 (17.68)	4.212 (20.06)	4.141 (16.42)	4.038 (14.42)
40%	4.230 (23.58)	4.110 (18.51)	4.188 (22.44)	4.281 (23.26)	4.198 (19.81)	4.314 (23.41)	4.243 (20.95)	4.395 (23.59)	4.302 (21.91)

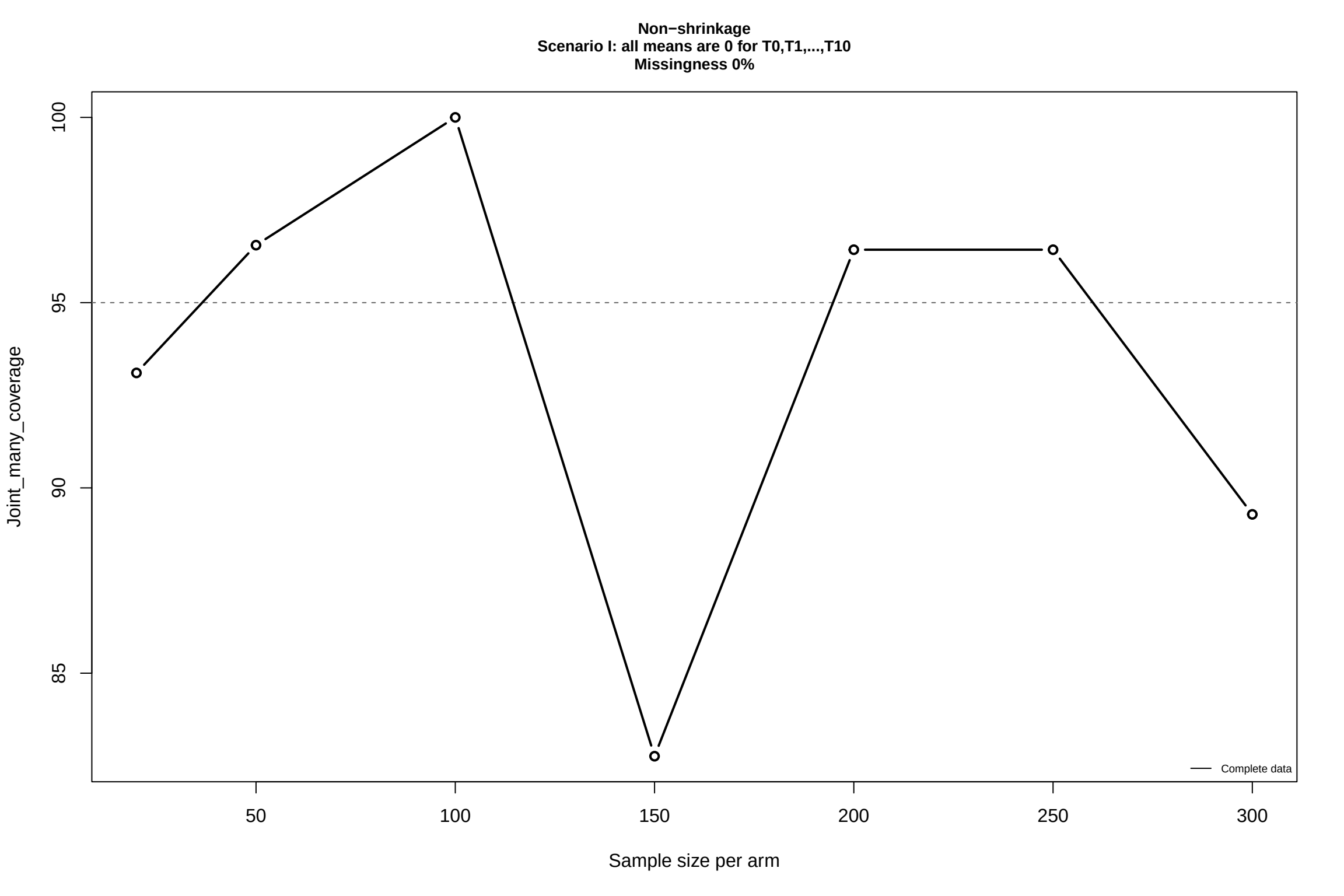
SampleSize_n50_10Treatments/table3_shrink_10_treatments_MAR_n50

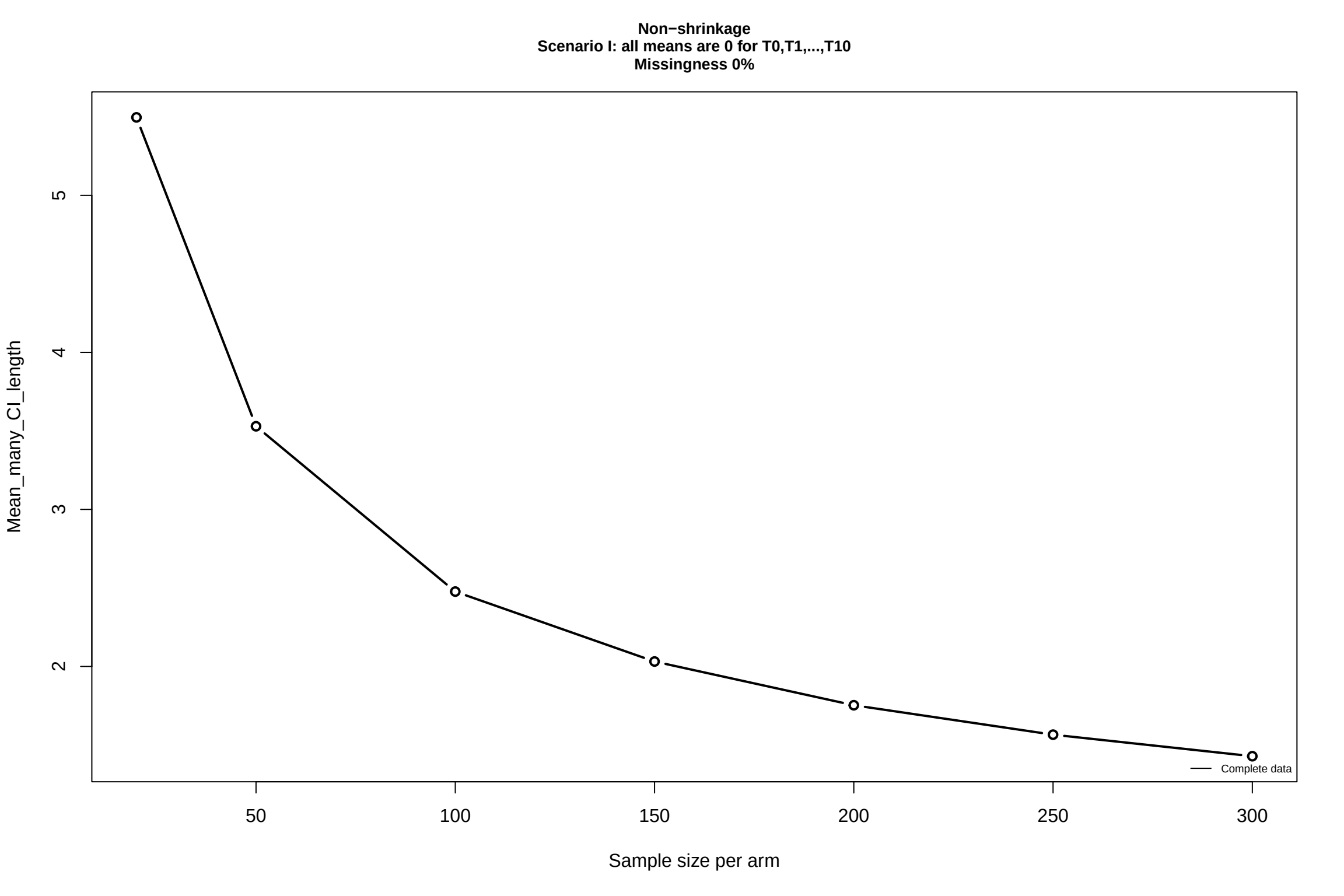
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
Scenario I: all means are 0 for T0,T1,...,T10											
0%	2.467	2.560	2.584	2.574	2.542	2.621	2.640	2.627	2.618	2.620	3.436
Imputation using linear regression											
10%	2.589 (4.96)	2.604 (1.71)	2.673 (3.45)	2.696 (4.75)	2.664 (4.77)	2.658 (1.42)	2.694 (2.04)	2.734 (4.06)	2.712 (3.59)	2.724 (3.98)	3.630 (5.67)
20%	2.764 (12.04)	2.772 (8.26)	2.798 (8.27)	2.769 (7.57)	2.852 (12.19)	2.803 (6.94)	2.859 (8.29)	2.880 (9.62)	2.855 (9.05)	2.793 (6.63)	3.819 (11.17)
30%	2.980 (20.79)	3.030 (18.36)	3.057 (18.30)	3.066 (19.11)	3.051 (20.00)	3.062 (16.83)	3.044 (15.33)	3.092 (17.68)	3.096 (18.27)	3.073 (17.30)	4.125 (20.07)
40%	3.400 (37.83)	3.318 (29.58)	3.330 (28.87)	3.425 (33.08)	3.413 (34.25)	3.447 (31.53)	3.404 (28.94)	3.430 (30.58)	3.400 (29.87)	3.484 (32.97)	4.696 (36.68)
Imputation using predictive mean matching method											
10%	2.530 (2.56)	2.558 (−0.08)	2.625 (1.59)	2.643 (2.68)	2.620 (3.06)	2.606 (−0.56)	2.647 (0.28)	2.679 (1.98)	2.669 (1.95)	2.676 (2.14)	3.550 (3.34)
20%	2.639 (6.97)	2.667 (4.17)	2.697 (4.36)	2.660 (3.35)	2.735 (7.56)	2.694 (2.79)	2.742 (3.89)	2.778 (5.74)	2.765 (5.63)	2.740 (4.58)	3.649 (6.20)
30%	2.804 (13.65)	2.835 (10.74)	2.838 (9.82)	2.874 (11.65)	2.856 (12.33)	2.830 (8.00)	2.851 (7.99)	2.905 (10.59)	2.922 (11.60)	2.843 (8.51)	3.895 (13.36)
40%	2.969 (20.35)	3.028 (18.26)	3.002 (16.15)	3.030 (17.70)	2.974 (16.98)	3.033 (15.73)	3.010 (14.03)	3.044 (15.87)	3.047 (16.39)	3.123 (19.23)	4.113 (19.72)
Imputation using propensity−score method											
10%	2.542 (3.06)	2.570 (0.39)	2.635 (1.96)	2.639 (2.52)	2.620 (3.03)	2.616 (−0.18)	2.641 (0.05)	2.677 (1.90)	2.668 (1.90)	2.681 (2.35)	3.571 (3.93)
20%	2.653 (7.54)	2.688 (5.00)	2.709 (4.83)	2.685 (4.31)	2.760 (8.57)	2.731 (4.22)	2.755 (4.35)	2.788 (6.13)	2.768 (5.74)	2.753 (5.10)	3.667 (6.74)
30%	2.780 (12.71)	2.812 (9.85)	2.886 (11.69)	2.843 (10.45)	2.901 (14.12)	2.855 (8.95)	2.896 (9.71)	2.934 (11.69)	2.905 (10.98)	2.868 (9.47)	3.871 (12.67)
40%	3.143 (27.39)	3.117 (21.73)	3.101 (20.00)	3.226 (25.31)	3.090 (21.53)	3.143 (19.93)	3.167 (19.98)	3.175 (20.87)	3.166 (20.93)	3.256 (24.27)	4.343 (26.42)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0											
0%	2.484	2.503	2.536	2.501	2.540	2.562	2.556	2.565	2.601	2.581	3.444
Imputation using linear regression											

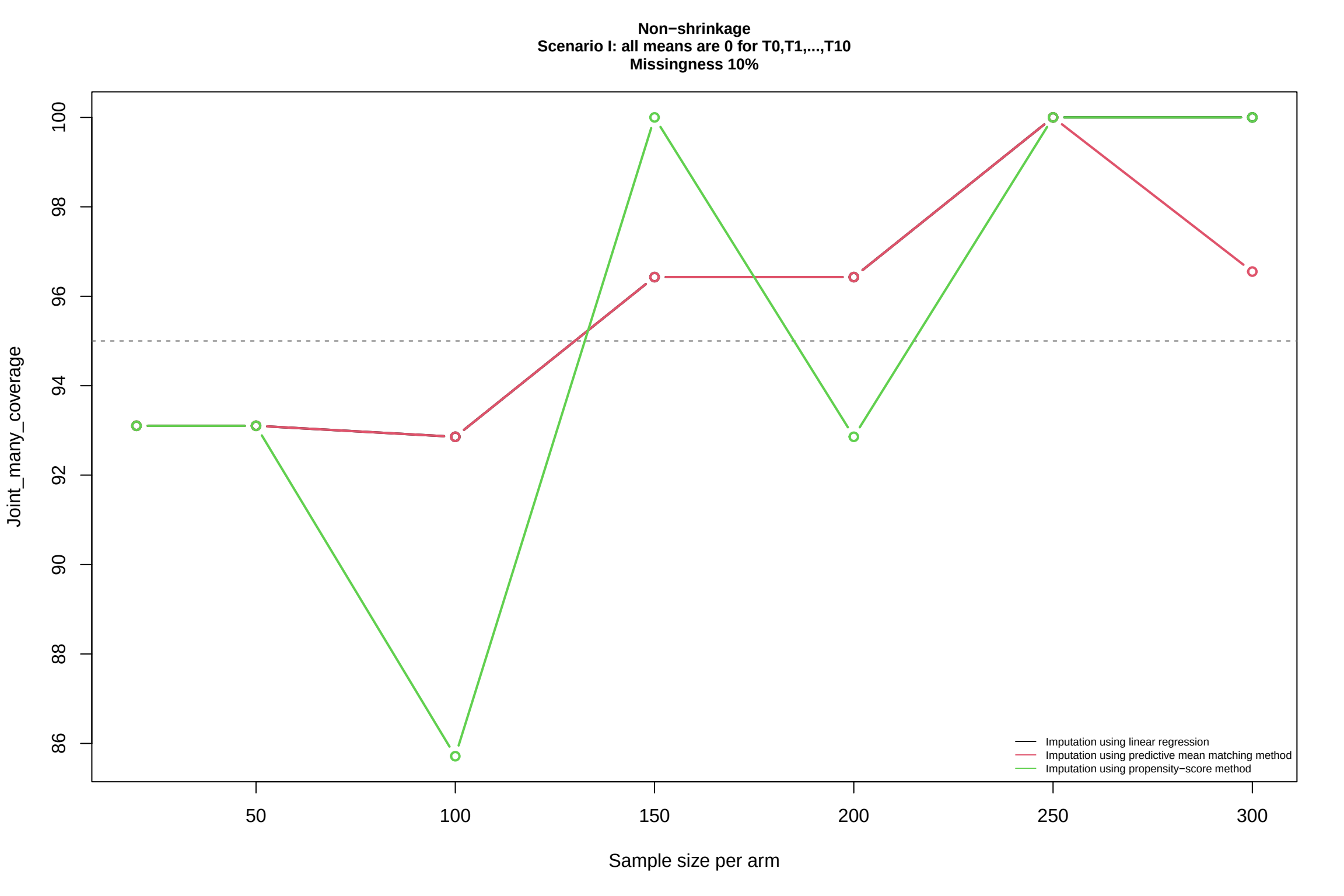
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	None_T6vs.T0	None_T7vs.T0	None_T8vs.T0	None_T9vs.T0	None_T10vs.T0	Many_T1vs.T0
10%	2.662 (7.17)	2.572 (2.75)	2.639 (4.06)	2.654 (6.13)	2.638 (3.87)	2.643 (3.16)	2.664 (4.22)	2.653 (3.41)	2.702 (3.91)	2.689 (4.20)	3.636 (5.57)
20%	2.687 (8.19)	2.769 (10.60)	2.763 (8.94)	2.739 (9.50)	2.790 (9.85)	2.805 (9.47)	2.807 (9.80)	2.833 (10.41)	2.854 (9.75)	2.853 (10.56)	3.794 (10.15)
30%	2.992 (20.47)	3.060 (22.23)	2.979 (17.44)	3.074 (22.92)	3.050 (20.11)	3.074 (19.99)	3.152 (23.31)	3.171 (23.61)	3.162 (21.58)	3.084 (19.51)	4.126 (19.80)
40%	3.248 (30.78)	3.316 (32.46)	3.289 (29.70)	3.305 (32.14)	3.350 (31.91)	3.399 (32.64)	3.384 (32.38)	3.375 (31.55)	3.469 (33.41)	3.381 (31.00)	4.515 (31.08)
Imputation using predictive mean matching method											
10%	2.601 (4.73)	2.511 (0.30)	2.579 (1.67)	2.594 (3.71)	2.583 (1.69)	2.581 (0.74)	2.606 (1.95)	2.596 (1.17)	2.638 (1.45)	2.637 (2.17)	3.559 (3.33)
20%	2.573 (3.62)	2.640 (5.44)	2.660 (4.89)	2.654 (6.13)	2.678 (5.44)	2.673 (4.32)	2.713 (6.12)	2.720 (6.01)	2.741 (5.39)	2.703 (4.73)	3.645 (5.83)
30%	2.838 (14.27)	2.836 (13.30)	2.790 (9.99)	2.906 (16.17)	2.875 (13.22)	2.901 (13.21)	2.947 (15.30)	3.018 (17.64)	2.946 (13.27)	2.855 (10.65)	3.895 (13.08)
40%	2.971 (19.63)	2.948 (17.75)	2.924 (15.27)	2.968 (18.68)	2.983 (17.46)	3.003 (17.20)	3.038 (18.86)	3.012 (17.39)	3.027 (16.40)	3.023 (17.12)	4.129 (19.87)
Imputation using propensity−score method											
10%	2.615 (5.28)	2.534 (1.24)	2.591 (2.14)	2.607 (4.24)	2.592 (2.07)	2.611 (1.89)	2.634 (3.03)	2.612 (1.83)	2.637 (1.40)	2.658 (3.00)	3.574 (3.76)
20%	2.626 (5.75)	2.683 (7.19)	2.706 (6.70)	2.693 (7.67)	2.731 (7.52)	2.719 (6.13)	2.749 (7.54)	2.765 (7.79)	2.767 (6.40)	2.758 (6.86)	3.705 (7.56)
30%	2.861 (15.18)	2.905 (16.05)	2.854 (12.54)	2.971 (18.78)	2.949 (16.13)	2.947 (15.03)	2.986 (16.81)	3.058 (19.19)	3.005 (15.57)	2.932 (13.63)	3.937 (14.29)
40%	2.997 (20.68)	3.063 (22.38)	2.976 (17.32)	3.032 (21.24)	3.100 (22.07)	3.038 (18.56)	3.123 (22.19)	3.072 (19.73)	3.183 (22.41)	3.115 (20.69)	4.168 (21.02)

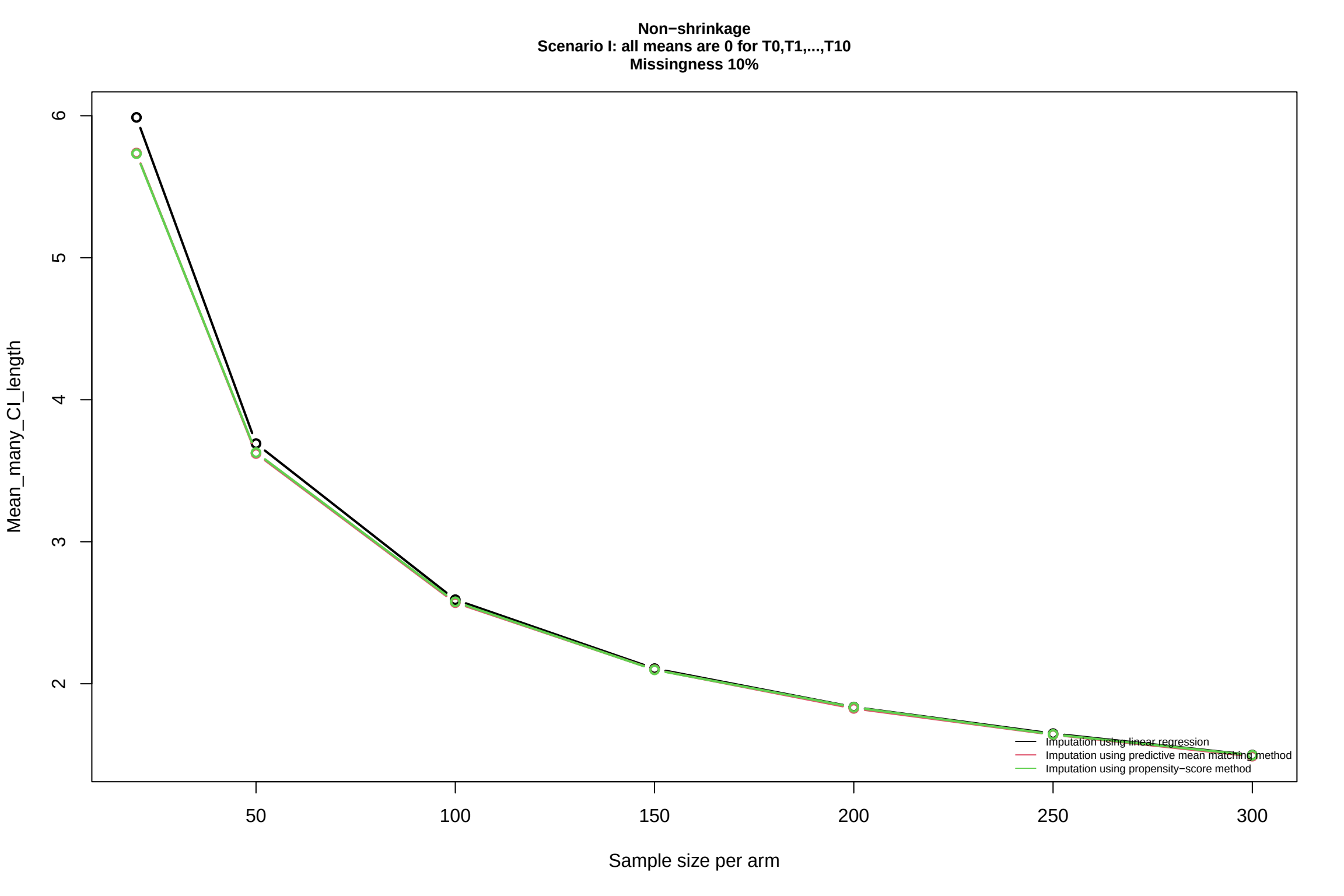
MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
Scenario I: all means are 0 for T0,T1,...,T10									
0%	3.460	3.478	3.481	3.464	3.495	3.501	3.494	3.493	3.502
Imputation using linear regression									
10%	3.620 (4.63)	3.650 (4.96)	3.688 (5.93)	3.656 (5.56)	3.661 (4.77)	3.653 (4.33)	3.689 (5.59)	3.671 (5.10)	3.684 (5.20)
20%	3.822 (10.48)	3.838 (10.35)	3.810 (9.46)	3.876 (11.91)	3.847 (10.10)	3.893 (11.19)	3.882 (11.12)	3.855 (10.36)	3.817 (8.98)
30%	4.172 (20.60)	4.197 (20.67)	4.202 (20.70)	4.196 (21.15)	4.217 (20.67)	4.181 (19.40)	4.228 (21.02)	4.232 (21.15)	4.212 (20.26)
40%	4.565 (31.96)	4.585 (31.85)	4.693 (34.80)	4.695 (35.54)	4.735 (35.50)	4.665 (33.24)	4.699 (34.50)	4.664 (33.51)	4.750 (35.62)
Imputation using predictive mean matching method									
10%	3.549 (2.59)	3.584 (3.04)	3.605 (3.56)	3.589 (3.61)	3.582 (2.51)	3.583 (2.33)	3.614 (3.44)	3.605 (3.22)	3.612 (3.13)
20%	3.678 (6.30)	3.692 (6.17)	3.652 (4.90)	3.708 (7.04)	3.688 (5.53)	3.726 (6.40)	3.738 (6.98)	3.727 (6.71)	3.725 (6.36)
30%	3.907 (12.94)	3.893 (11.93)	3.942 (13.23)	3.925 (13.33)	3.901 (11.62)	3.903 (11.48)	3.965 (13.48)	3.987 (14.14)	3.896 (11.25)
40%	4.161 (20.27)	4.133 (18.83)	4.145 (19.06)	4.091 (18.09)	4.168 (19.27)	4.119 (17.65)	4.164 (19.18)	4.176 (19.57)	4.249 (21.32)
Imputation using propensity−score method									
10%	3.572 (3.25)	3.598 (3.47)	3.609 (3.67)	3.594 (3.75)	3.602 (3.06)	3.583 (2.34)	3.618 (3.54)	3.608 (3.31)	3.629 (3.61)
20%	3.707 (7.16)	3.711 (6.69)	3.691 (6.03)	3.748 (8.21)	3.747 (7.22)	3.738 (6.75)	3.757 (7.53)	3.735 (6.94)	3.747 (6.98)
30%	3.894 (12.56)	3.980 (14.45)	3.913 (12.41)	4.004 (15.58)	3.950 (13.04)	3.983 (13.75)	4.022 (15.12)	3.981 (13.97)	3.950 (12.77)
40%	4.281 (23.74)	4.267 (22.69)	4.406 (26.57)	4.252 (22.75)	4.317 (23.54)	4.333 (23.75)	4.341 (24.25)	4.343 (24.33)	4.428 (26.44)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0									
0%	3.447	3.453	3.448	3.464	3.454	3.466	3.467	3.470	3.467
Imputation using linear regression									

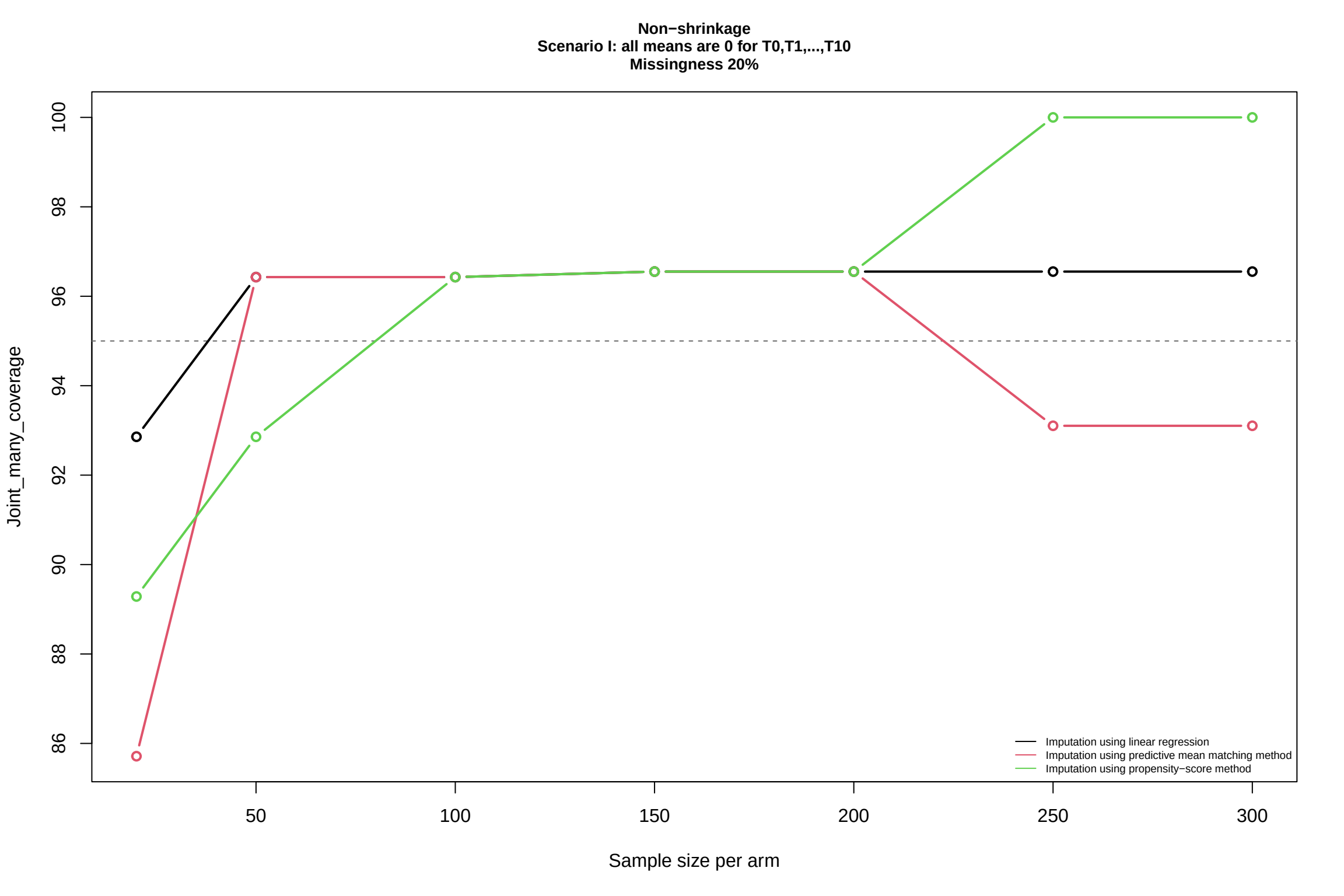
MR	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0	Many_T6vs.T0	Many_T7vs.T0	Many_T8vs.T0	Many_T9vs.T0	Many_T10vs.T0
10%	3.583 (3.95)	3.624 (4.96)	3.621 (5.02)	3.624 (4.60)	3.611 (4.53)	3.630 (4.75)	3.623 (4.49)	3.657 (5.36)	3.631 (4.71)
20%	3.838 (11.36)	3.821 (10.66)	3.806 (10.36)	3.858 (11.35)	3.865 (11.88)	3.865 (11.52)	3.899 (12.44)	3.892 (12.15)	3.909 (12.74)
30%	4.222 (22.48)	4.103 (18.85)	4.206 (21.96)	4.188 (20.88)	4.223 (22.25)	4.282 (23.56)	4.315 (24.47)	4.289 (23.58)	4.232 (22.04)
40%	4.599 (33.42)	4.574 (32.47)	4.563 (32.32)	4.635 (33.78)	4.684 (35.59)	4.665 (34.60)	4.655 (34.25)	4.762 (37.21)	4.663 (34.47)
Imputation using predictive mean matching method									
10%	3.506 (1.73)	3.546 (2.69)	3.539 (2.65)	3.551 (2.50)	3.533 (2.28)	3.545 (2.28)	3.554 (2.51)	3.567 (2.78)	3.557 (2.58)
20%	3.672 (6.53)	3.688 (6.82)	3.690 (7.01)	3.718 (7.33)	3.689 (6.79)	3.744 (8.03)	3.745 (8.01)	3.736 (7.66)	3.715 (7.14)
30%	3.914 (13.56)	3.835 (11.07)	3.967 (15.04)	3.938 (13.66)	3.980 (15.21)	4.000 (15.42)	4.088 (17.92)	3.975 (14.53)	3.910 (12.77)
40%	4.100 (18.95)	4.078 (18.11)	4.104 (19.01)	4.132 (19.26)	4.142 (19.90)	4.184 (20.72)	4.156 (19.86)	4.156 (19.76)	4.171 (20.29)
Imputation using propensity−score method									
10%	3.531 (2.45)	3.561 (3.13)	3.562 (3.31)	3.563 (2.83)	3.568 (3.29)	3.577 (3.22)	3.573 (3.06)	3.567 (2.78)	3.585 (3.40)
20%	3.722 (7.99)	3.734 (8.16)	3.730 (8.18)	3.777 (9.04)	3.745 (8.42)	3.787 (9.27)	3.801 (9.63)	3.762 (8.39)	3.780 (9.02)
30%	4.020 (16.64)	3.934 (13.95)	4.061 (17.77)	4.047 (16.81)	4.054 (17.37)	4.064 (17.27)	4.155 (19.85)	4.073 (17.35)	4.026 (16.12)
40%	4.252 (23.36)	4.142 (19.95)	4.193 (21.59)	4.290 (23.83)	4.195 (21.44)	4.305 (24.22)	4.244 (22.40)	4.371 (25.95)	4.303 (24.08)

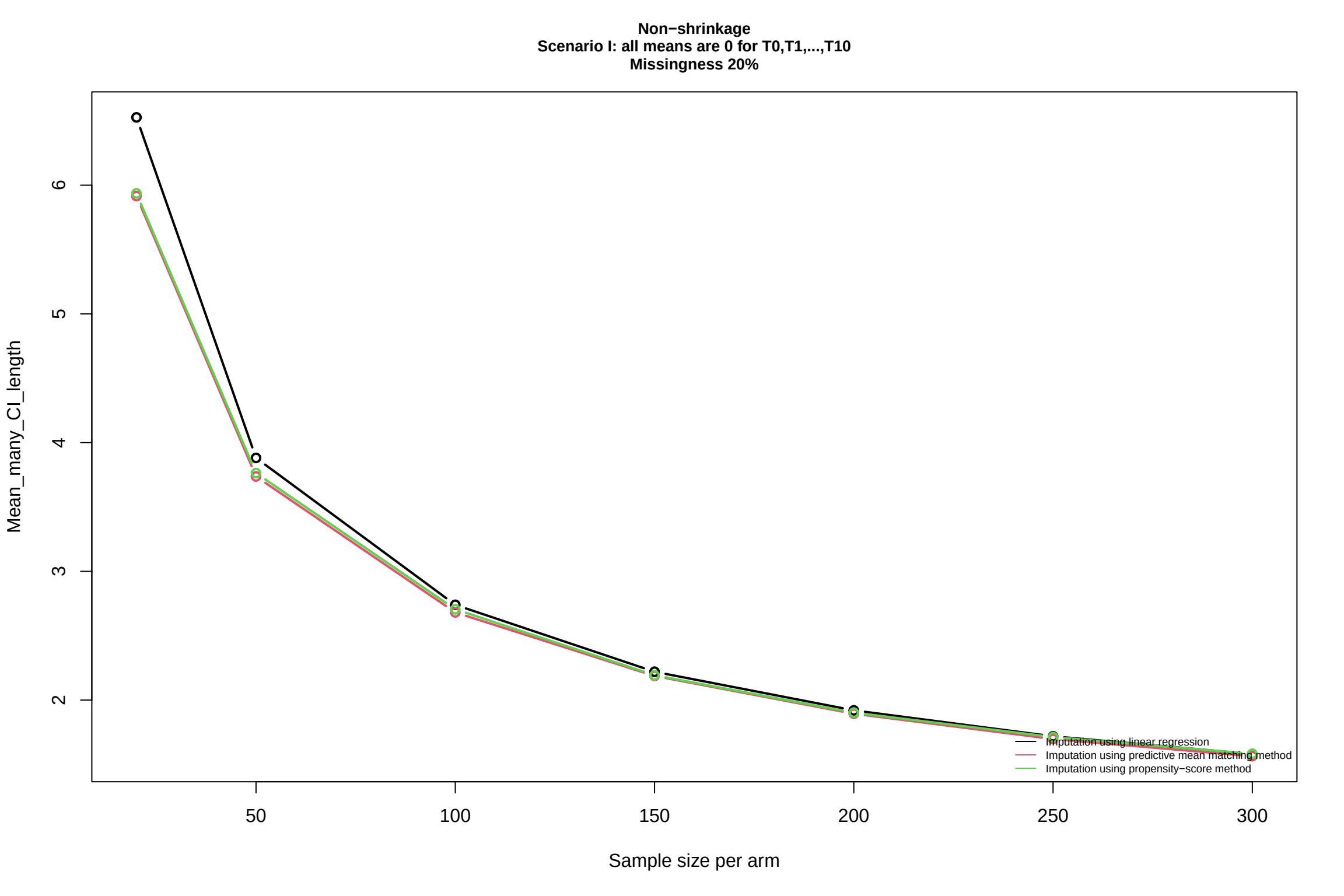

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 0%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Complete data

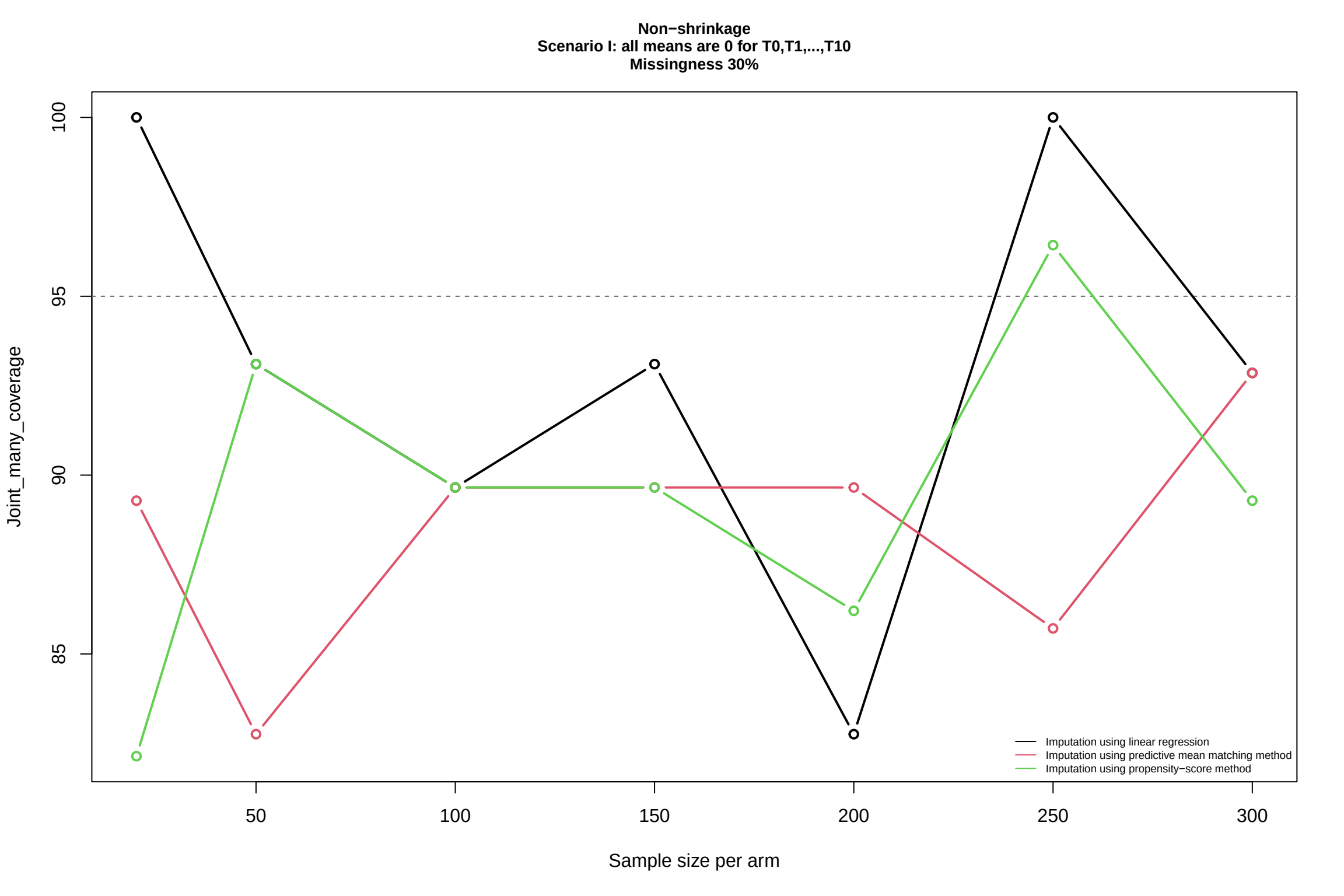

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

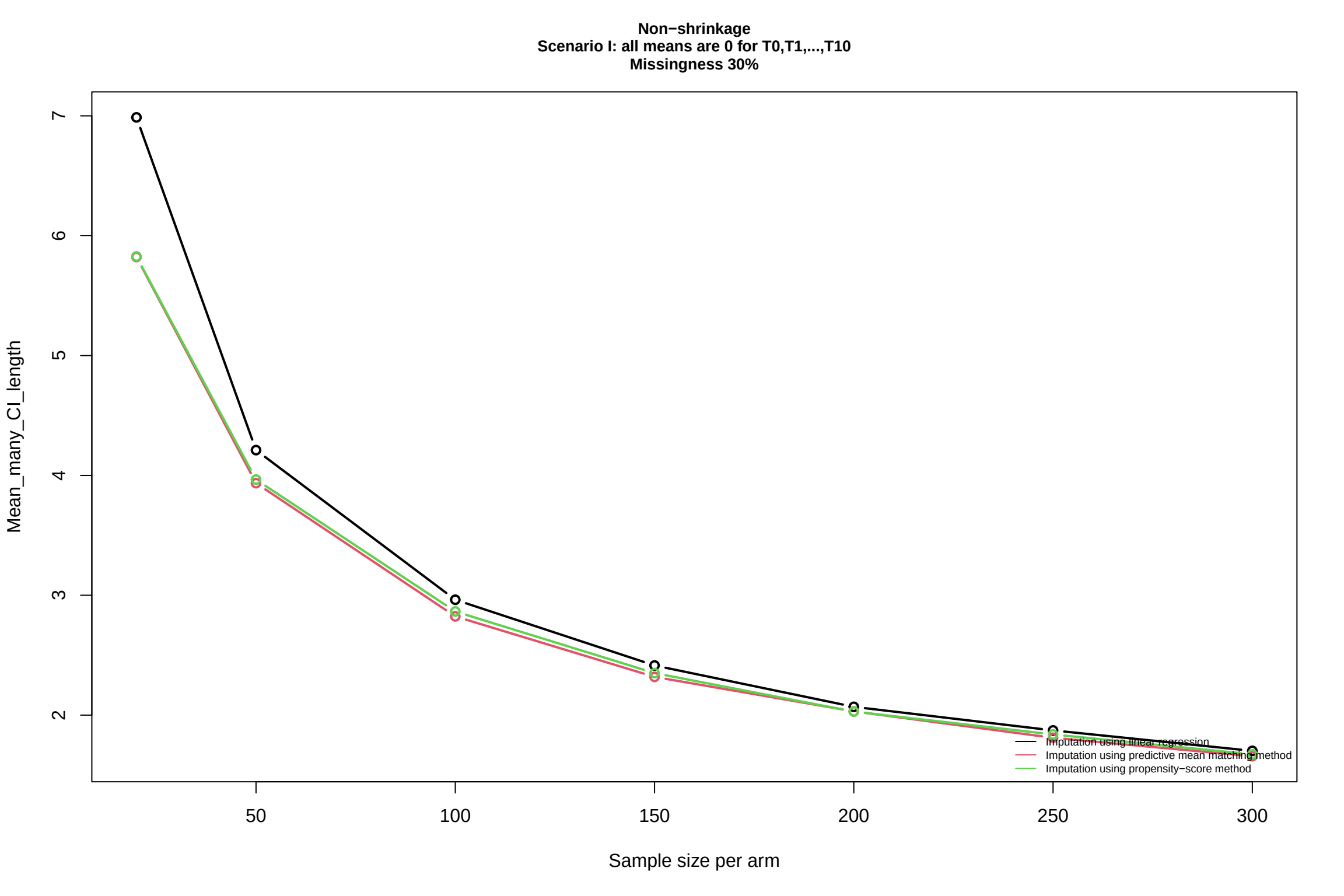

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

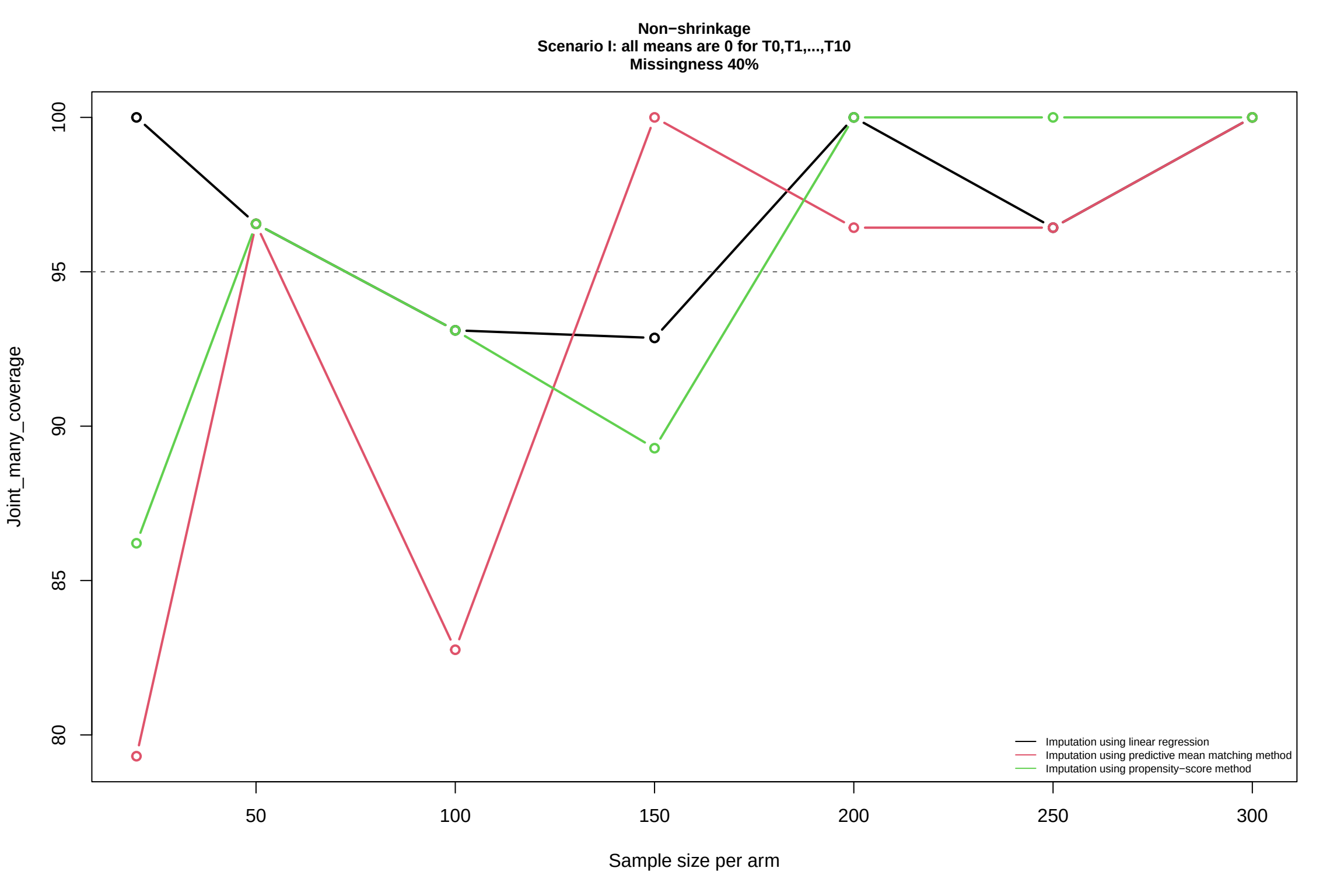

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 10%
Mean_many_CI_length
2
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6
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100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

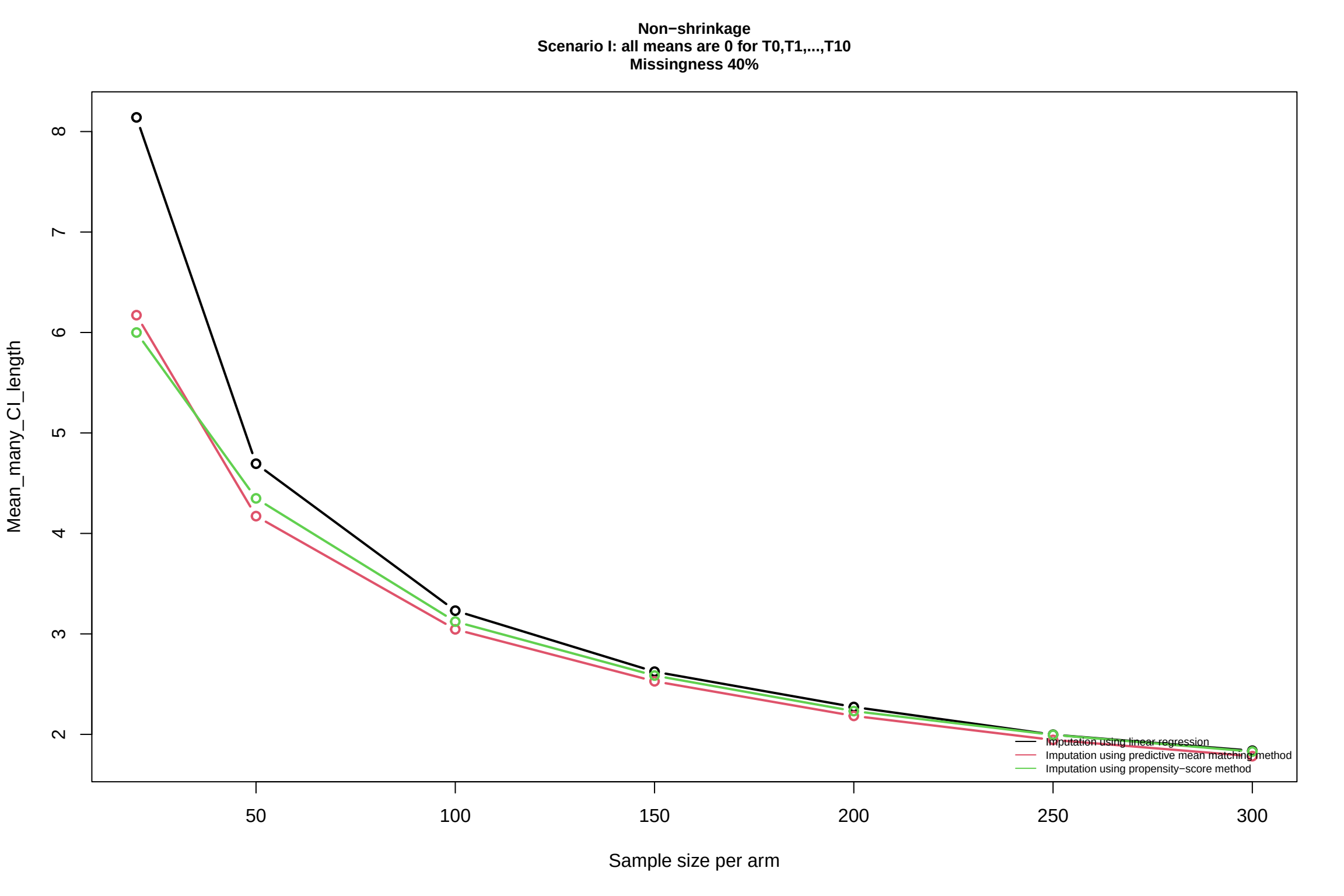

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 20%
Joint_many_coverage
100
98
96
94
92
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88
86
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300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

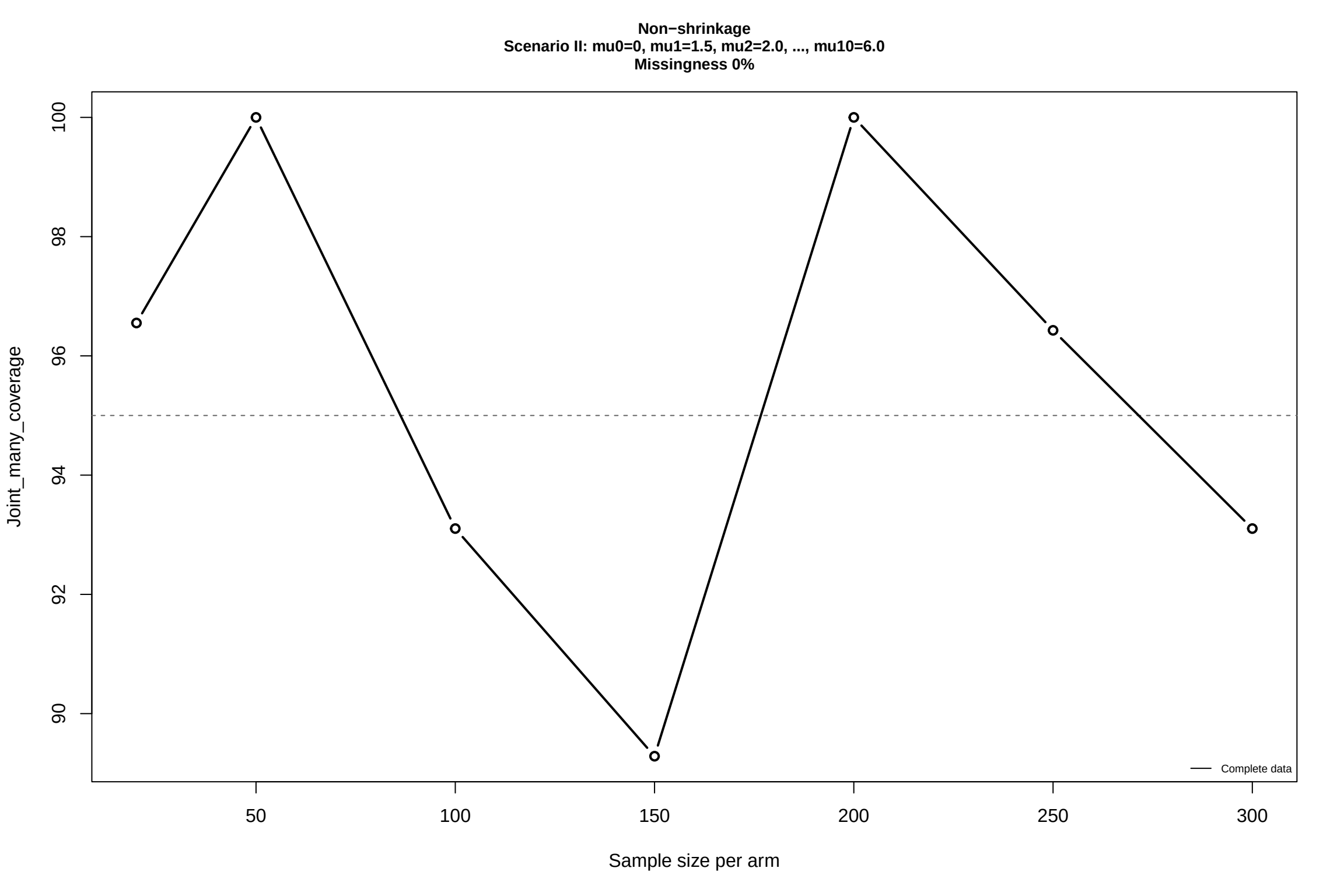

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 20%
Mean_many_CI_length
2
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100
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250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

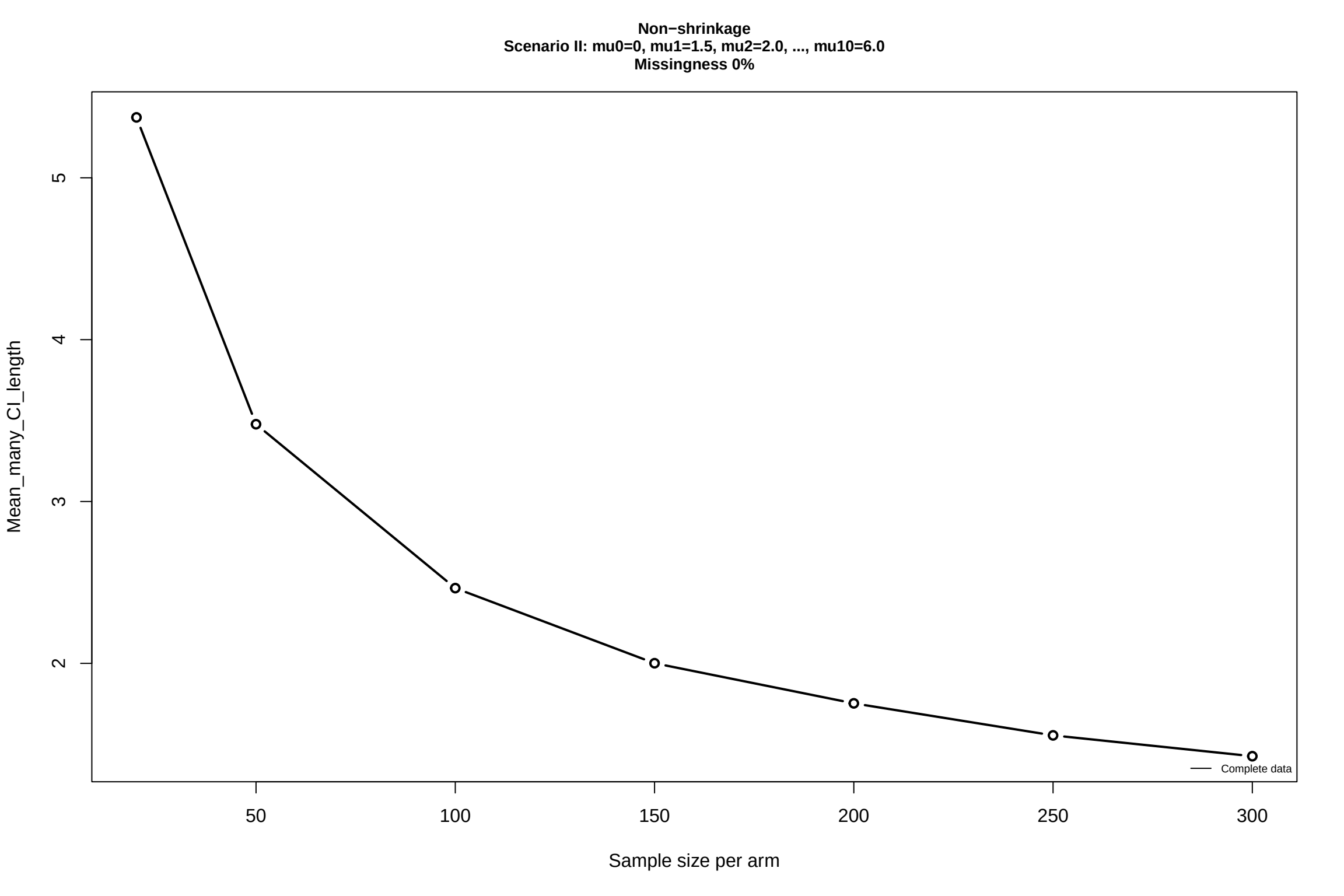

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

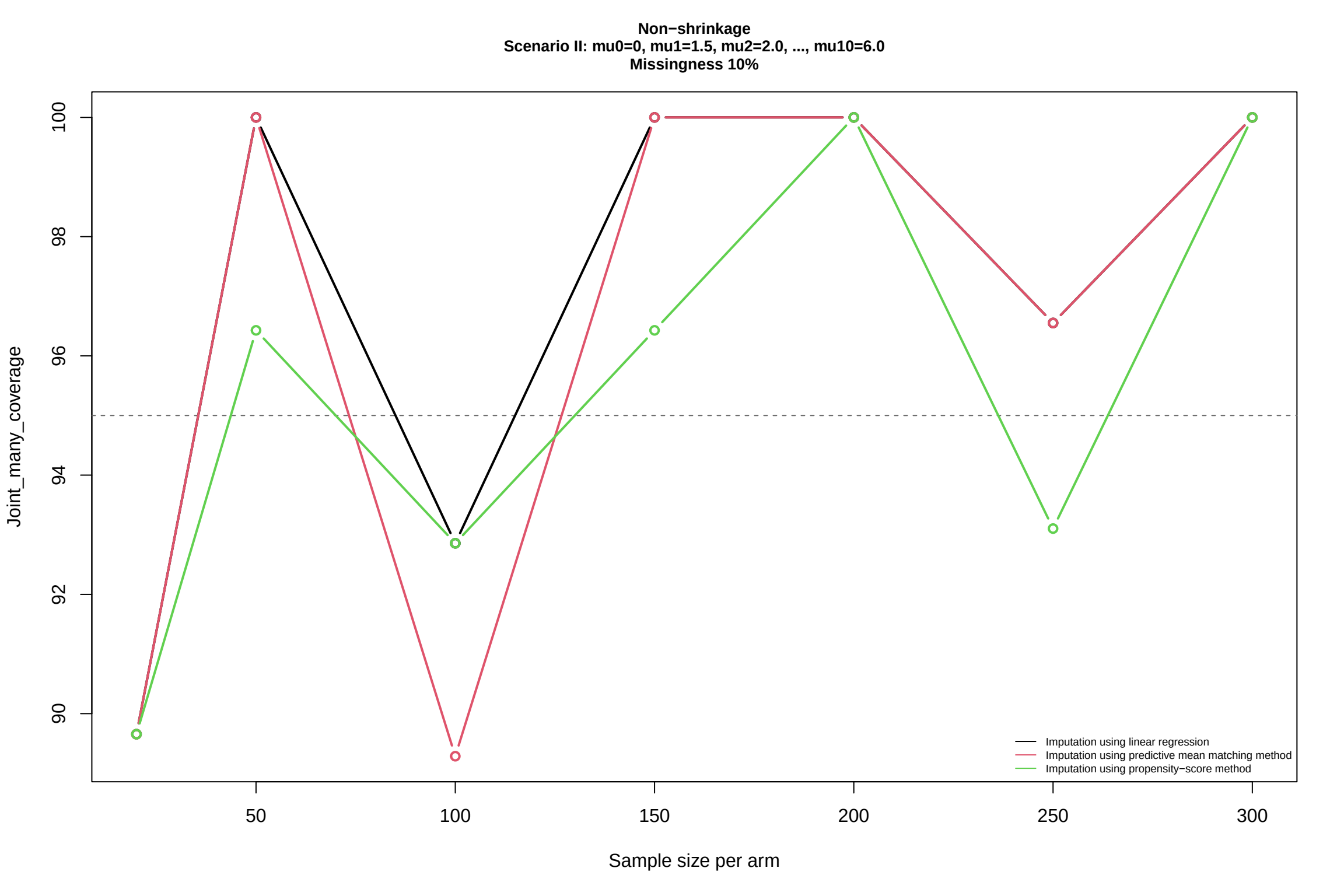

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 30%
Mean_many_CI_length
2
3
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6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

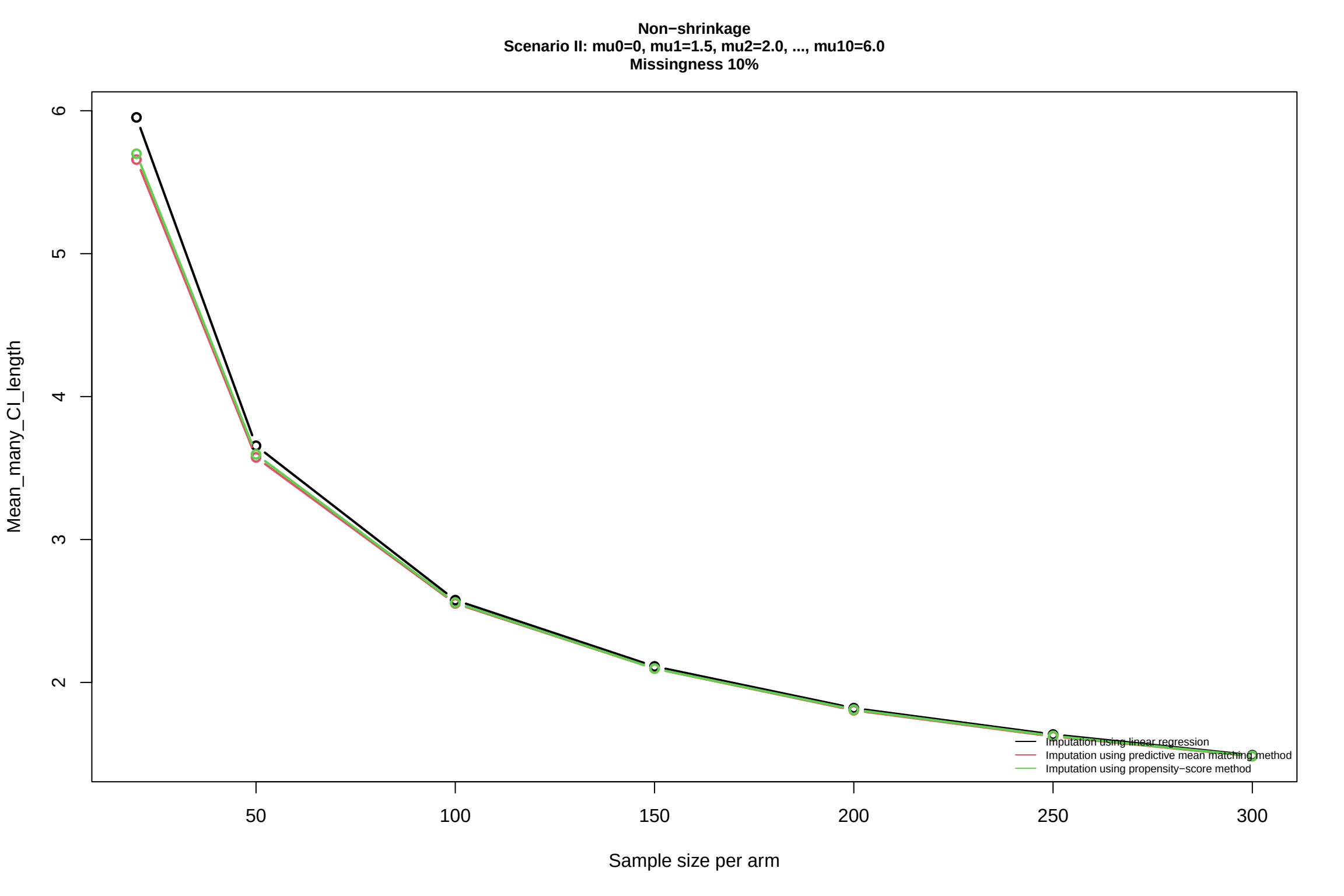

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 40%
Joint_many_coverage
100
95
90
85
80
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100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

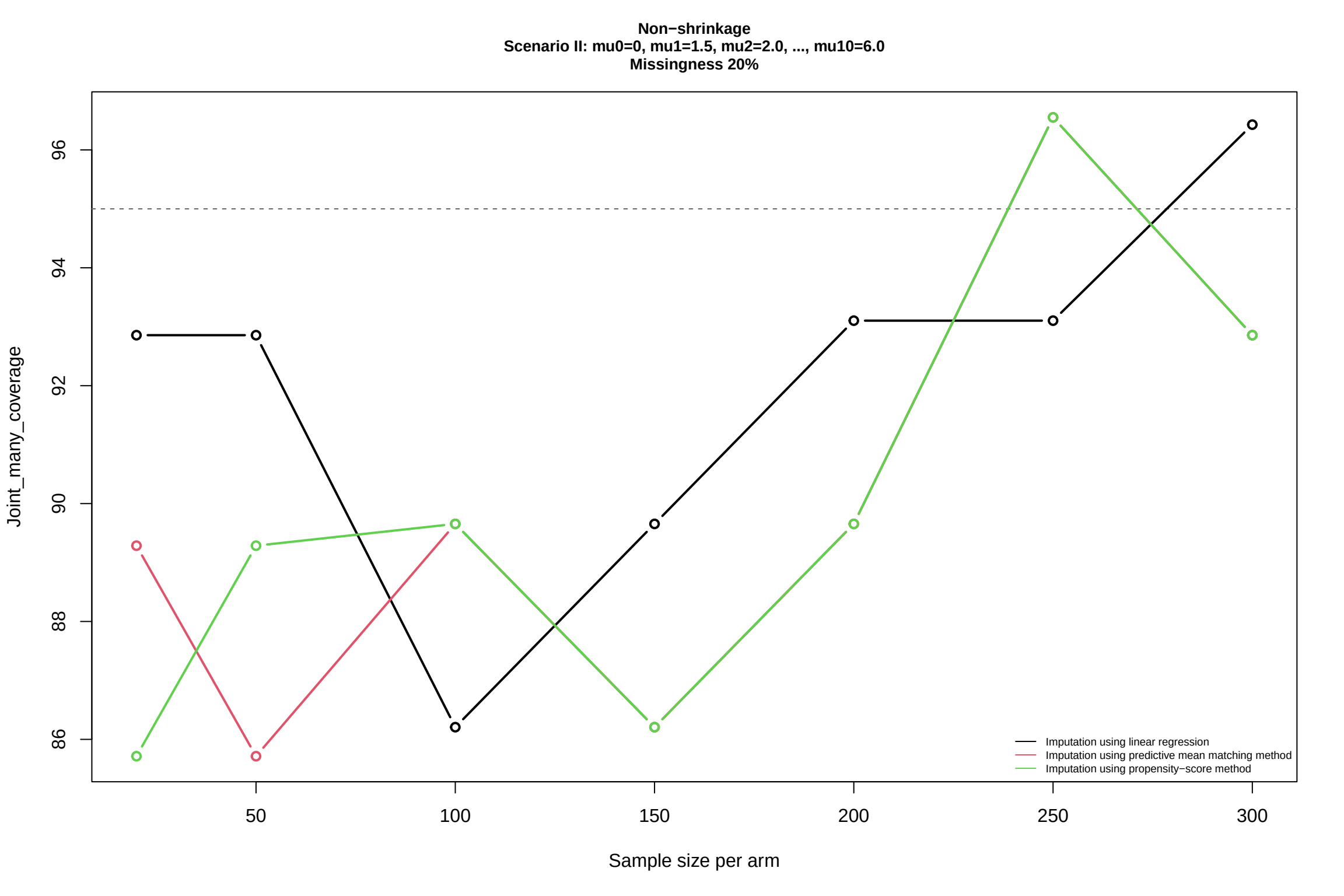

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
8
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

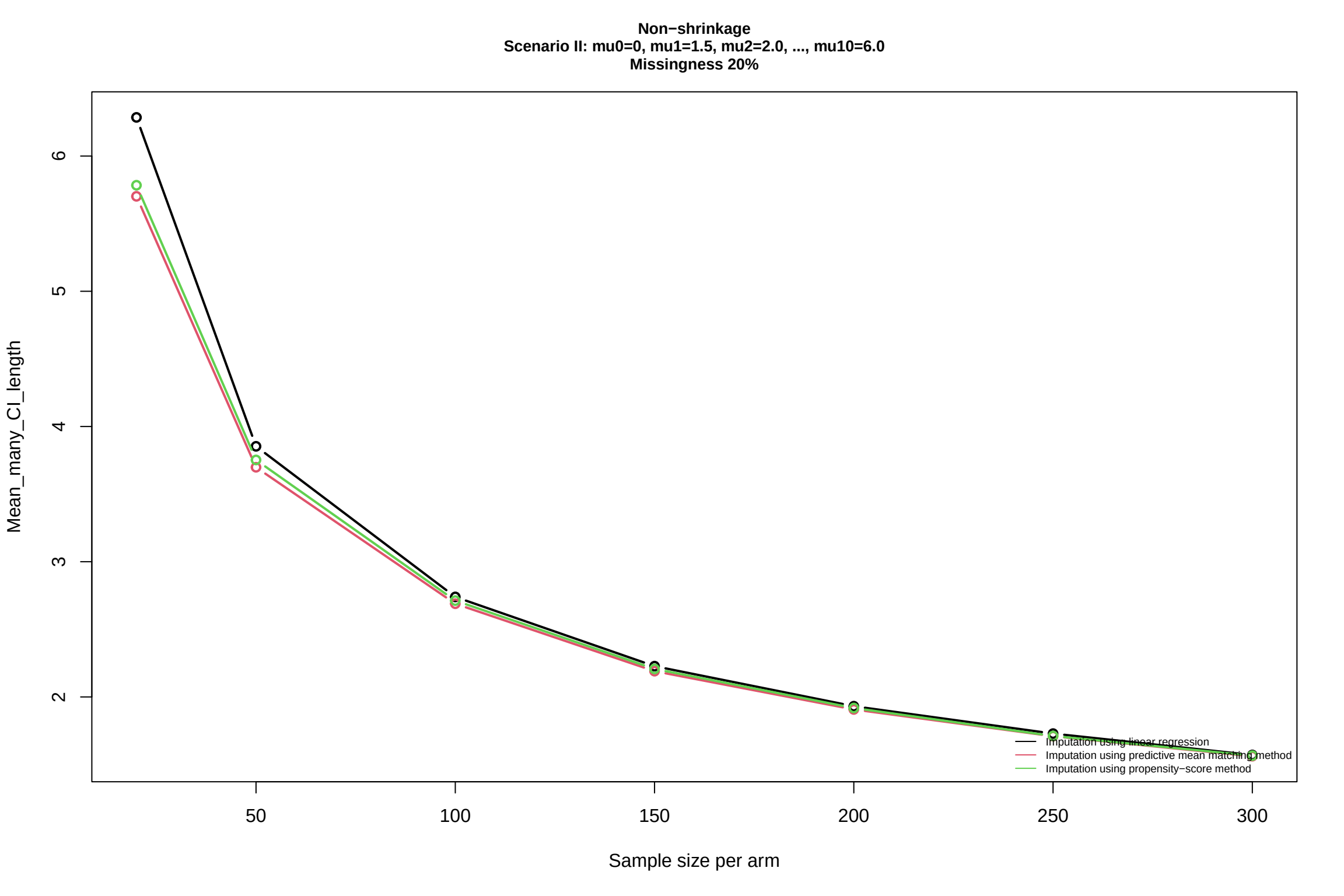

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 0%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Complete data

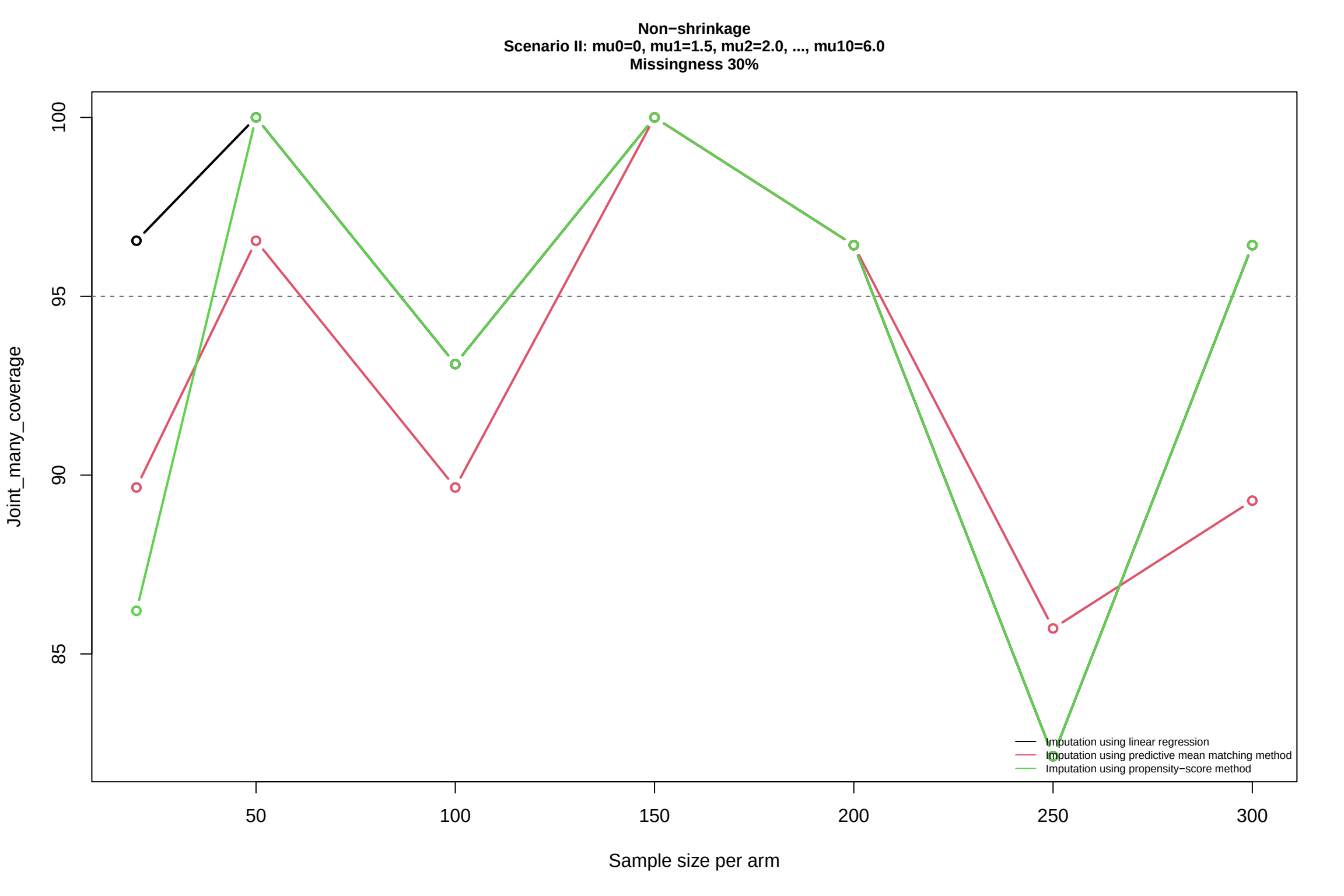

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

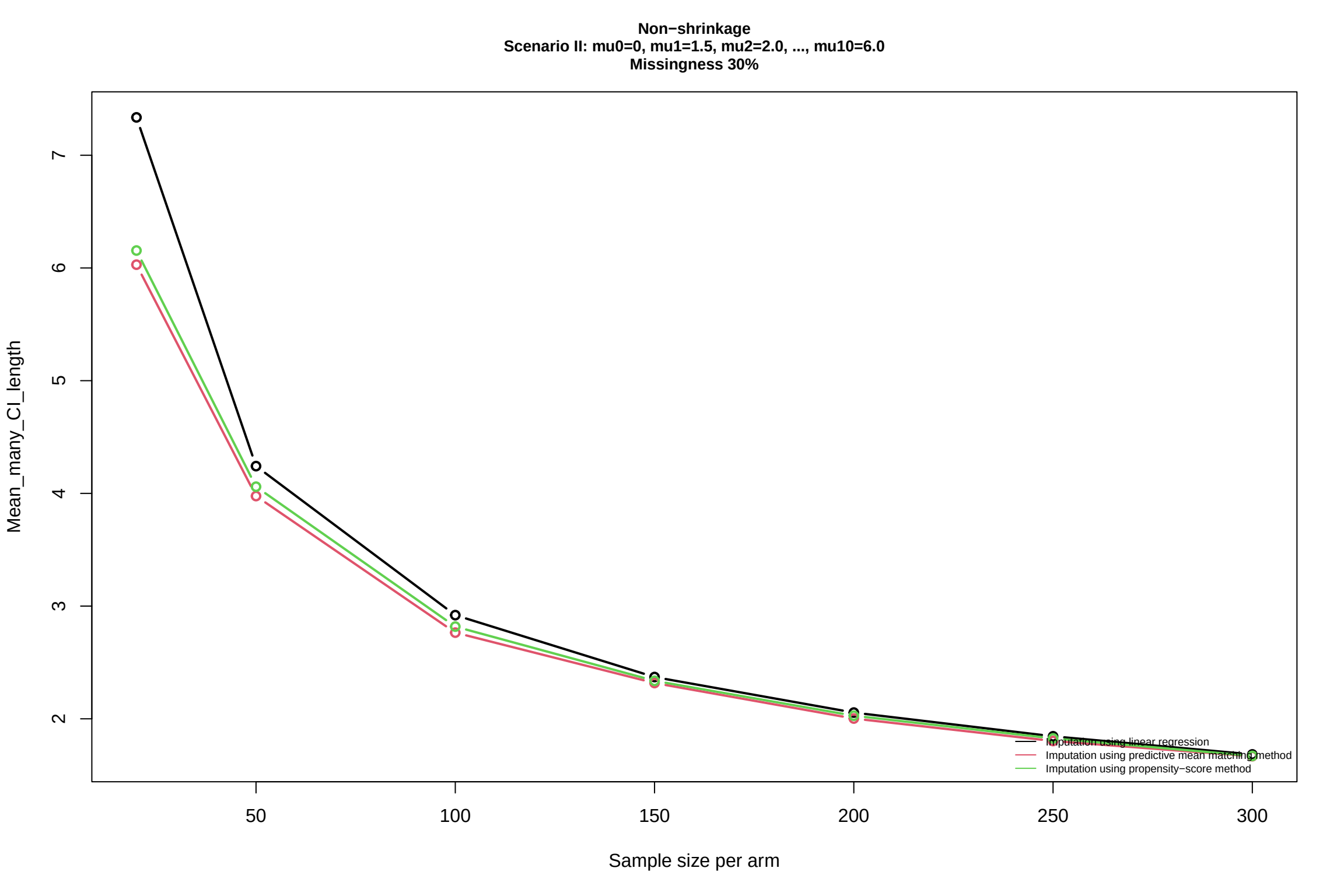

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
50
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150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

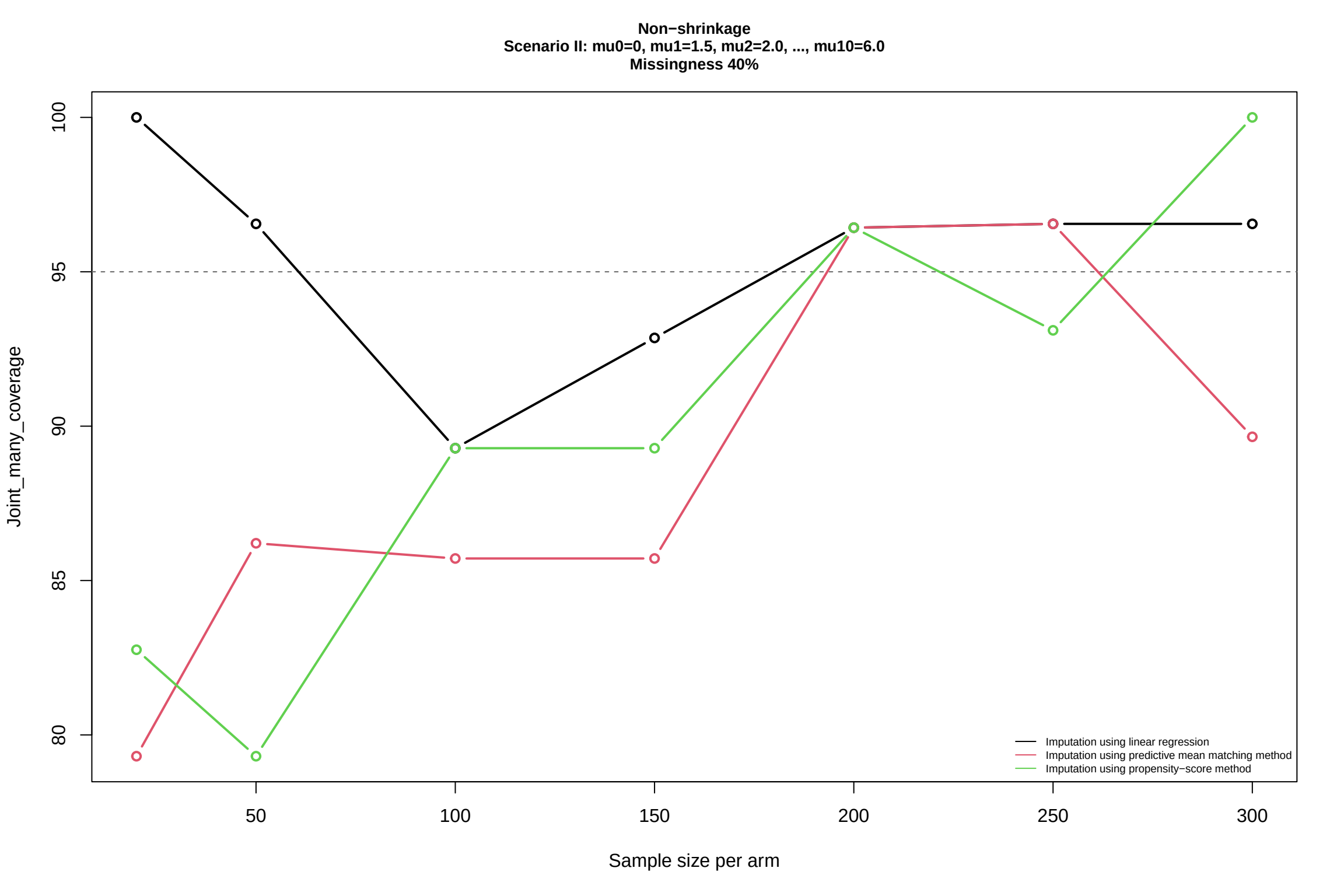

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 10%
Mean_many_CI_length
2
3
4
5
6
50
100
150
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250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

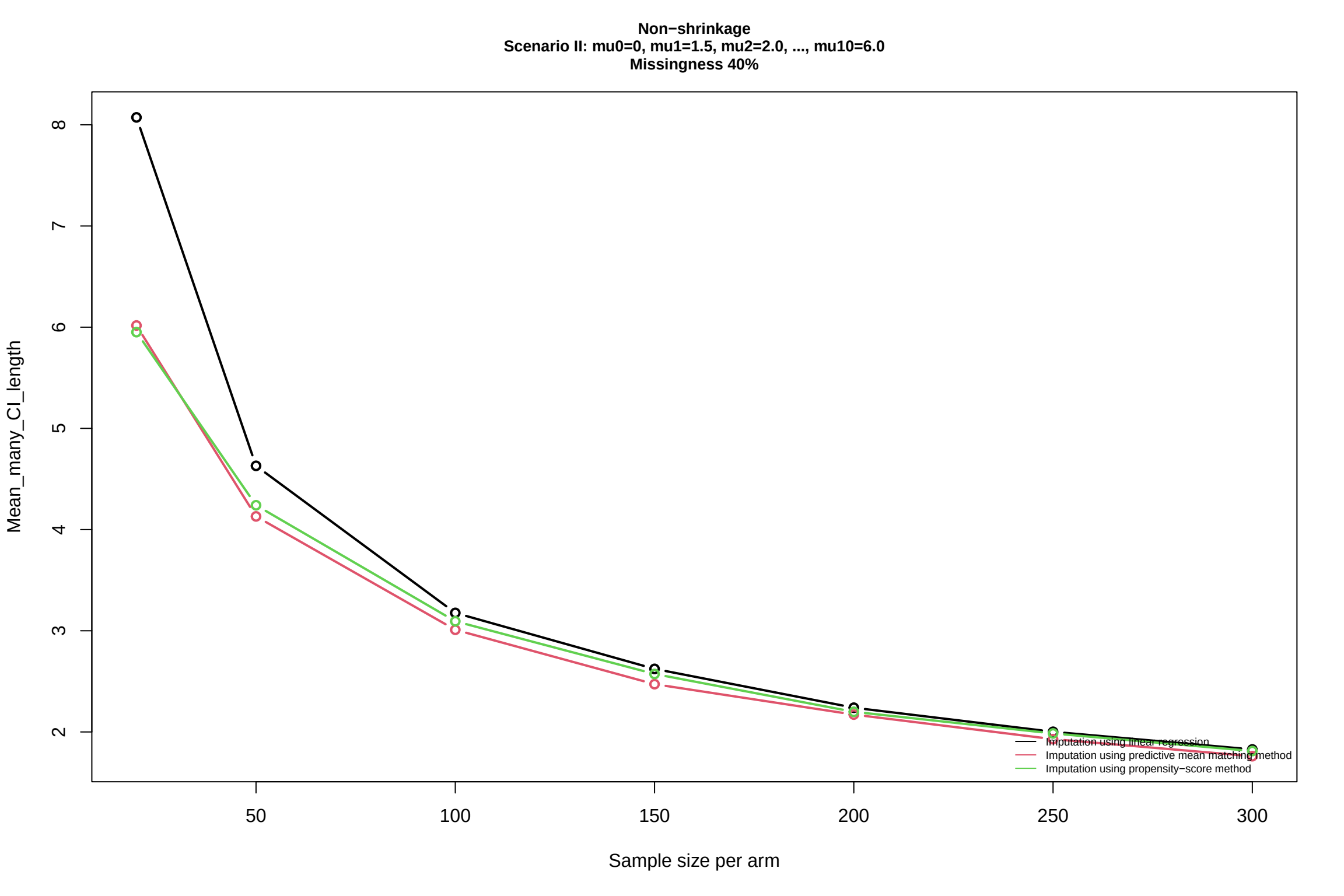

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 20%
Joint_many_coverage
86
88
90
92
94
96
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

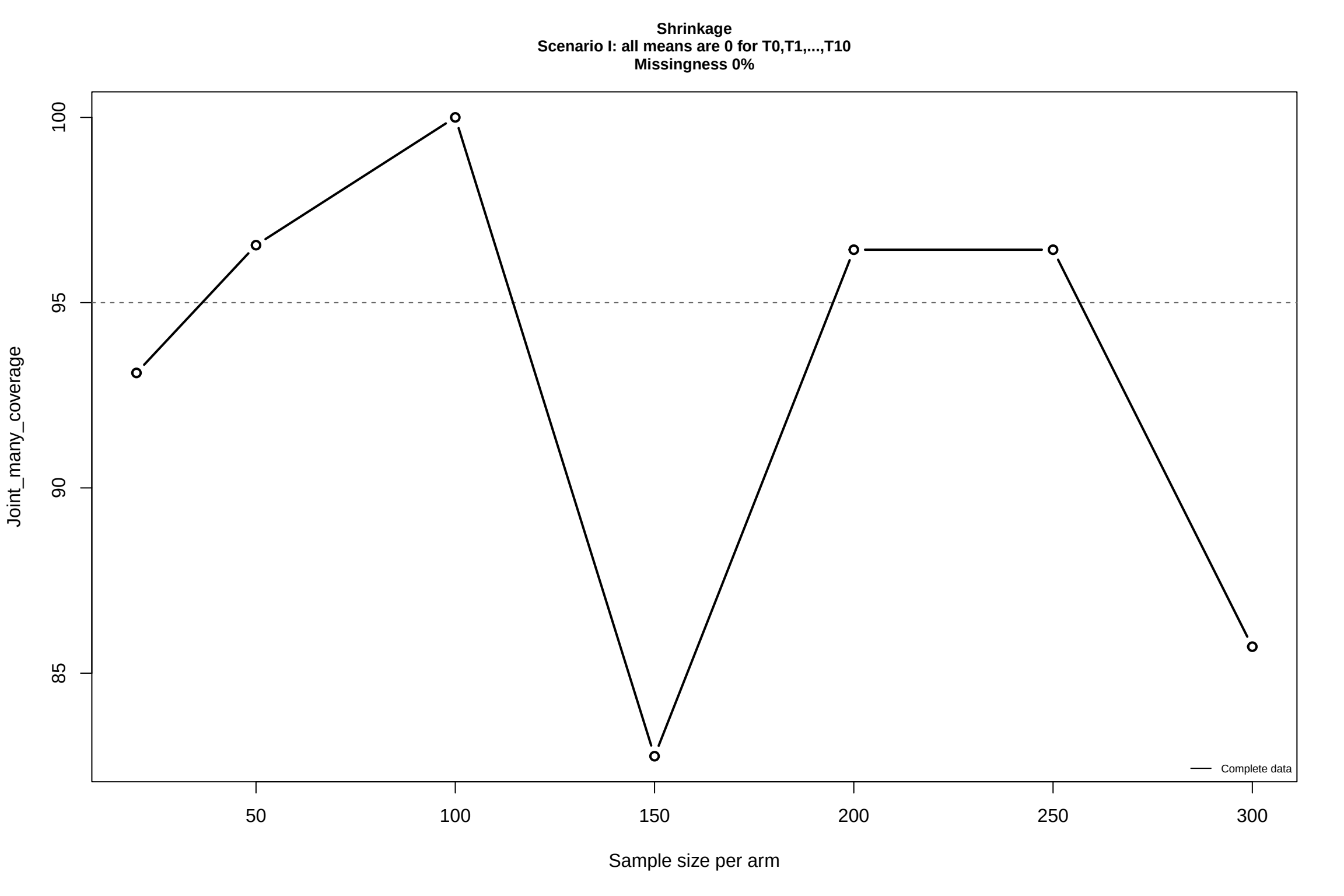

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

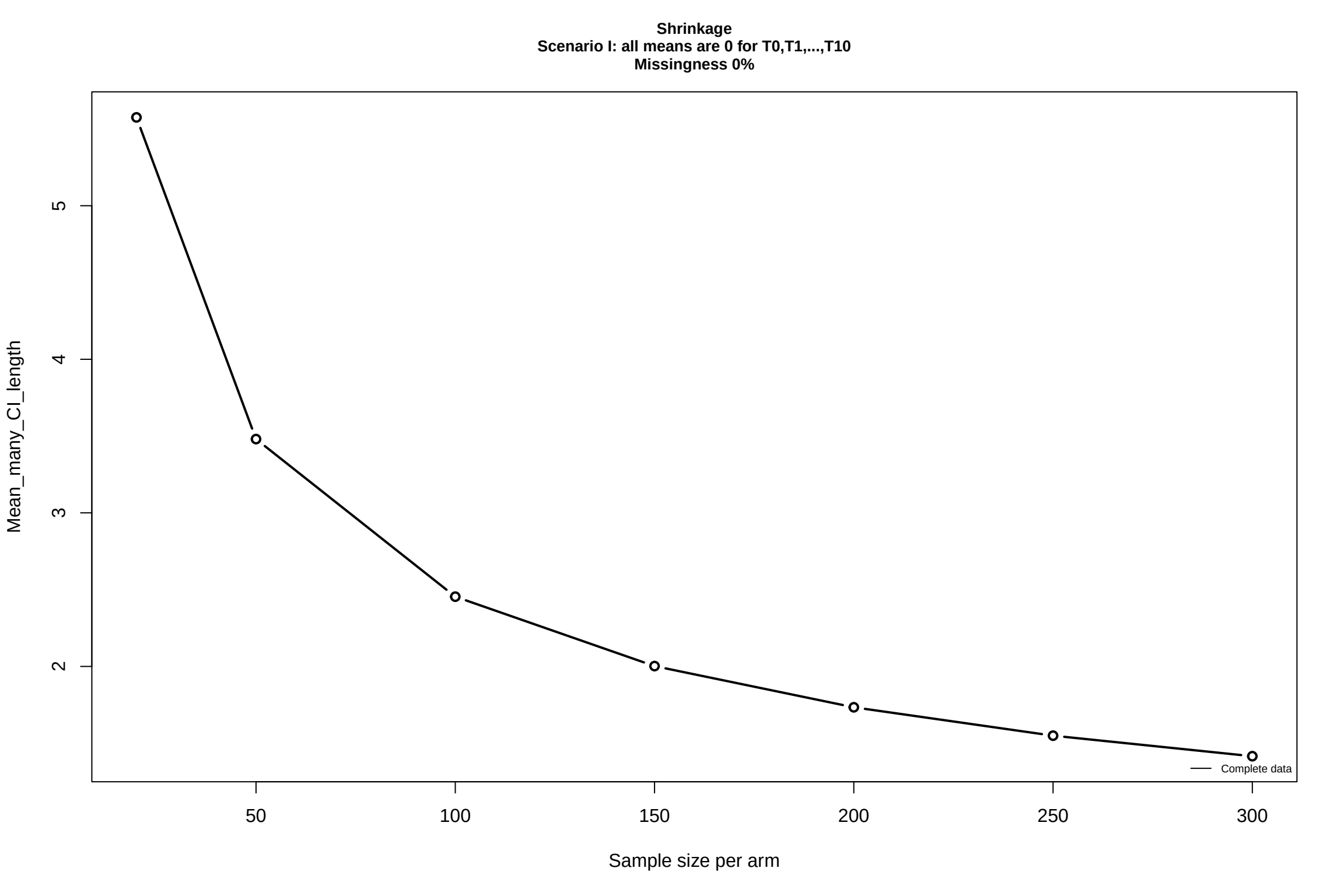

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

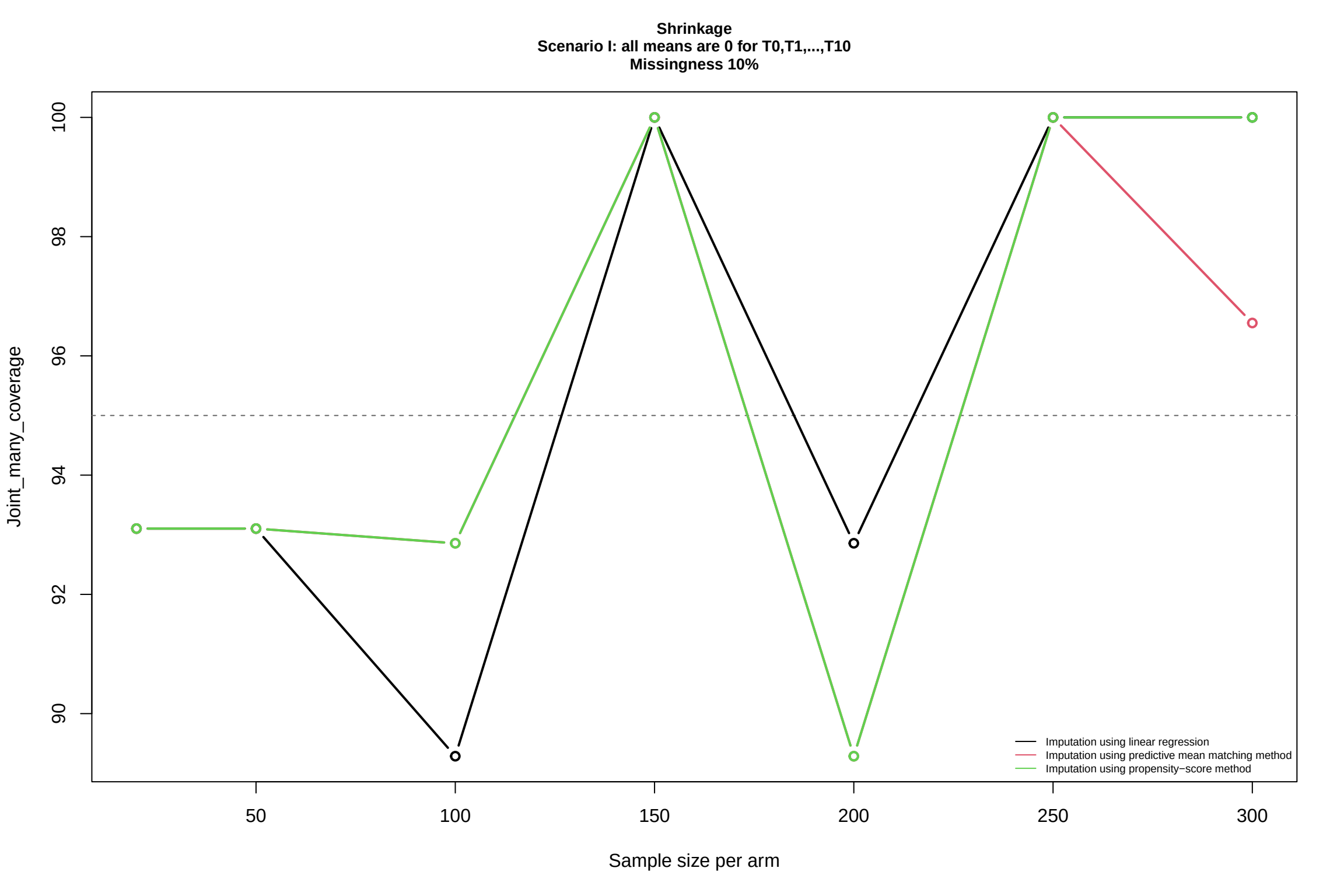

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 30%
Mean_many_CI_length
2
3
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100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

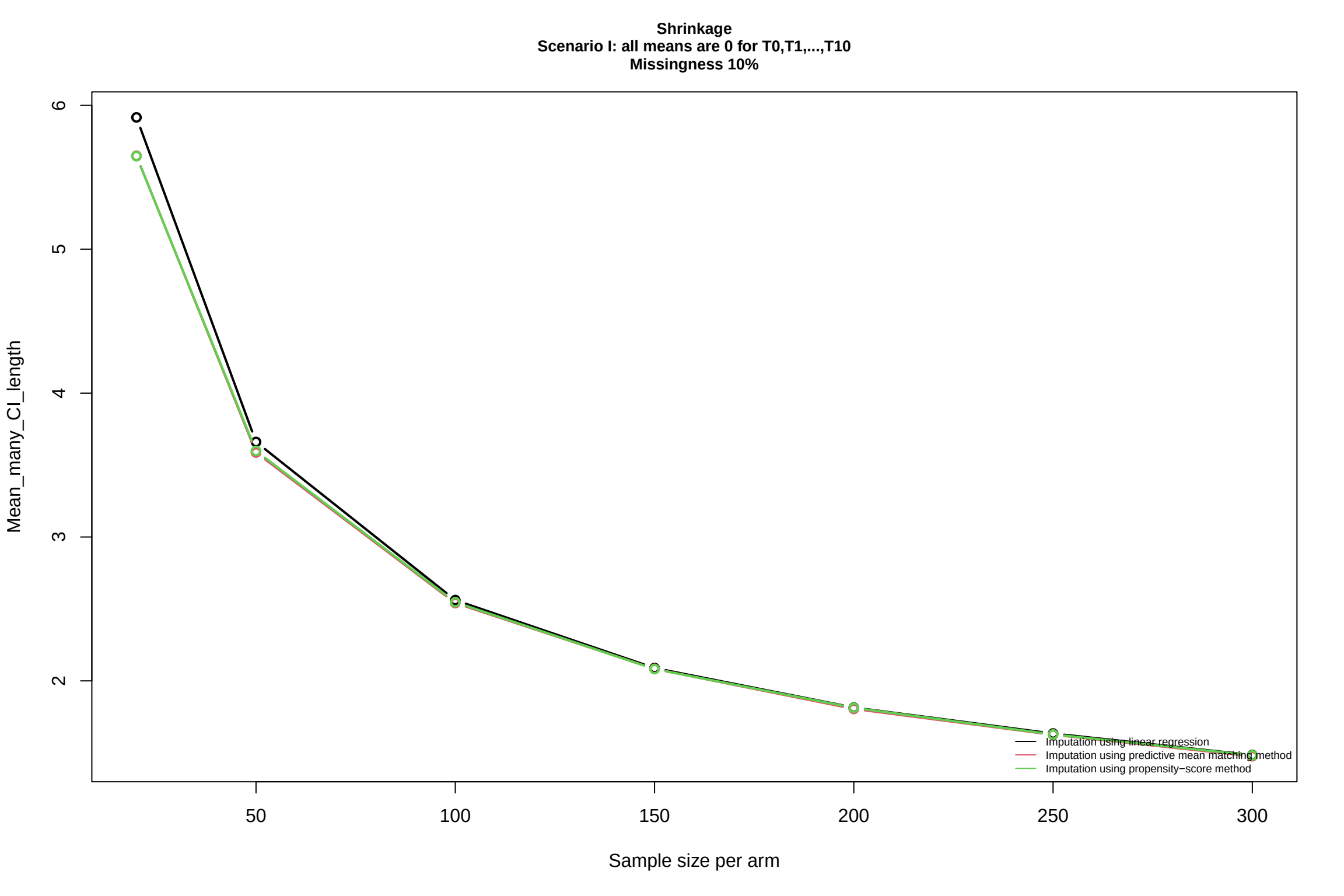

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 40%
Joint_many_coverage
100
95
90
85
80
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200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

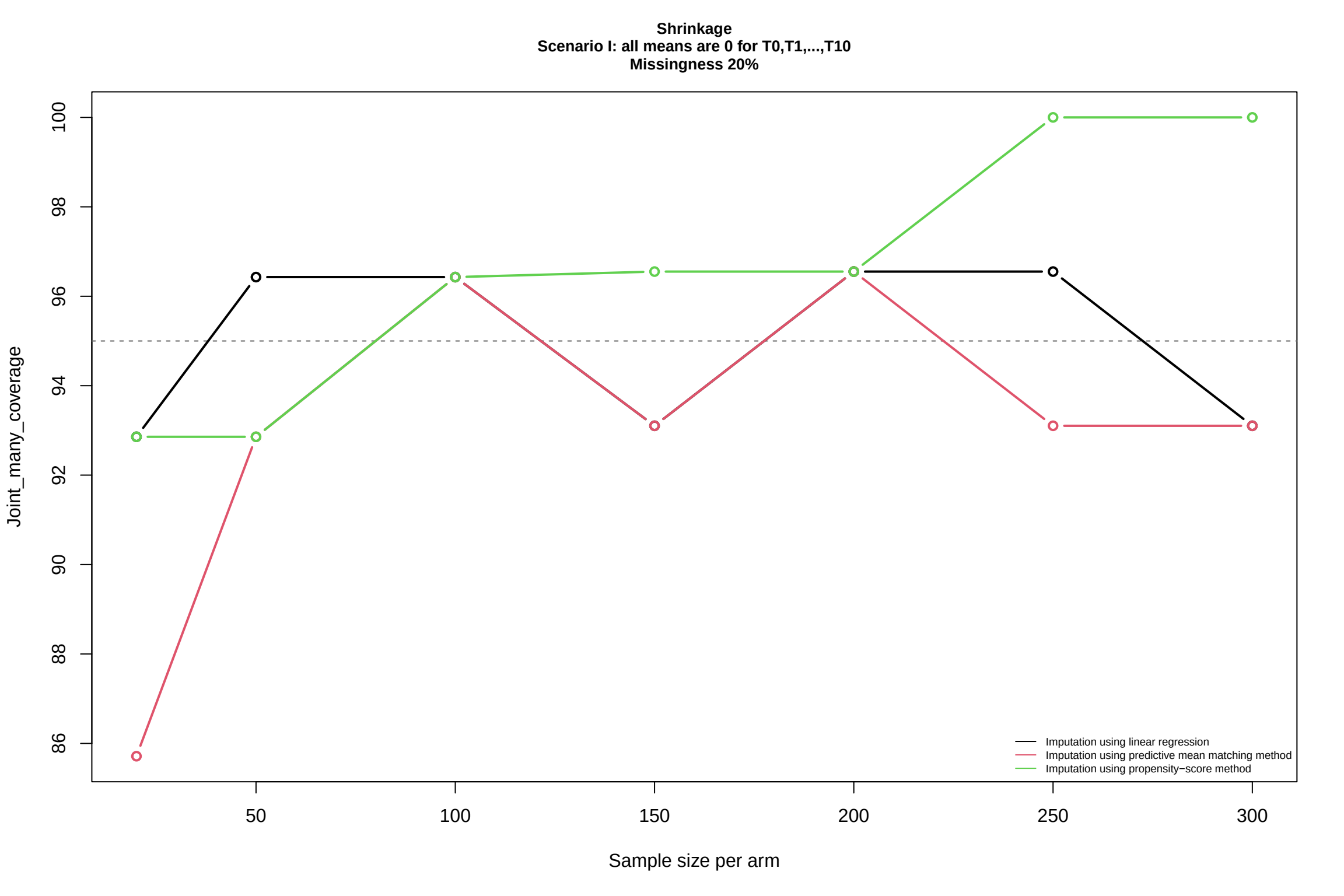

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 40%
Mean_many_CI_length
2
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7
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150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

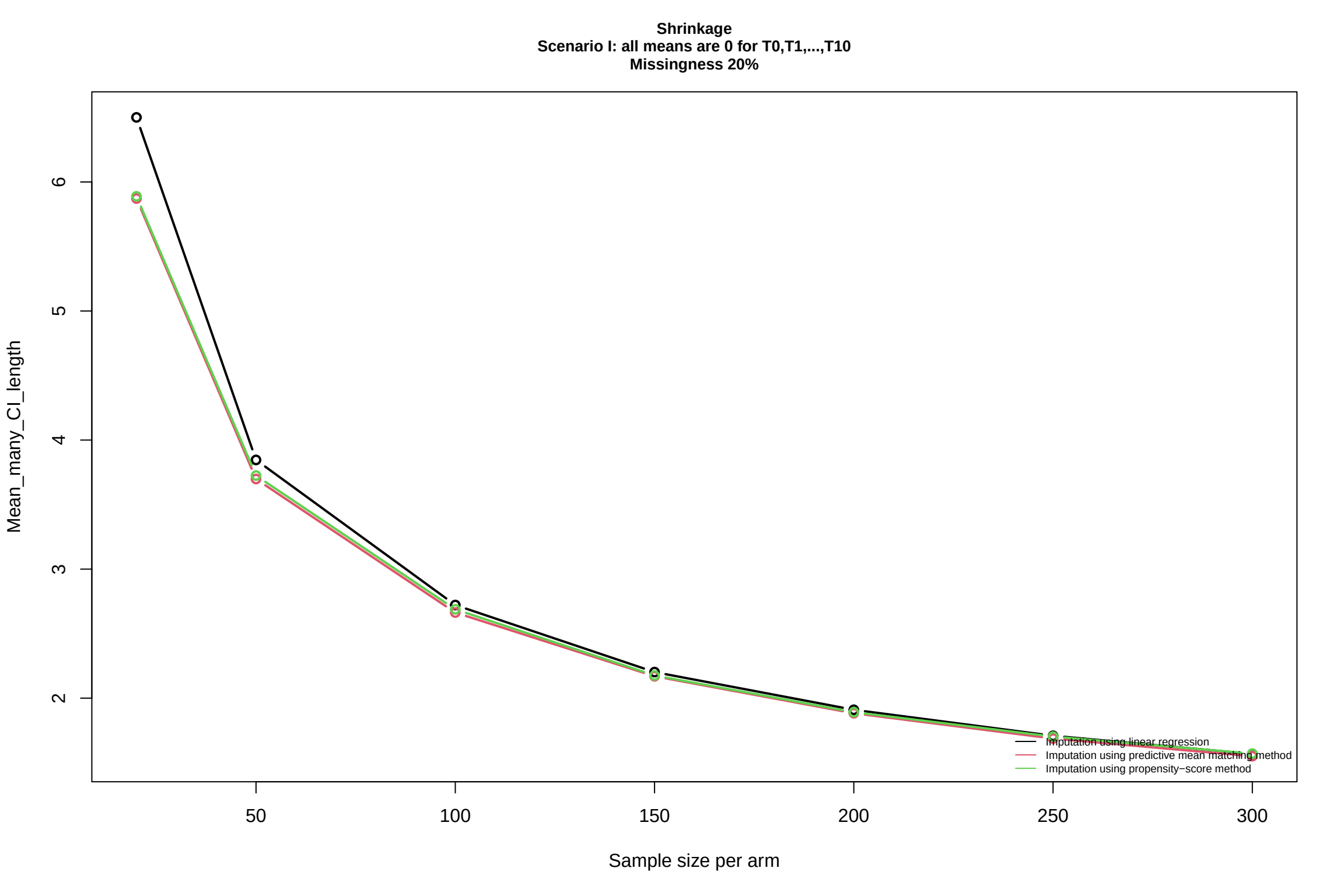

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 0%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Complete data

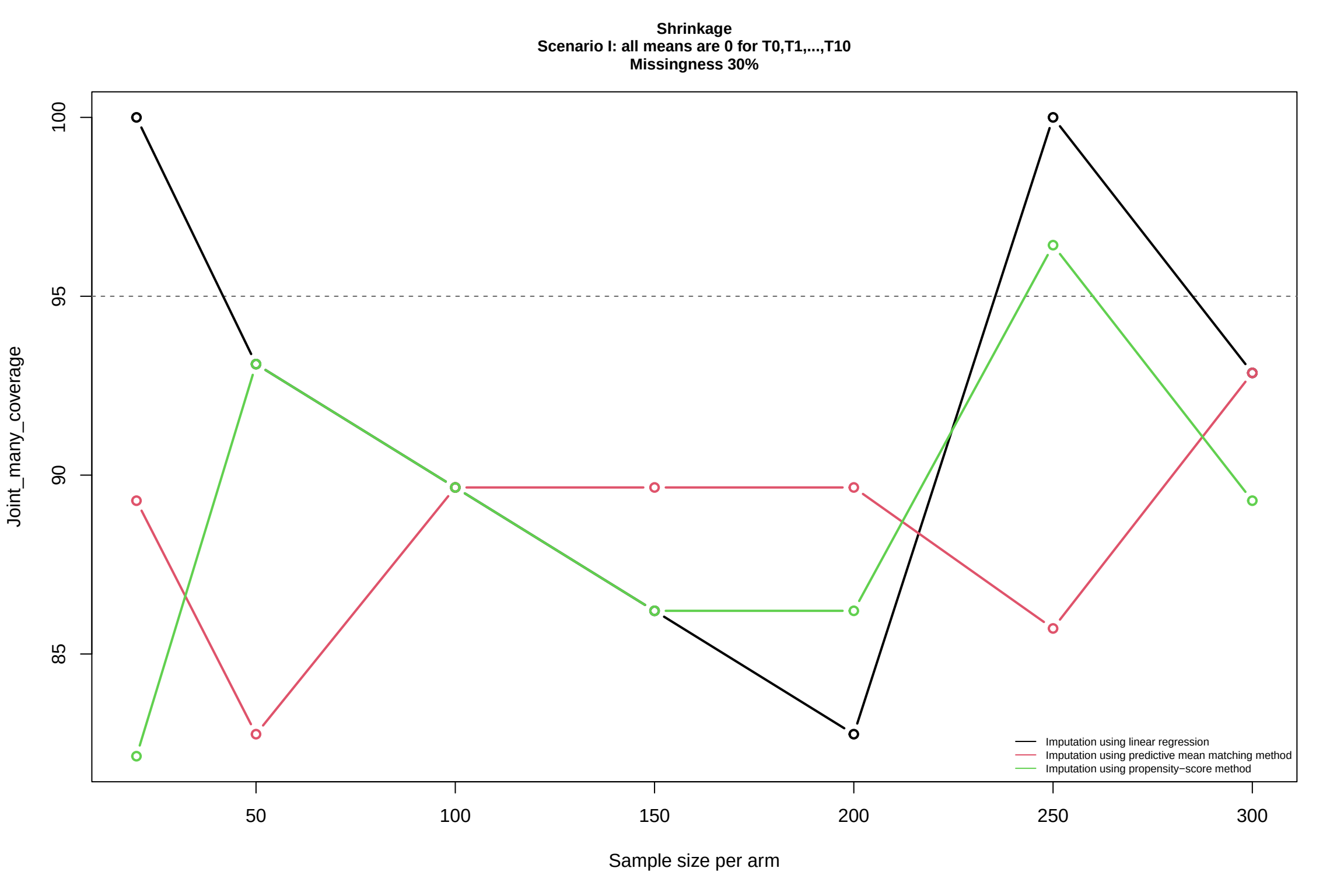

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

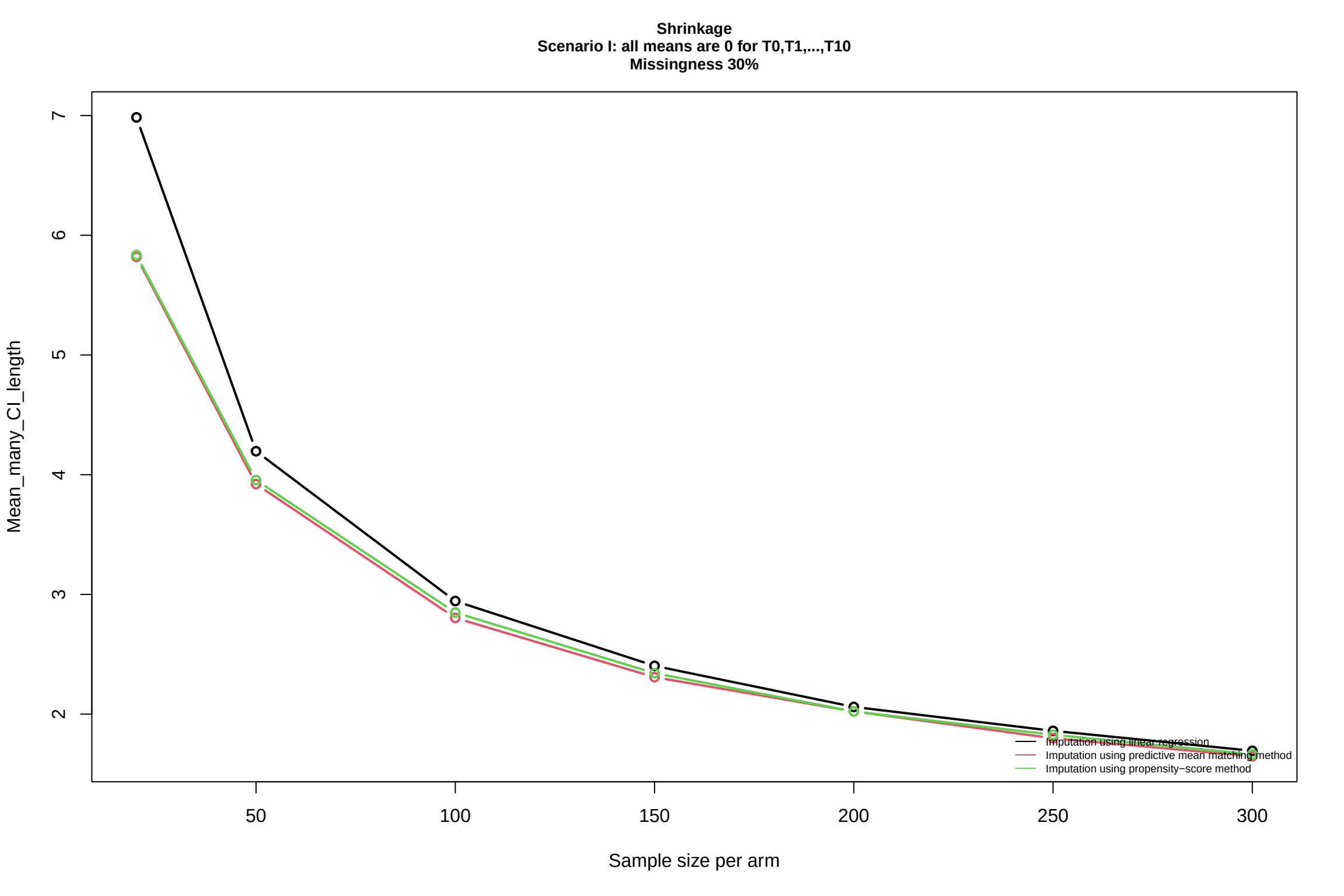

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

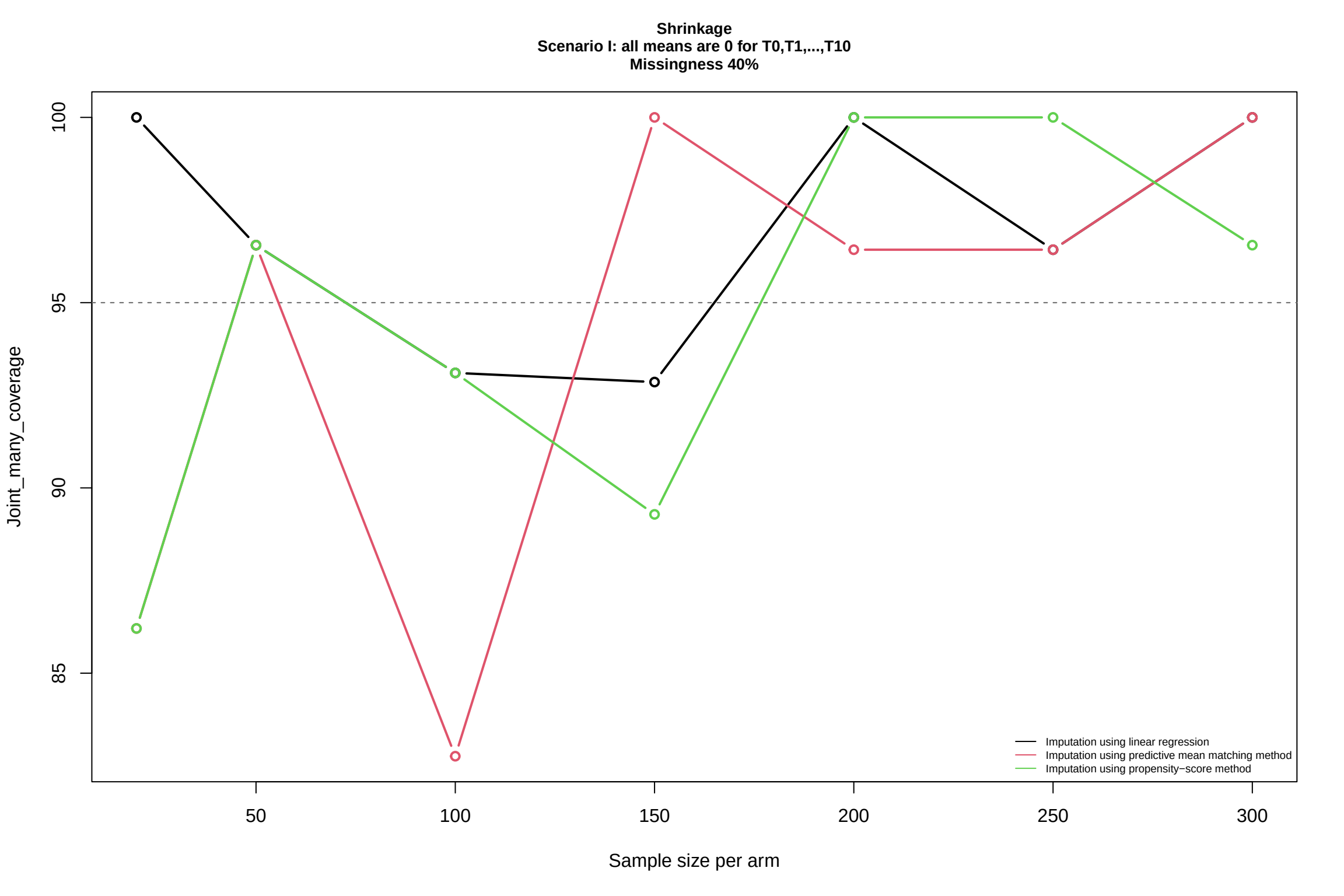

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 10%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

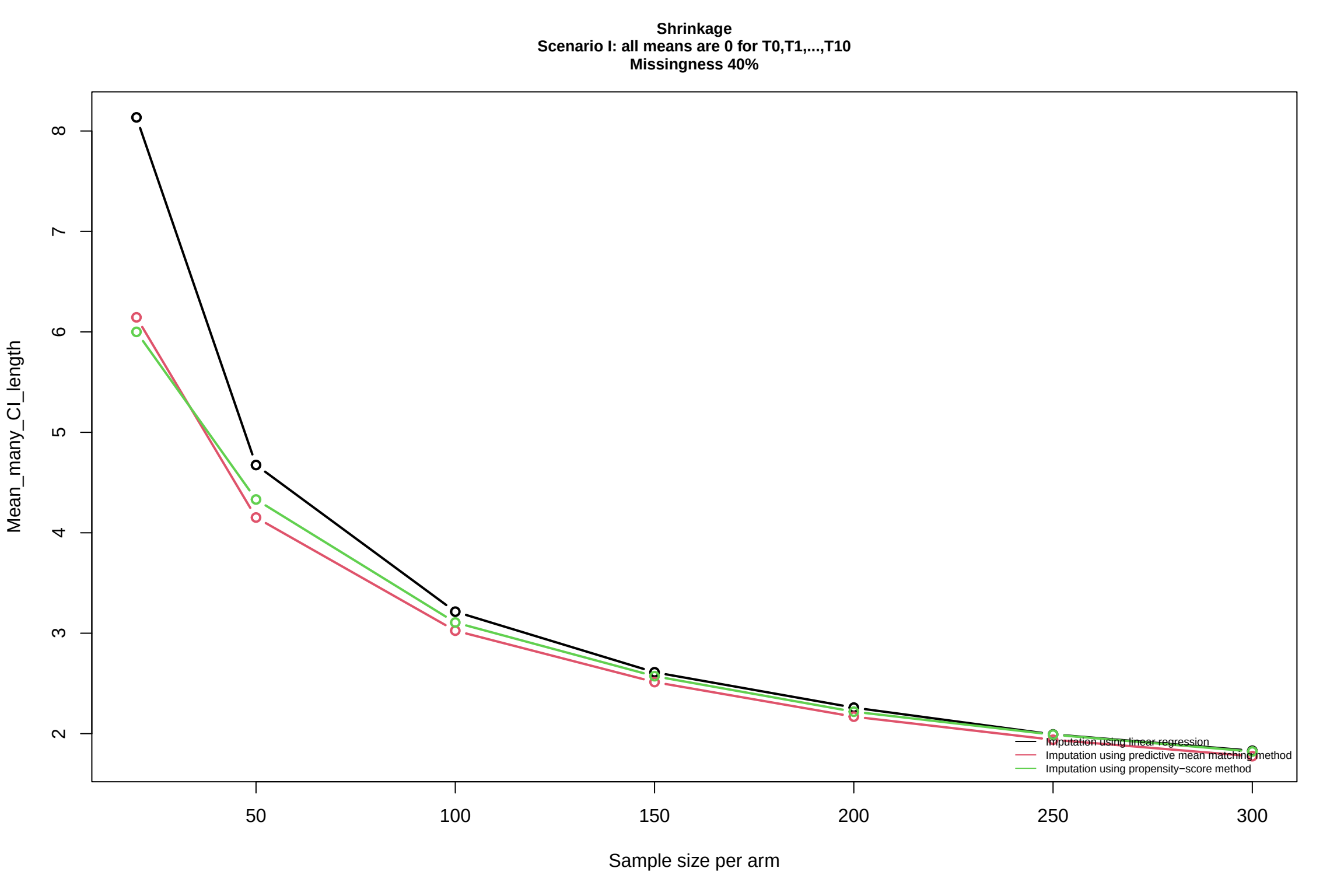

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 20%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

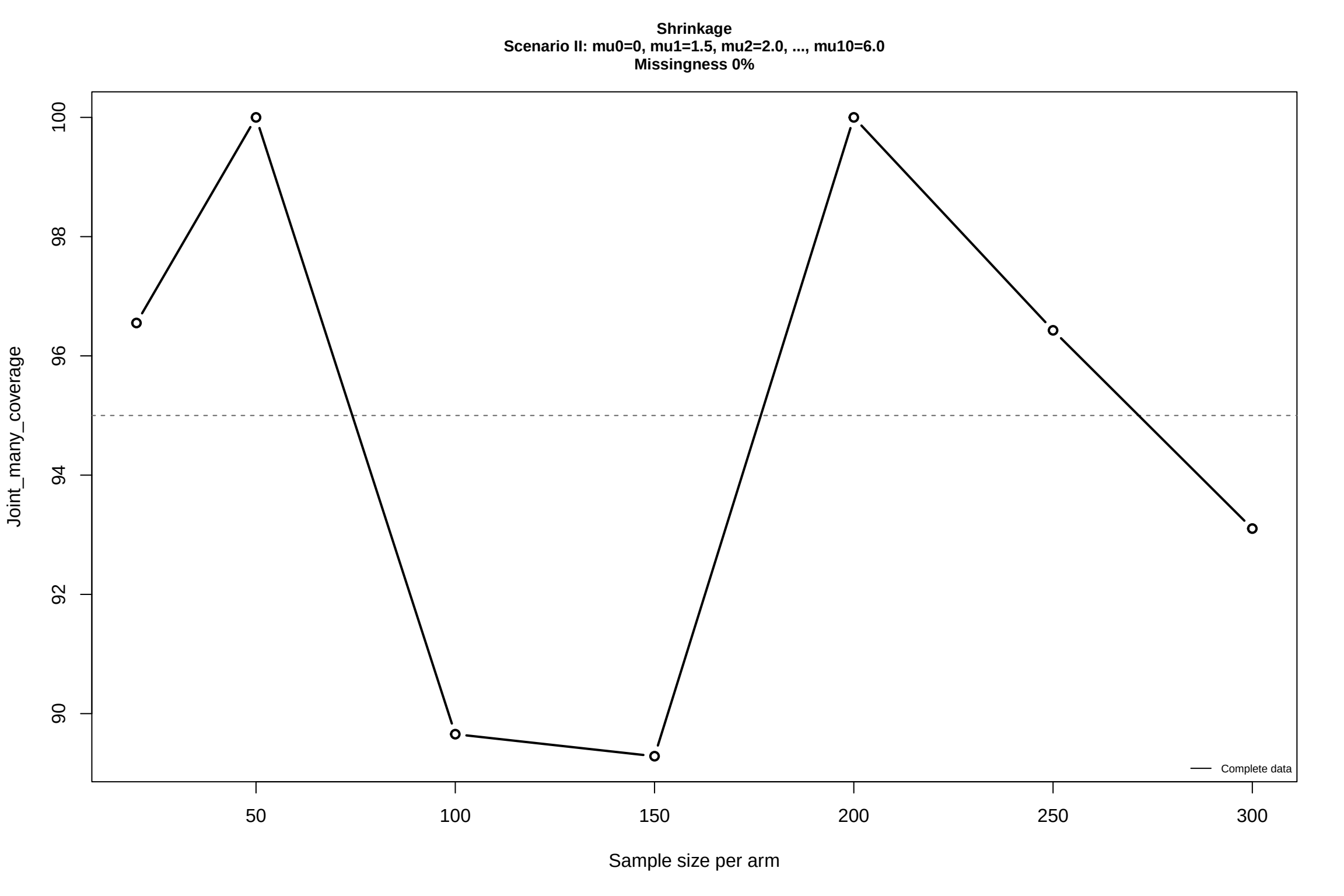

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 20%
Mean_many_CI_length
2
3
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50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

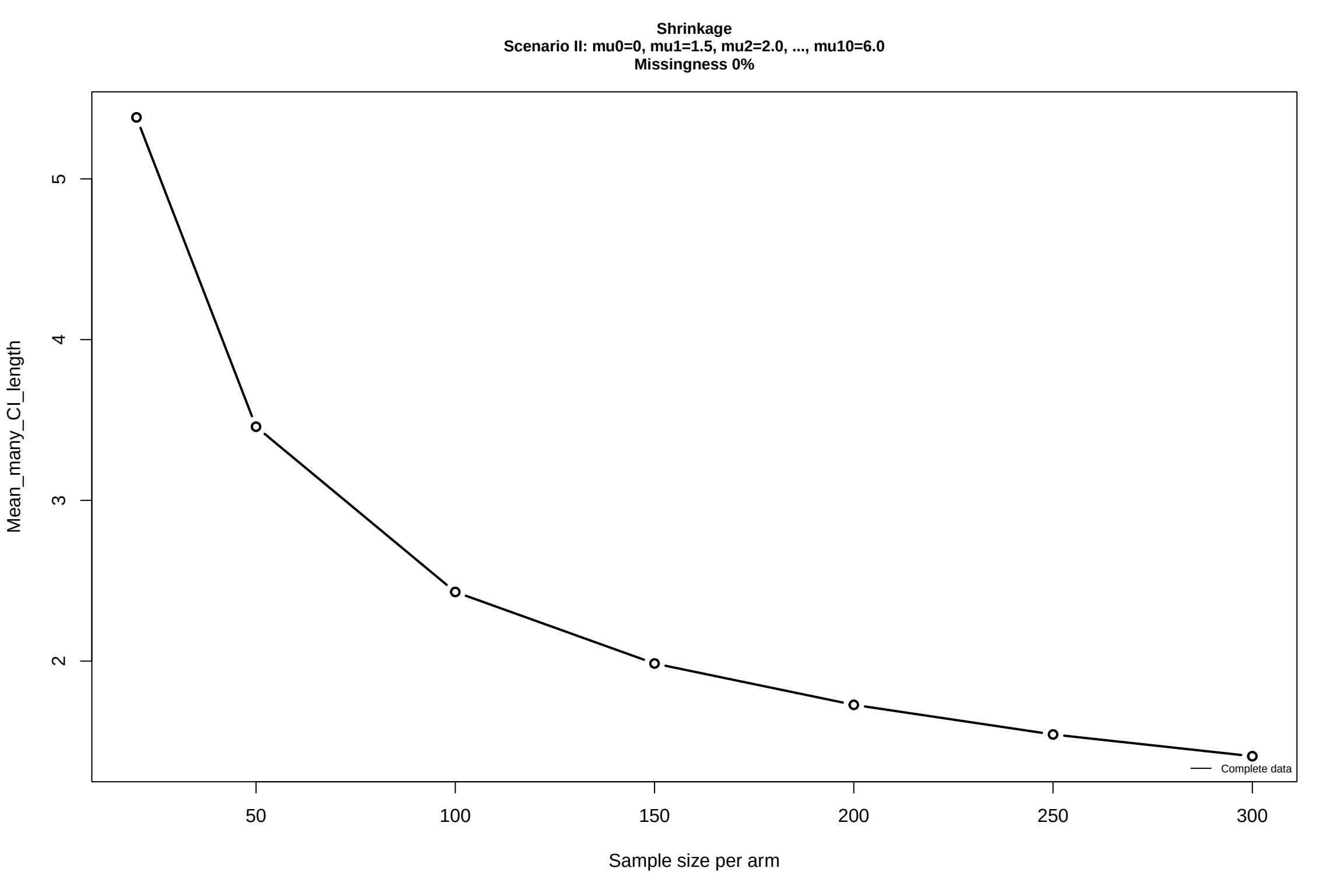

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

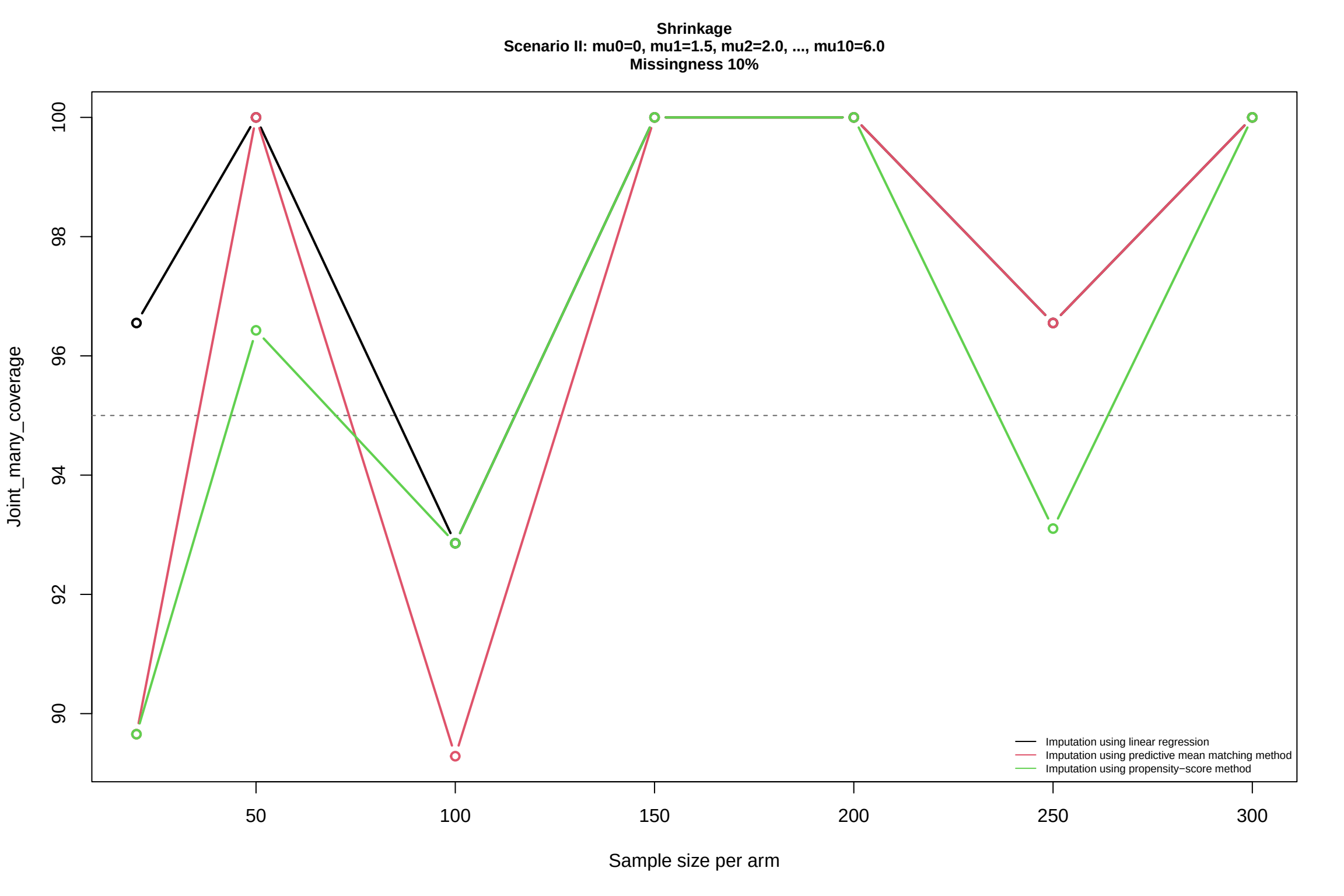

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

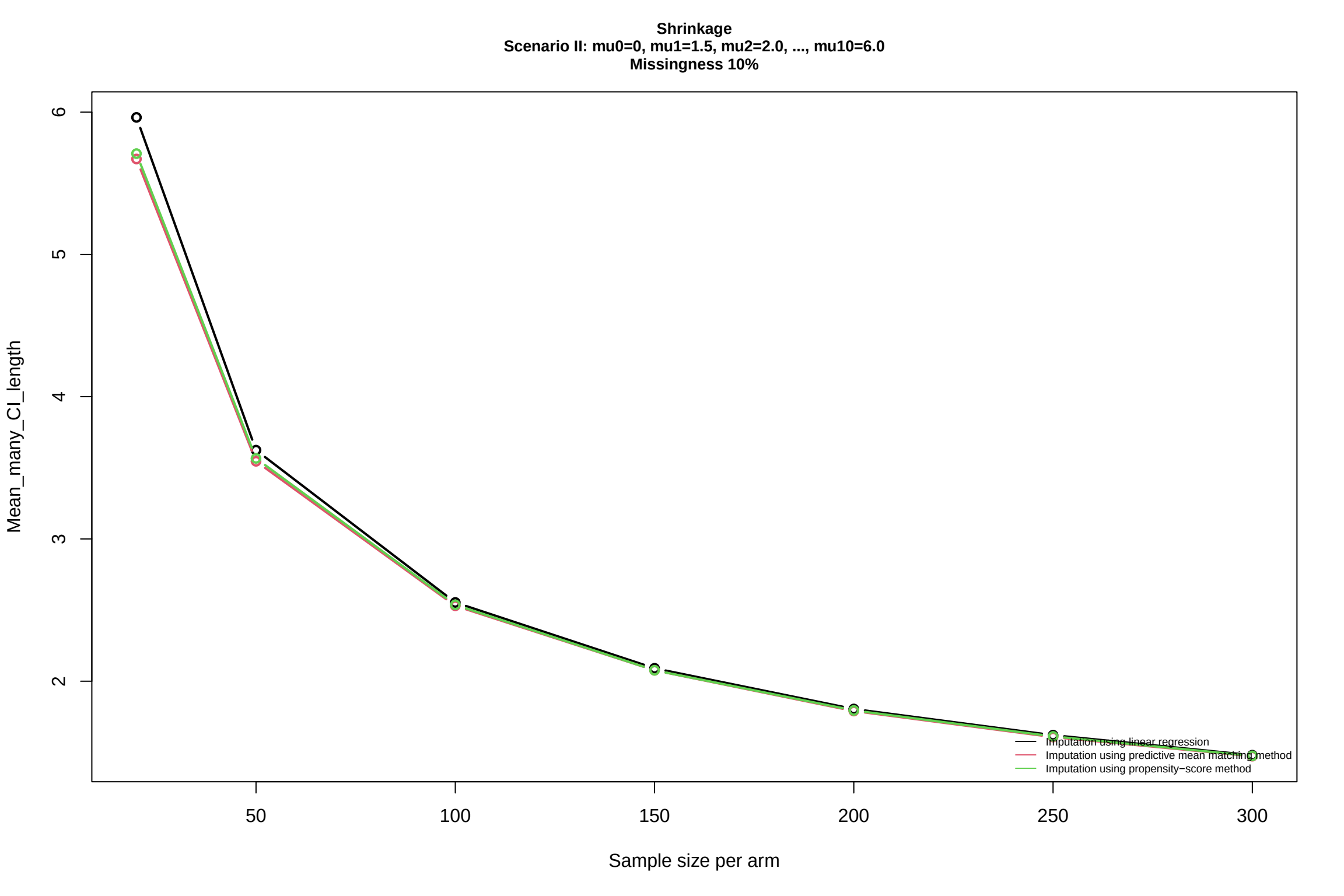

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 40%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

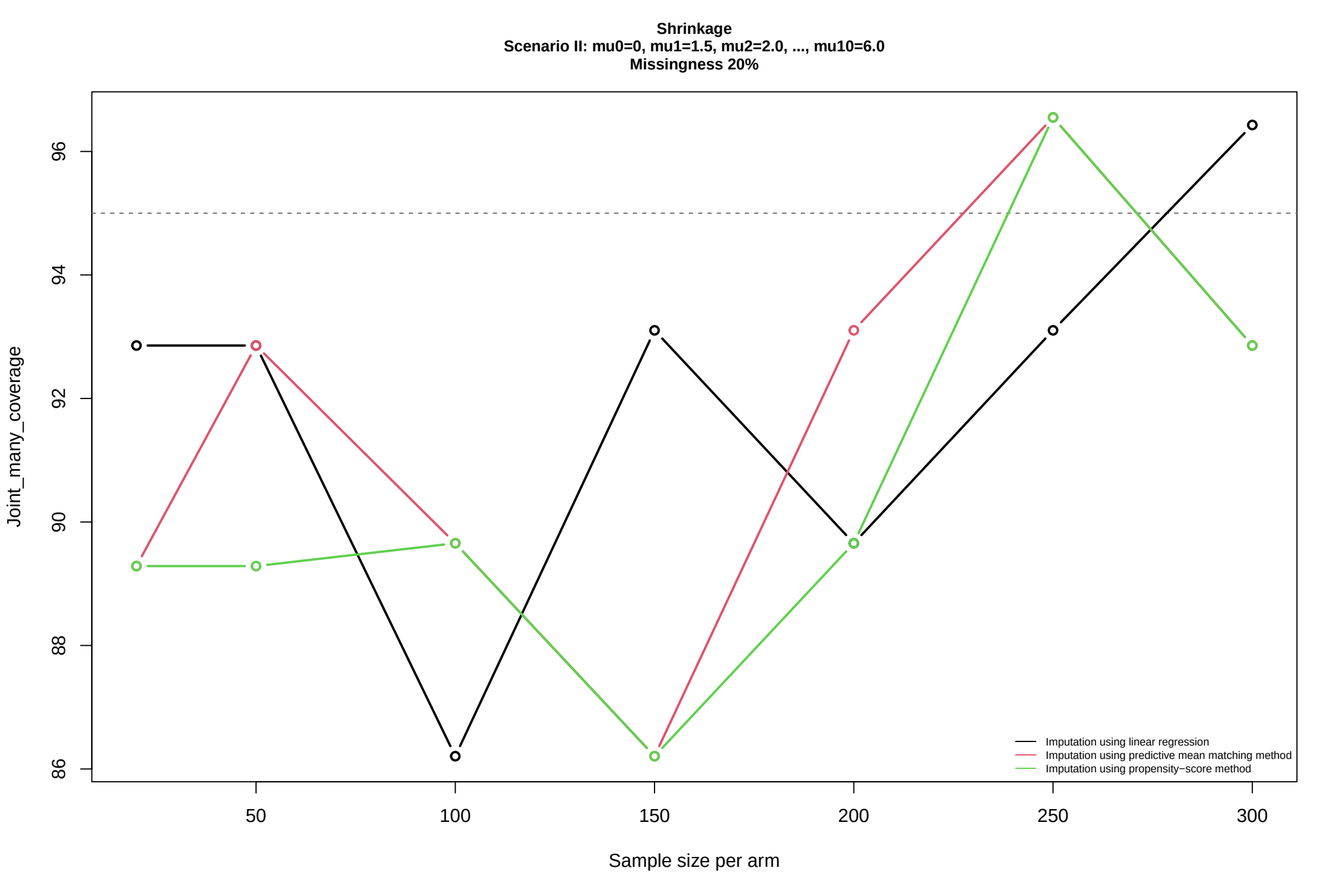

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T10
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
8
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

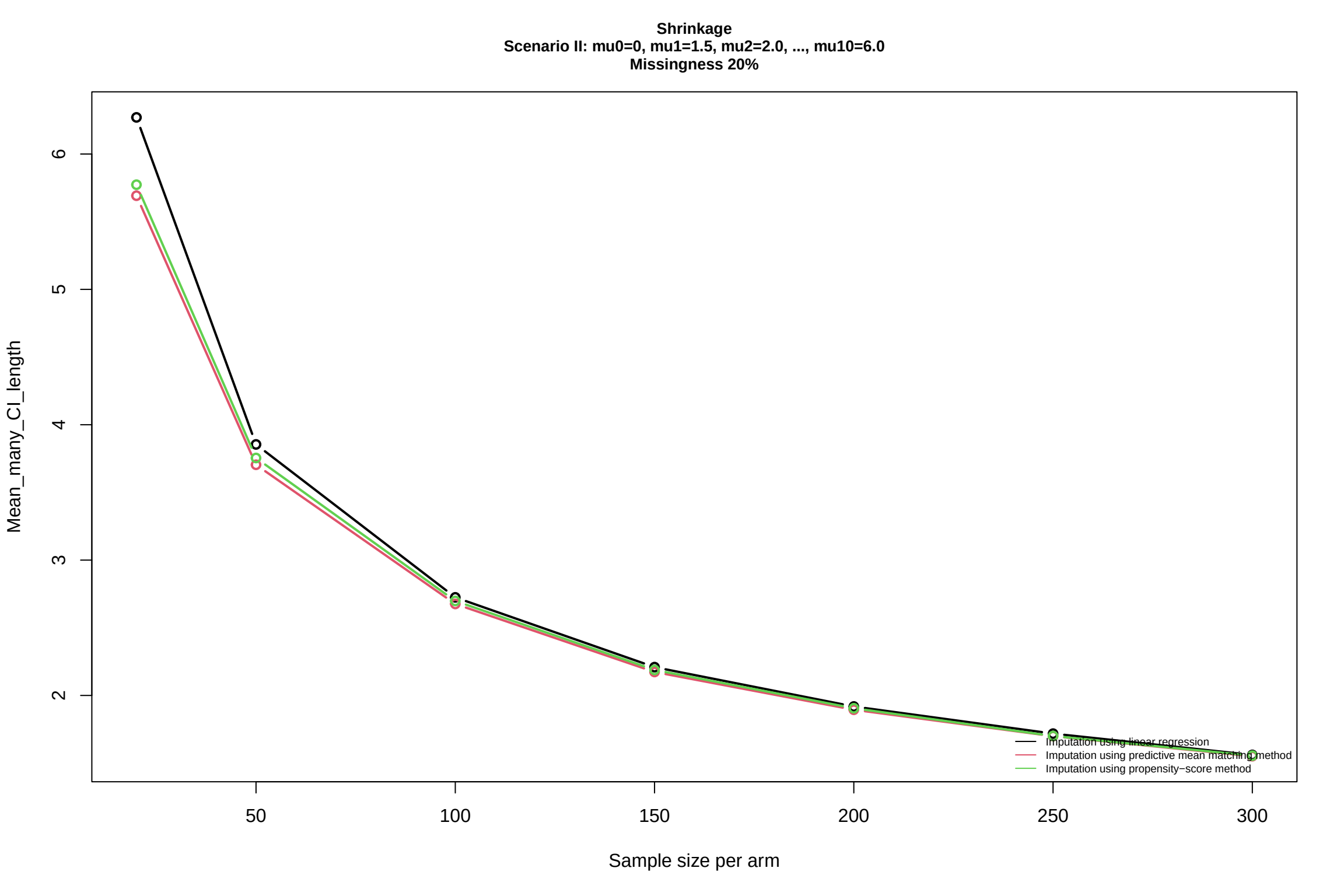

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 0%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Complete data

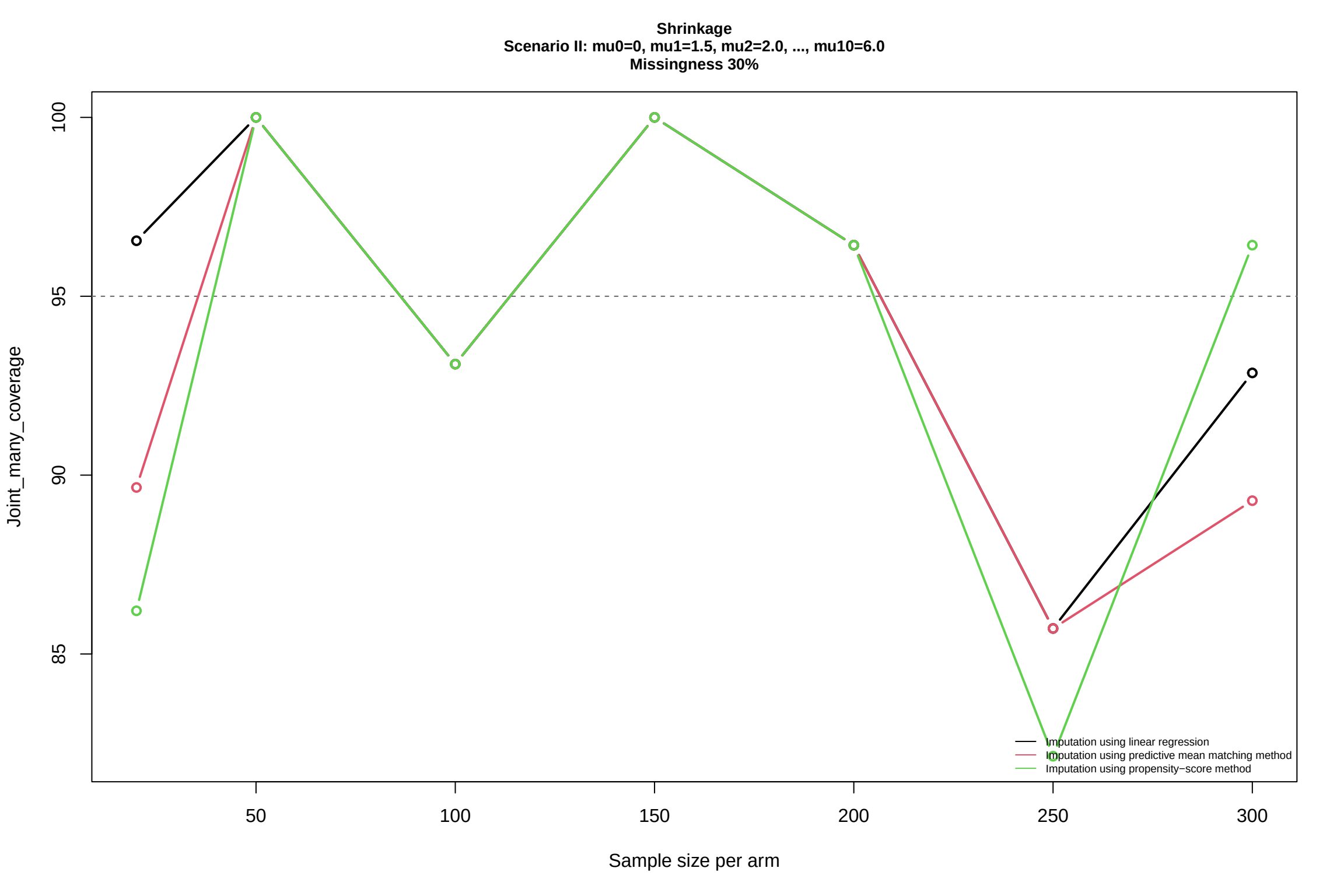

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

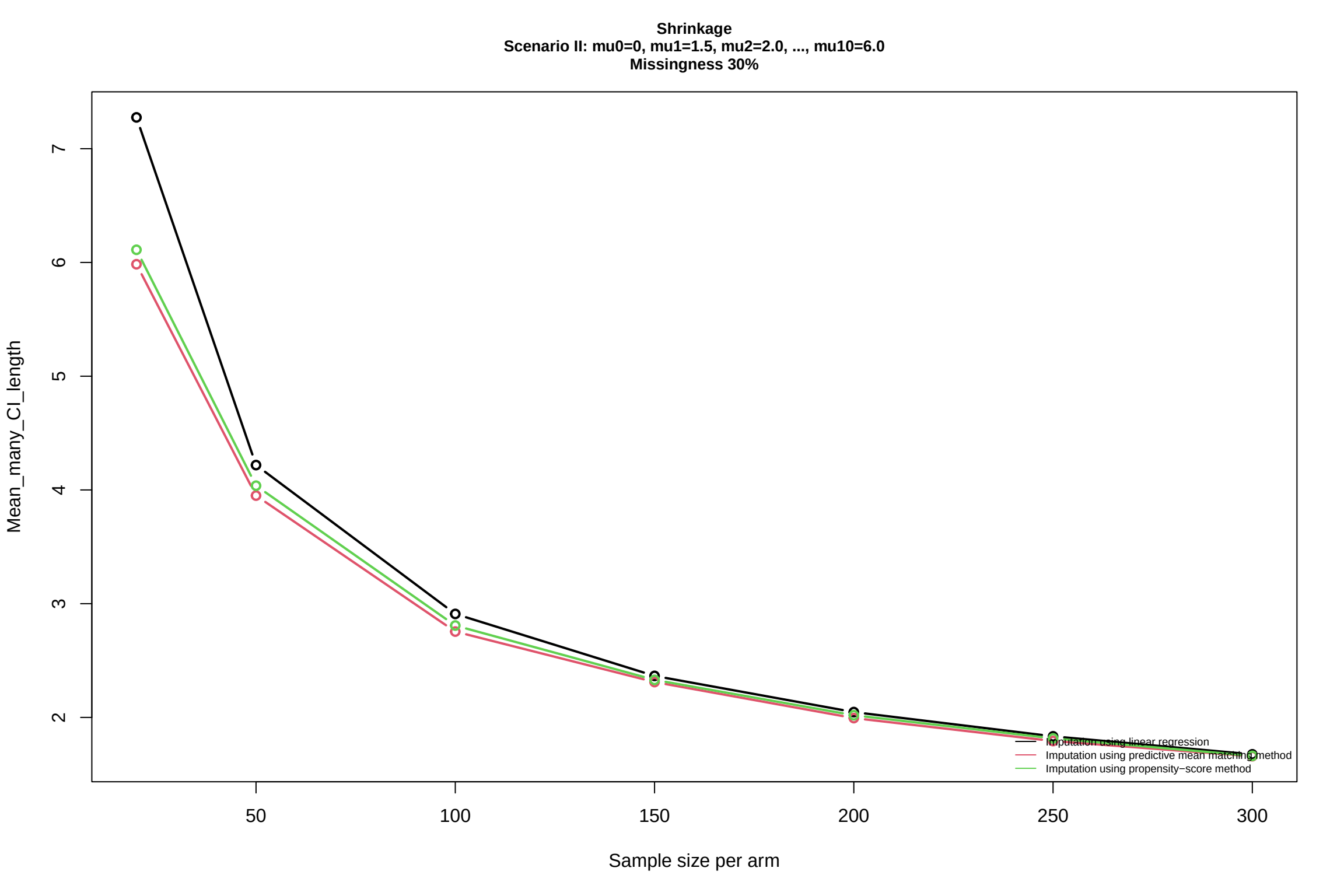

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

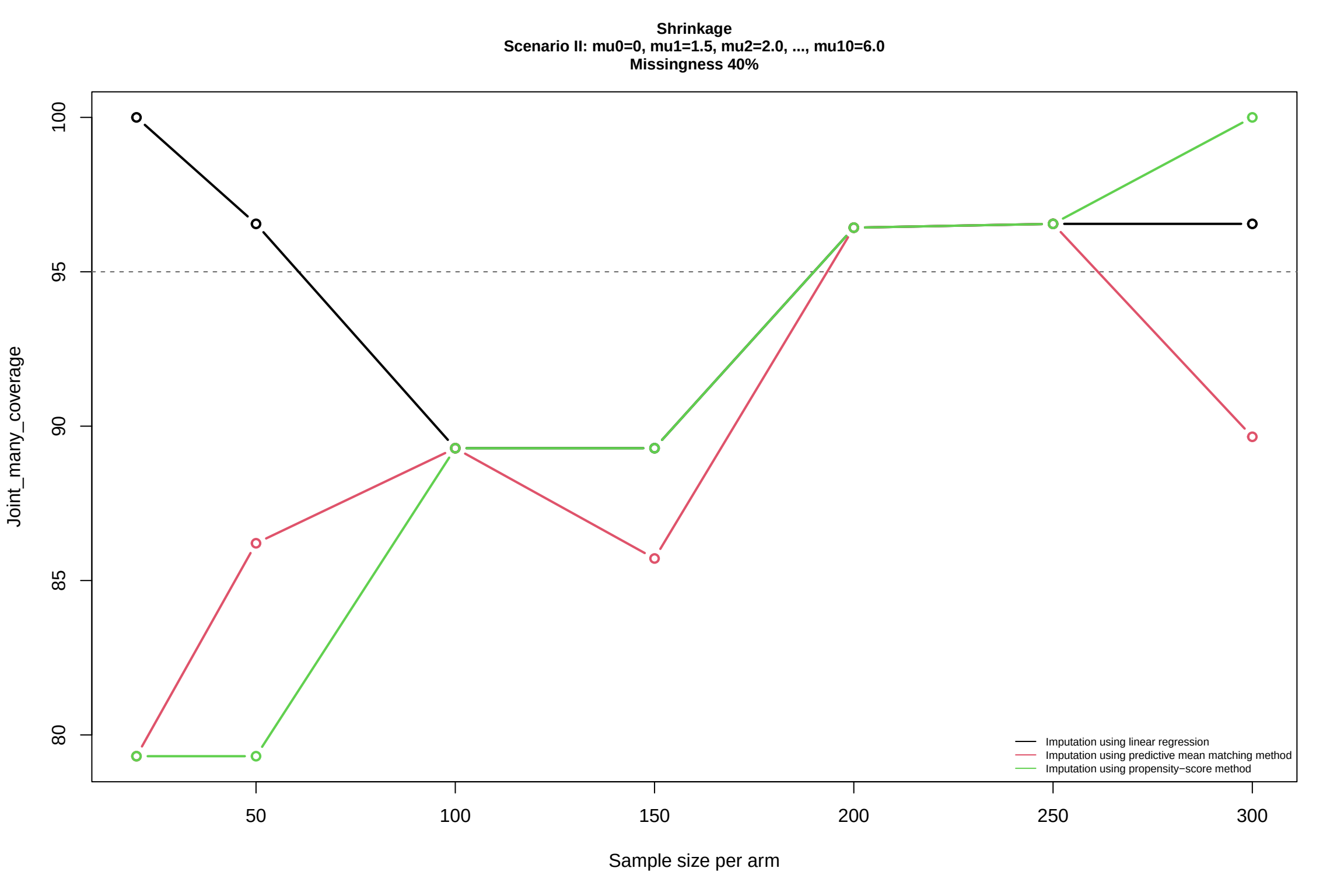

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 10%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

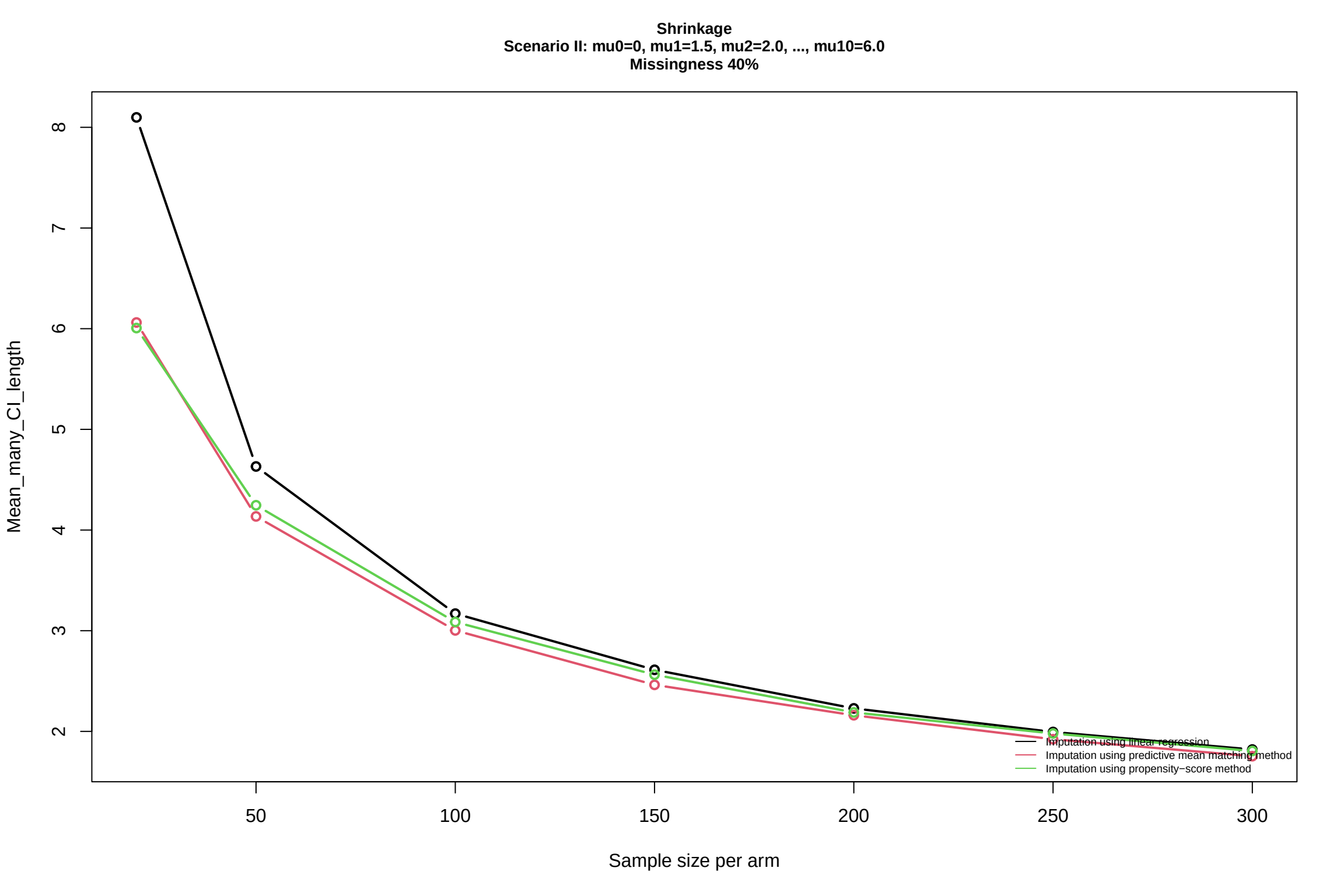

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 20%
Joint_many_coverage
86
88
90
92
94
96
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 40%
Joint_many_coverage
100
95
90
85
80
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0
Missingness 40%
Mean_many_CI_length
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3
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Sample size per arm
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